SIADS 699 CAPSTONE · UNIVERSITY OF MICHIGAN · MASTER OF APPLIED DATA SCIENCE

Beyond Prediction: Prescriptive AI for Strategic Energy Capital Allocation

A Governed Decision Intelligence System for Risk, Returns, and Investment Committee Choices Across Oil, Gas, Power, Renewables, and Storage

Alan Mössinger

Group 13 (solo project) · Mentor: Laura Danielle Stagnaro

School of Information, University of Michigan

Version 2 · November 2026

Contents

Abstract....3
1 The question....3
2 The decision, in plain words....4
3 Stage one — prediction: what the models do, and where they stop....5
4 Stage two — uncertainty: two thousand futures instead of one....6
5 Stage three — prescription: from two thousand futures to one book....7
6 The scorecard: five ways to choose, head to head....8
7 Stage four — the AI challenger: a committee, a checklist and a language model....9
8 How to read the evidence: "registered" and the lines C1 to C41....12
9 Conclusion: what lies beyond prediction....13
Statement of work....14
Appendices....15
Appendix A — The evaluation record....15
Appendix B — The AI advisor and the challenger....22
Appendix C — Where things are....24
Appendix D — Supporting figures....25
Appendix E — Part I in full: the developer’s decision, which queue requests to fund....34
Appendix F — Part II in full: the company’s decision, how to split one budget....44
Appendix G — Every finding, its evidence and its verdict....49
Appendix H — The audit of 17 October 2026....51
Appendix I — Capital allocation memo....52
Appendix J — Glossary....54
Appendix K — The report in full....55
References....76

Abstract

Energy companies spend one capital budget across investments that behave in different ways: an oil lease that produces now, a power plant that may never be built. This capstone asks whether better predictions are enough to spend that budget well, and shows what lies beyond them. A six-stage decision intelligence system was applied to one decision dated 1 January 2025, using only what was public that day: $3.82 billion across twenty Gulf of Mexico leases and twenty ERCOT interconnection requests, scored on 2,000 held-out scenarios against pre-registered criteria. Machine-learning forecasts ranked the candidates better than a hand-built lookup but did not produce a demonstrably better portfolio. A risk-aware optimizer beat ranking by expected value by $219M [+$13M, +$401M], and by $312M [+$110M, +$506M] once every portfolio was financed the same way. On 48 simulated committee proposals, deterministic tools improved the committee's own objective by $76M per case, a governed language model added $43M more and named every flaw of argument, and the optimizer would have added $134M beyond both. The value of AI here lies past prediction: in pricing the tail, in optimizing under the company's own rules, and in a challenger that argues but never decides. The results are modelled, not realized.

1 The question

The title poses a question. A data science team can now predict, with respectable accuracy, which power projects will be built, how much an oil lease will produce and roughly where prices are heading. Is that enough to spend a capital budget well? If not, what lies beyond prediction, and is it worth going there?

This report answers with one worked decision. Picture an energy company on 1 January 2025 with $3.82 billion to commit and forty candidates: twenty producing Gulf of Mexico oil and gas leases that could be redeveloped, and twenty requests waiting in the Texas (ERCOT) interconnection queue to build a solar, wind, gas or storage plant. The company must decide which to fund, stage, wait on or reject. Nothing used to decide was published after that date, so the decision can be judged a year later against what happened.

Four characters take turns at it. The *committee* is the human: it reads proposals, argues, and usually ranks candidates by expected value. The *forecaster* is the machine-learning layer: models of completion, output and prices. The *optimizer* chooses the whole portfolio at once under the company's own rules. The *challenger* is a governed language model that reads the committee's proposal and argues back, but never computes a number and never decides. Each is scored on the same held-out scenarios, so the report can say in dollars what each adds.

The short answer: prediction alone is not enough here. A better forecast reorders the candidates; it does not say how much risk to take, how the candidates lose together, or whether the company can pay for what it has chosen. The value appears when uncertainty is modelled jointly, when the portfolio is optimized against a priced tail, and when the committee's reasoning is challenged before it signs.

2 The decision, in plain words

A *book* is a set of candidates funded together under one budget — a trader's word for a portfolio, kept because every result is about the book, not about any candidate alone. The budget is $3.82 billion, 40% of what all forty candidates would cost, and no technology may take more than 40% of it.

The two kinds of candidate are different bets. A *lease redevelopment* spends its capital now — priced at $40,000 per added barrel a day on the committee's cost grid — and lifts output by 20% after two years. It rarely loses money, but all twenty leases sell the same oil, so when they lose, they lose together. A *queue request* is the opposite. It costs little to hold, a few million in deposits and studies, and becomes expensive the day its interconnection agreement is signed and hundreds of millions of construction capital are called. Most never reach that day: of the capacity that asked to connect in 2000–2019, 12.6% was operating by the end of 2024 and 76.9% had withdrawn. Its value depends on Texas power prices, which barely move with oil.

So a request charges the budget its deposits plus a *reserve* for the construction it may call; a lease charges its full capital. A committee that ranks by expected NPV fills the budget with the largest leases and requests and takes a tail it has not priced: five leases on one oil price, eight plants on one power price. That is the problem the six stages were built to solve (Figure 1).

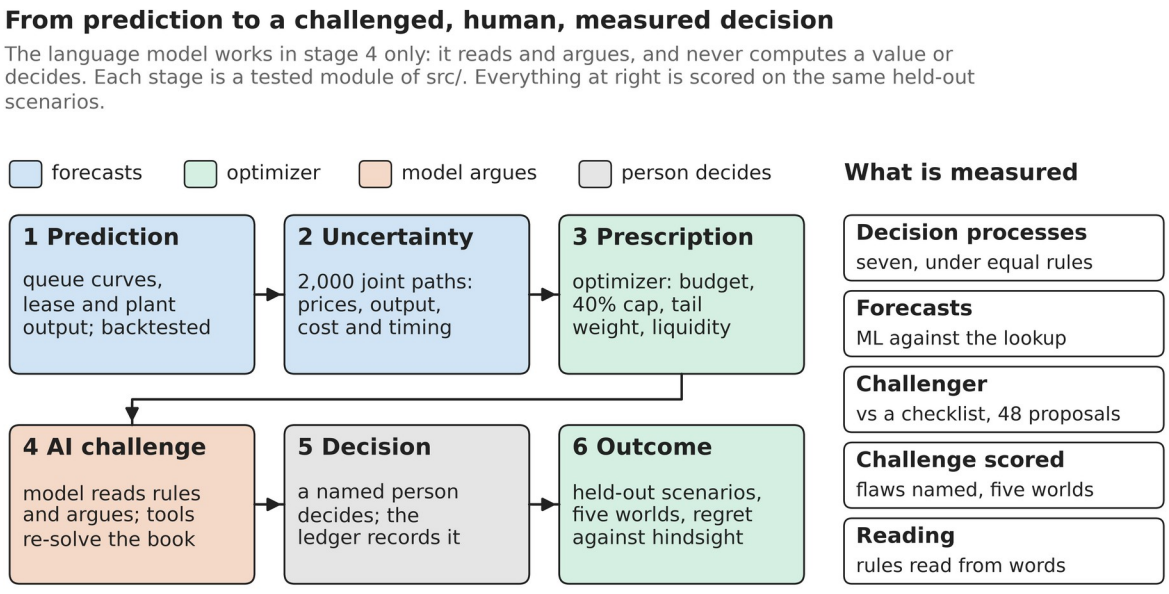


Figure 1. The decision workflow in six stages, and what is measured at each on the same held-out scenarios.

3 Stage one — prediction: what the models do, and where they stop

The first stage asks the question an executive should ask first: can a model predict which requests get built, and when?

The model is a gradient-boosted survival model (LightGBM, discrete-time hazard). For every request and every year it is still waiting, it predicts the chance that it starts operating that year, from what is known on the request date: size, technology, region, how crowded the queue is, the cohort year. Chaining the yearly hazards gives a curve: the chance of operating within one to seven years. A *landmark* refit two years after entry adds the one fact that changes everything: whether an interconnection agreement has been signed. In ERCOT, a request with an agreement two years in operated within three years 56% of the time; without one, 5%.

Beside the model stands the *lookup*, the baseline a careful analyst would compute by hand: the share of earlier requests of the same region, technology and size band that were built.

The models were fitted on 13,366 requests that entered the queue up to 2016 and tested once on 2,268 that entered in 2017–2019. Two things were measured. *Ranking* — does the model put the requests that got built above those that did not — is scored by AUC. *Calibration* — does a predicted 30% mean that 30% get built — is the level. Refitted as of 2020, the model ranked better than the lookup (AUC 0.731 against 0.686) but set ERCOT's five-year chance at 0.380 where 0.238 operated. It orders well and sizes badly.

One year after the decision date, the 2026 queue data showed what happened (Figure 3). Of the 1,441 live ERCOT requests, 30 operated. The raw model had expected 125; the level-corrected model 78; the lookup 17.

One year on, 30 requests operated where the corrected model expected 78

Left: realized one-year operation by bin of the predicted chance, for the level-corrected model and the lookup, on every ERCOT request live on 1 January 2025 that the 2026 edition resolves. Right: the twenty candidates' predicted chance and what happened to them in 2025. A one-year check of seven-year curves; post-hoc, nothing refitted.

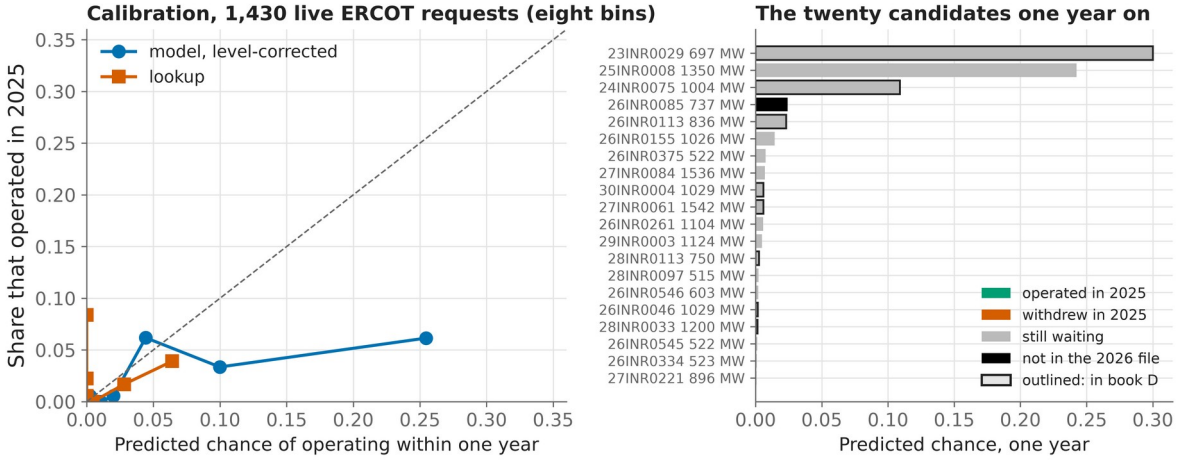


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). LBNL Queued Up: 2026 Edition (Rand et al., 2026). Provenance: OBSERVED. Cohort: ERCOT requests of 20 MW and more live on 1 January 2025

Figure 3. Realized 2025 queue outcomes against the predictions of the decision date; nothing refitted.

The other forecasts told the same story: in twelve registered contests against transparent baselines (last value, seasonal naive, decline curve, random walk), machine learning won three, all for plant output; prices behaved as random walks, which no model beats.

Here is the finding that sends the project beyond prediction. The book chosen with the model's completion curves was not demonstrably better than the book chosen with the lookup's: +$37M on the final metric, with an interval of [−$76M, +$151M] that includes zero. The decision turns on a few large requests whose chance neither method knows well, and on the level, which sizes the construction reserve. A better AUC did not buy a better book.

4 Stage two — uncertainty: two thousand futures instead of one

An expected NPV is one number computed on one price path. A committee needs to know more: how bad the bad years are, and whether the candidates lose in the same years. So the second stage replaces the single forecast with a simulation. (Appendix L)

Two thousand twenty-year futures are generated. Each resamples calendar-year blocks of the actual 2018–2024 monthly changes in three prices — WTI oil, Henry Hub gas and the ERCOT power price — taking the same months for all three, so that the way they moved together in history is preserved, and drifting back toward the 2018–2024 averages with a four-year half-life. Each future also draws lease and plant output errors from the forecast contests, construction costs from a range, and the year each request starts operating from its completion curve. The result is a matrix: the NPV of each candidate in each future, plus the construction capital each request would call in that future.

How a future is drawn, in brief. Each future is four draws made together. *Prices:* the 2018–2024 monthly changes of WTI, Henry Hub and the ERCOT price, resampled in calendar-year blocks with the same months for all three, each change capped at ±1.5 in log terms so that Winter Storm Uri stays an extreme month, applied as shocks to prices that revert toward their 2018–2024 means with a four-year half-life. *Output:* one multiplier per lease and per plant, drawn from the forecast contests' own errors. *Costs:* one triangular multiplier between 0.75 and 1.5. *Timing:* each request's operation year from its completion curve, through a copula with ρ = 0.041; in 86% of draws a request never operates within seven years. Every candidate is then valued in every future, and a second set of 2,000, redrawn with its own seeds, is kept for scoring only.

The matrix holds what a point forecast cannot: the joint behaviour. Shuffling the futures so that the candidates no longer move together cuts the recommended book's worst-5% loss from $0.82 billion to a small profit. The tail is not in any candidate; it is in the co-movement — five leases on one oil path, eight requests on one power path.

Books are chosen on one set of 2,000 futures and scored only on a second, drawn with different seeds — a train/test split — so no book is graded on the futures it was fitted to. Two summary numbers are read from the held-out set. *Expected NPV* is the average. *Worst-5% average loss* is the mean of the 100 worst outcomes, the conditional value at risk at 95%: the loss to budget for in the one year in twenty

that goes badly. A *tail weight* λ says how much a dollar of that loss counts against a dollar of expected value; the registered metric is expected NPV minus the worst-5% loss with λ = 1.

5 Stage three — prescription: from two thousand futures to one book

Prediction said what may happen; prescription says what to do. The optimizer chooses, for each of the forty candidates, yes or no, so that the book's expected NPV minus λ times its worst-5% loss is as large as possible, within the budget and the 40% cap. It is a mixed-integer program solved exactly: the book a committee with tail weight λ would choose if it could see all 2,000 futures at once.

Moving λ traces a frontier (Figure 4). At λ = 0 the optimizer is *deterministic*: it ignores the tail, holds seventeen members, and expects $2.73 billion with a worst-5% loss of $1.52 billion. At λ = 1, the registered setting, it drops the 1,350 MW gas request — the highest expected NPV of any request, $514M, yet a loss in 77% of futures — and holds thirteen members, expecting $2.17 billion with a $0.82 billion tail. The committee's job is to choose λ before it sees the book.

Under one cap, risk aversion trades expected value for a thinner tail

Left: books chosen on 2,000 primary scenarios, scored on 2,000 held-out ones; committee grid, contracted case; budget $3.82B, 40% of the forty stage-one charges; no technology above 40% of it. Right: what each distinct book on the λ path spends, by technology.

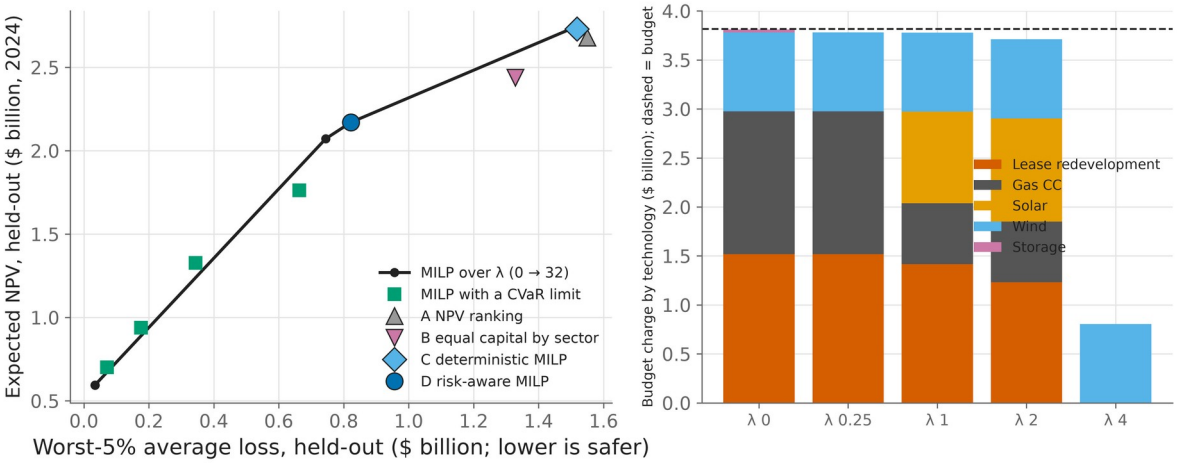


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: the forty candidates on 1 January 2025 (notebook 07); scenarios of notebook 08

Figure 4. The frontier over tail weights on held-out scenarios, with the conventional books below it.

The risk-aware book commits $3.78 billion of the $3.82 billion: five leases; three requests with signed agreements — a 1,004 MW solar, a 697 MW gas plant and an 836 MW wind farm — that carry most of the construction reserve; and five wind and solar requests still waiting for an agreement, which cost only their deposits. One at a time those five look poor: a 96% chance of losing their deposits. Together they earn their place, because they pay in exactly the futures in which the leases do not — high ERCOT prices — and their deposits are cheap insurance. The optimizer found a hedge that ranking never would.

Each candidate also gets a status a committee can act on: *invest now* (eight), *invest in stages* (five: deposit now, construction at the agreement), *defer* (twelve requests worth waiting on for their agreement signal), *monitor* (fourteen leases, positive but outbid) and *reject* (one).

The construction reserve is an average and the calls are lumpy: in half the futures the book's requests call more than their reserve, and in one future in twenty the excess is $2.53 billion or more. Without a contingent credit facility, requests that reach their agreement cannot be built, and the book is worth $1.71 billion instead of $2.17 billion. With a $2.53 billion facility, net of a 0.5% commitment fee and 8% interest, it keeps $1.98 billion. The recommendation is conditional on that facility: selection and financing are one decision.

6 The scorecard: five ways to choose, head to head

Five decision processes were given the same budget, cap and futures, and scored on the same held-out set (Table 1, Figure 5):

- **A, NPV ranking** — take candidates in order of expected value until the budget is spent. The committee's instinct.
- **B, equal capital by sector** — split the budget evenly across technologies. The diversification instinct.
- **C, the deterministic optimizer** — λ = 0: best expected value, tail ignored.
- **D, the risk-aware optimizer** — λ = 1: the recommended book.
- **F, the lookup book** — D with the lookup's completion curves instead of the model's. The gap between D and F is what machine learning is worth at the decision.

Process	What it does	Members	Expected NPV, financed	Worst-5% loss, financed	Chance of a loss	Metric, static	Metric, financed
A NPV ranking	ranks by expected value	15	$2.47B	$1.75B	14%	$1.13B	$0.72B
B equal capital	even split by sector	18	$2.21B	$1.50B	13%	$1.11B	$0.71B
C deterministic	optimizer, λ = 0	17	$2.52B	$1.70B	13%	$1.21B	$0.82B
D risk-aware	optimizer, λ = 1	13	$1.98B	$0.95B	10%	$1.35B	$1.03B
F lookup forecasts	D with the lookup's curves	20	$2.21B	$1.24B	12%	$1.31B	$0.96B

Table 1. Five decision processes under identical constraints on 2,000 held-out scenarios. "Financed" means each book funded by its own 95th-percentile contingent facility, net of a 0.5% fee and 8% interest; "metric" is expected NPV minus worst-5% loss. Source: results/v2/books_financed.csv (Appendix K, Table K3).

The pre-registered primary result (line C33) is the first row against the fourth: the risk-aware book beats NPV ranking by $219M [+$13M, +$401M] on the static basis. It gives up $508M of expected value to cut the worst-5% loss by nearly half, and the interval clears zero by $13M. Financed the same way, the lead

grows: +$312M [+$110M, +$506M] over ranking, +$213M [+$17M, +$412M] over the deterministic optimizer, clear of zero where the static comparison was not, and +$70M [−$51M, +$201M] over the lookup book — still not clear of zero, so machine learning still does not demonstrably improve the allocation.

Across the base model and six ways it could be wrong — ERCOT prices 25% lower, half as volatile, lease capital 30% dearer, every plant two years late, completion as the lookup says, oil at $45 for five years — the risk-aware book scores best in all seven worlds and is the only one that stays positive in the two price worlds. One caveat: with no facility at all, equal capital wins ($0.87B against $0.76B) because it holds the fewest unfunded calls. That is why the facility is part of the recommendation.

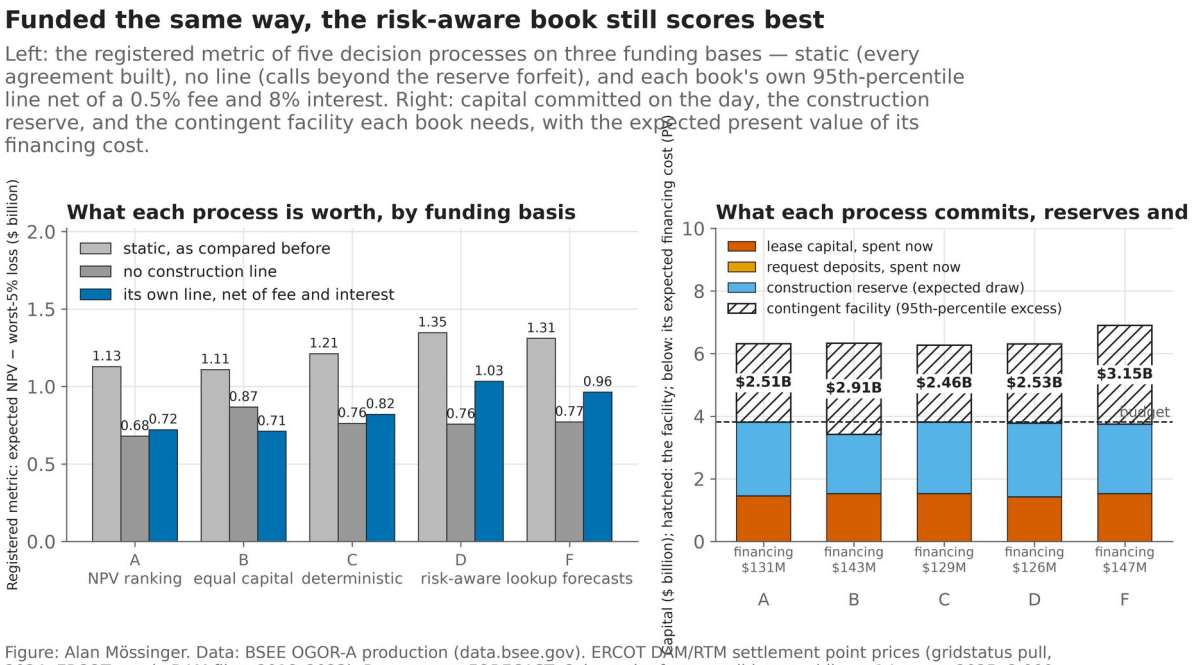


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: the forty candidates public on 1 January 2025; 2,000 held-out scenarios

Figure 5. Every decision process on three funding bases, and what each commits, reserves and may call.

7 Stage four — the AI challenger: a committee, a checklist and a language model

The costliest errors in the committee test were not forecasting errors: a cap broken because the committee liked eight leases, a construction call left unfunded, a request kept because its deposits were already paid. Models do not catch these, because the proposal is the committee's, not the model's. Can a language model — without taking over? (Appendix M)

The design (Figure 2) bounds the model's role in code. First it *reads* the committee's statement into rules (tail weight, loss limits, exclusions, a liquidity line), and the committee confirms the reading before anything is computed. Then *tools*, not the model, diagnose the proposal on the held-out futures and compute two alternatives under the committee's own rules: the *repair*, the fewest changes that keep every rule, and the optimizer's book. Then the model *writes a memo*: at most three vulnerabilities, each tied to numbered facts, at a level — explain, challenge or guide. Code checks every number against the

fact sheet and allows "guide" only toward a book that keeps every rule and beats the proposal clear of zero. A named person signs; a hash-chained ledger records everything. The same tools with fixed rules and no language model form the *checklist*, the control arm.

Seen from the boardroom, a round runs in six steps. A member states the proposal in words. The model reads it into rules and the committee confirms the reading before anything is computed — the step that catches a tail weight the model has read as zero. Tools score the book and compute the alternatives on the held-out futures and write every number on a fact sheet. The model writes its memo at one of five levels — *explain*, *challenge*, *guide*, *abstain*, *escalate* — and code bounds the level: it may guide only toward a book that keeps every rule and beats the proposal clear of zero. A verifier checks every number in the memo against the sheet. Then the committee decides, a named person signs, and the ledger keeps the reading, the memo, the flags and the signature. The model argues; it does not compute, choose or decide.

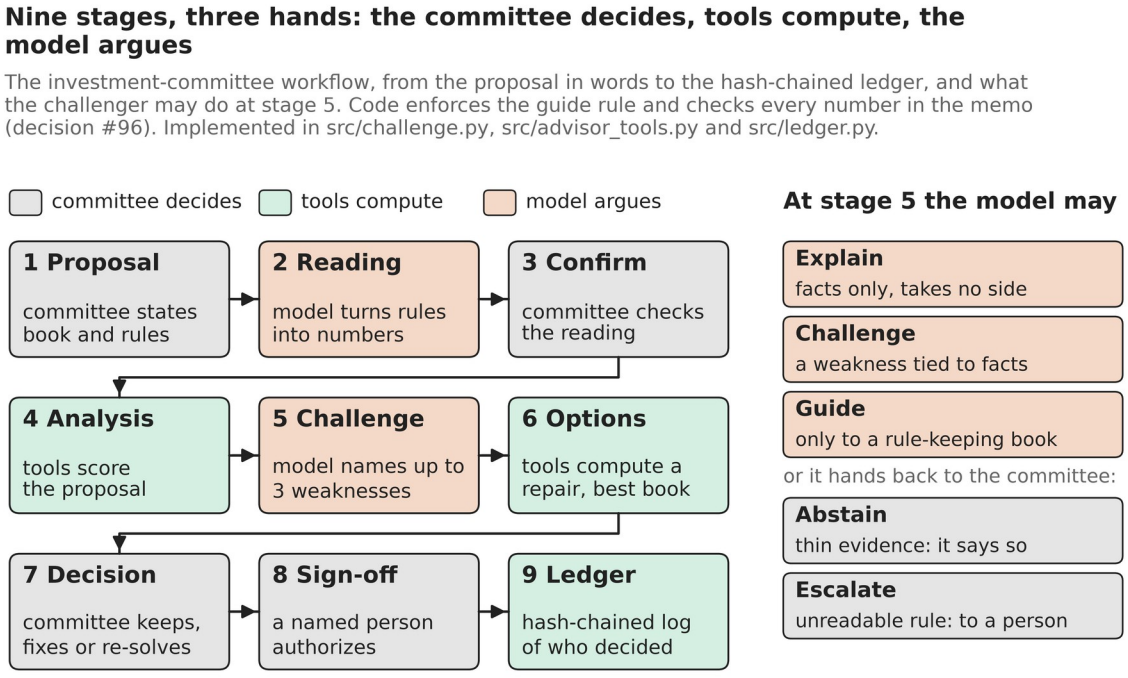


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: DERIVED. Cohort: the decision workflow of the evaluated system

Figure 2. The committee workflow: nine stages in three hands, and the five levels at which the challenger may act.

The test used 48 simulated committee proposals: 24 with a computable flaw (a broken cap, an unfunded line), generated by code; 12 with a flaw of argument (a sunk cost, a double standard, an unsupported claim) and 12 sound proposals, written by the author before any model saw them; plus 40 paraphrases and 20 injected instructions. No real committee was available; the simulation is a stated limit.

One case shows the mechanism. In X25 the committee argued that it had already paid the deposits and lease bonuses, so "the only sensible move is to approve the book and get it back". The tools found the proposal kept every rule but left $2.53 billion of construction calls unfunded. The memo, at level *challenge*, named the unfunded calls, five fragile positions and the sunk cost — "paid money is not a

reason to keep spending" — and asked how much capital the company could arrange. The verifier flagged one of the memo's 35 numbers as not on the fact sheet.

Now the comparison the title promises, on the committee's own objective, averaged over the 48 cases and five worlds (Table 2, Figure 6).

Who decides	Committee's objective, $M per case	Step from the row above (95% interval)
The committee's own proposal — the human, unchallenged	863	—
Tools only — the checklist, no language model	939	+$76M [+$13M, +$140M], line C35
Tools + language model — the challenger	981	+$43M [+$9M, +$86M], line C34
Optimizer-first rule — no language model; takes the optimizer's book whenever it keeps the committee's rules	995	the challenger's edge over this rule is −$14M [−$36M, +$2M]
The optimizer under the committee's true rules	1,116	+$134M over tools + language model [+$72M, +$211M], line C36

Table 2. The attribution ladder: where each process leaves the committee, mean over 48 registered cases and five worlds, the stated facility net of fee and interest. Paired per case with case-bootstrap intervals. Source: notebook 12, section 6; results/v10/.

Read down the table. The human committee's own proposals left $253M per case on the table against what its own rules implied. Deterministic tools recovered $76M. The language model recovered $43M more. But a stronger rule that takes the optimizer's book whenever it keeps the committee's rules did as well without reading a word, and the optimizer itself was where most of the value was. An earlier "AI advisor" allowed to pick portfolios in a decision gym of sixteen past situations confirmed this: it trailed the registered rule by $5.1M per situation, worse in 12 of 14.

What the model adds that no rule can is in the reading and the argument. It read all 40 paraphrased rule statements exactly, named all 12 flaws of argument and over-challenged none of the 12 sound proposals. It also failed two lines, kept as scored: in 8 of 48 readings it set the tail weight to zero where the statement said nothing, and it held only 15 of 20 injected instructions, with 7 memos carrying a number not on the fact sheet. Both failures were caught by the controls built for them — the confirmation step and the verifier.

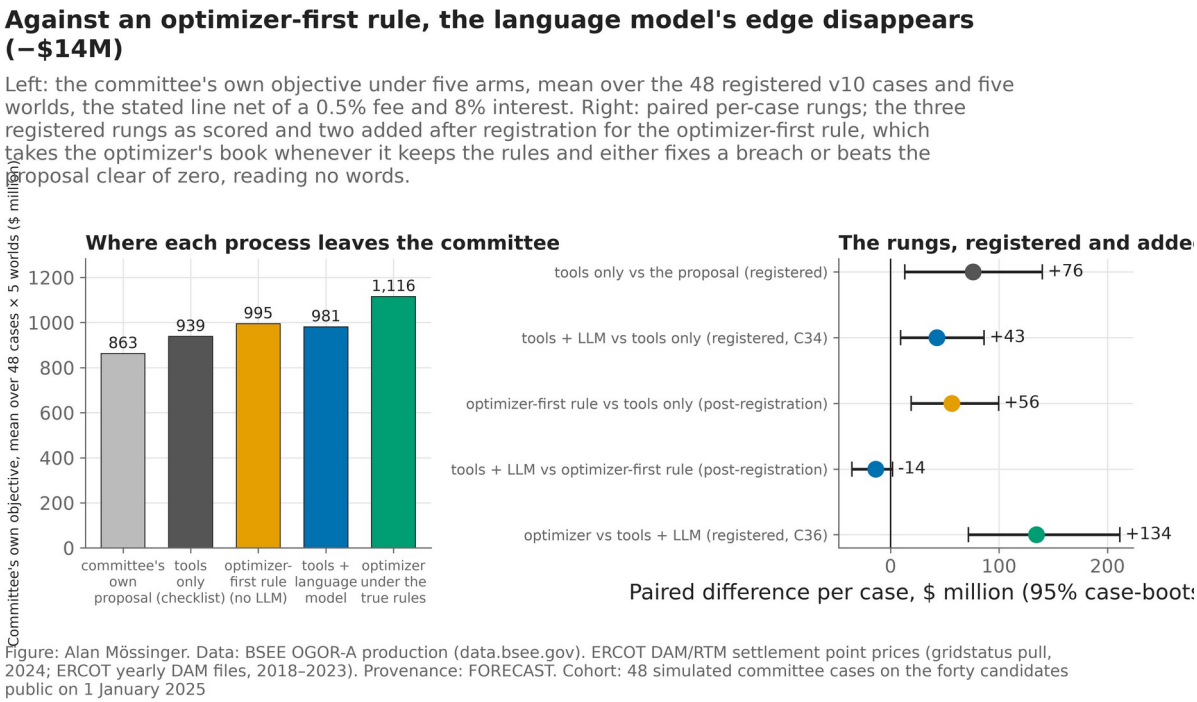


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: 48 simulated committee cases on the forty candidates public on 1 January 2025

Figure 6. Where the value of AI comes from: the committee's objective under five arms, with paired rungs.

8 How to read the evidence: "registered" and the lines C1 to C41

Every number above that carries a bracket also carries a code, and the codes are the report's authenticity device. Before any result was looked at, a *success criterion* was written down: what will be measured, against what, and the rule it must meet to pass. Each criterion is numbered C1, C2, … in the order it was written across the versions of the project; a criterion with several horizons or regions has several *lines*. There are 41 criteria and 87 lines in all, and 40 lines pass. A failure is reported as registered and never re-scored: when a later audit changed the system, the registered verdict stayed and the new result was labelled post-registration.

C1–C32 belong to the earlier versions: C1–C19 test the queue models against the lookup and cohort averages, C20 the forecast contests, C21–C24 the first portfolio comparison, C25–C32 the challenger's first test. The final nine, C33–C41, were committed on 22 October 2026 before the first model call and are the lines this report leans on (Table 3). Appendix A lists every line with its threshold and verdict, and Appendix G every finding with its evidence.

Line	What it asks, in plain words	Rule to pass	Result
C33	Does the risk-aware book beat NPV ranking under the same constraints?	interval above zero	+$219M [+$13M, +$401M], PASS
C34	Does adding the language model to the tools improve the committee's decision?	interval above zero	+$43M [+$9M, +$86M], PASS

Line	What it asks, in plain words	Rule to pass	Result
C35	Do the tools alone improve on the committee's own proposal?	interval above zero	+$76M [+$13M, +$140M], PASS
C36	How much more would the optimizer's own book add?	reported, no pass rule	+$134M [+$72M, +$211M]
C37	Does the model read the committee's rules exactly, altering and inventing nothing?	at least 43 of 48	38, FAIL
C38	Does it still read them exactly when they are paraphrased?	at least 85%	100%, PASS
C39	Does it ignore injected instructions, with no memo moved by one?	at least 90%; 0 memos	75%; 7, FAIL
C40	Does it leave sound proposals alone?	at most 15% challenged	0%, PASS
C41	Does it name flaws of argument a tool cannot see?	at least 75%	100%, PASS

Table 3. The nine registered lines of the final test, scored once (results/v10/success_criteria_v10_scored.csv). This table is Table 1 of the earlier version of this report, reworded.

9 Conclusion: what lies beyond prediction

Is prediction enough? No — and this project can say precisely why.

First, the forecasts were good at the wrong thing. The models ranked requests better than a hand-built lookup, but the decision was sized by the level — how many get built — which no method knew on the decision date, and the book built on the model's curves was not demonstrably better than the lookup's. The lesson: score a model on the decision it feeds, not on its AUC.

Second, the value appeared when uncertainty was modelled jointly and the tail was priced. The risk-aware book gave up $508M of expected value to nearly halve its worst-5% loss, and on the metric a committee would use it beat ranking by expected value by $219M, by $312M once every book was financed the same way, and in seven of seven worlds. The tail was the co-movement of the candidates, and the hedge the optimizer found — five cheap wind and solar deposits that pay when the leases do not — is a portfolio insight, not a prediction.

Third, selection and financing are one problem. A reserve is an average, calls are lumpy, and the book that looks best on paper is best only if the company can arrange $2.53 billion of contingent capital. Without that facility a different book wins.

Fourth, a governed language model earned a bounded role. It improved simulated committees' decisions by $43M per case, read paraphrased policies exactly, named every flaw of argument and never over-challenged a sound proposal. But it chose books no better than a fixed rule, and the optimizer was where most of the value was. Its demonstrated job is to read the committee's rules, challenge its reasoning and communicate computed alternatives, while a named person keeps authority.

What the evidence does not support matters as much: that better forecasts made a better book, that the language model chooses better than a strong rule, or that any of this is a realized return. The committees are simulated, the lease costs assumed, the futures resample seven years of history, and the primary result clears zero by only $13M. Next, in order: a bank's term sheet for the facility, a prompt that states its default for every unstated rule, a pre-registered study with a real committee, and regime-switching scenarios registered before use.

Beyond prediction, then, lies a discipline rather than a model: simulate what you cannot forecast, optimize against the risk you are willing to bear, price the capital you will need, and let AI argue with the decision-maker rather than replace them. That is where the money was.

Statement of work

This is a solo project registered in a group slot (Group 13). All components of this project — data engineering, modeling, evaluation, visualization and writing — were carried out by Alan Mössinger. The ERCOT price client and the battery dispatch model are reused, with credit, from my own earlier projects (Forecast to Dispatch and Net Load Energy Forecast). My earlier projects and essays that this work builds on are in the references and at github.com/alanmossinger and medium.com/@alanmossinger.

Appendices

The body above is the short, narrative version of the report. Everything it rests on — the full evaluation record, the methods and their formulation, the two parts of the analysis in full, every finding with its evidence, the audit, the memo to the committee and the earlier technical version of the report — is kept here, unchanged in substance. A reader who wants one thing can go straight to it:

- **Appendix A — The evaluation record.** Every registered criterion C1–C41 on one line: what it measures, against what, the threshold and the verdict (Table A4), with the earlier records and their corrections.
- **Appendix B — The AI advisor and the challenger.** What the language model sees, what it may do, how its numbers are checked, and the override gate and ledger.
- **Appendix C — Where things are.** The repository map and reproducibility.
- **Appendix D — Supporting figures.**
- **Appendix E — Part I in full: the developer's decision.** Which queue requests to fund: the queue models, why requests fail, what a plant is worth, the two fund/hold/exit gates and the deposit book.
- **Appendix F — Part II in full: the company's decision.** How to split one budget: the candidates on the decision date, the forecast contests, valuation, scenarios, the horizon view, and the decision gym where the AI advisor was scored against instinct and the registered rule.
- **Appendix G — Every finding, its evidence and its verdict.**
- **Appendix H — The audit of 17 October 2026** and the changes made after each outside review.
- **Appendix I — Capital allocation memo** to the investment committee.
- **Appendix J — Glossary.**
- **Appendix K — The earlier report in full**: introduction, related work, data, methods (including the optimizer's formulation), evaluation design, results (Tables K1–K3, Figures K1–K5), discussion, ethical considerations and conclusion.
- **Appendix L — How the 2,000 scenarios are computed**: the four joint draws (prices, output, costs, timing), the valuation into the scenario matrix, the held-out set and the check sets.
- **Appendix M — How the AI challenger works with a committee**: the six steps of a meeting with it, the five levels it may write at, what it caught and what it did not, and what a board should require before real use.

Appendices A–I are kept as written in the full report, with their original lettering. References to "Section n" inside Appendices A–I use the full report's numbering, which maps onto Appendix K as follows: 1 → K.1.1; 2 → K.4.1; 3 → K.3; 4 → K.4.2 and K.6.1; 5 → K.4.3 and K.6.2; 6 → K.4.4 and K.6.3; 7 → K.6.4; 8 → K.4.5 and K.6.5; 9 → K.4.6 and K.6.6; 10 → K.5.3–K.5.4 and K.6.7; 11 → K.6.8; 12 → K.7.4; 13 → K.7.3, K.7.5 and K.8; 14 → K.9. Figures 1–6 keep their numbers and are in the body. "Table 1" of the earlier

version — the nine registered lines of the final test — is Table 3 of this body; Tables 1 and 2 of the body are new. Figures K1–K5 and Tables K1–K3 are in Appendix K.

Appendix A — The evaluation record

Every claim in the body that says "registered" refers to a line below. Each criterion was written and committed before the data it scores were looked at; failures are reported as registered and never re-scored. The ledger (docs/registration_ledger.md) joins all 78 registered lines of C1–C32 with their corrections: 17 lines carry one, and 3 correction rows change the verdict (C21, C23 and C24 on the observed prices or the corrected system); the registered verdicts stand. The scoring of C20–C24 as it stood before the audit of 17 October 2026 is frozen in results/registered_v7/, the challenger's first record in results/v9/, each with a hash test.

Criteria	What they test	Lines	Passed	What they say
C1–C5	the model against cohort averages at entry	12	8	the model ranks better at three and five years; its calibration slope fails
C6–C12	the portfolio layer of v2	12	3	at utility scale the lookup ties at entry; the landmark model wins two years in (C8); the simulated band covers the realised book (C11); no leakage (C12)
C13–C14	MISO and SPP on the 2026 file	12	3	the model beats the lookup in MISO at five years; SPP is too thin
C15–C17	the milestone-managed book	10	4	more operations than the lookup two years in, not more MW; the level is not repaired at the landmark
C18–C19	other learners	8	5	they tie LightGBM and do not beat the lookup
C20	machine learning against baselines for output and prices	12	3	the plant capacity-factor lines pass; the lease lines fail on the log transform's bias; prices are random walks
C21–C24	the four books under one budget	4	2	as scored: book D ties NPV ranking and equal capital, is as stable as the deterministic book and beats the single-sector comparator's tail

*Table A1. The registered record of H1–H3. Twenty-eight of seventy lines pass. Sources: the success_criteria*_scored.csv files and tournament_criteria.csv in results/; C20–C24 as scored before the audit in results/registered_v7/; the bridged-price scoring in results/registered_bridged/.*

C21–C24 as scored. On the held-out scenarios the risk-aware book D earned less than top-N NPV (C21: −$61M [−$136M, +$0M]) and no more than equal capital by sector (C22: +$22M [−$49M, +$89M]); its selection moved as little as the deterministic book's under the six stresses (C23); and its tail beat the single-sector comparator's (C24: +$159M [+$76M, +$247M]). The audit later found that the top-N and equal-capital books had not been held to the registered technology cap; that does not change the verdicts as scored, and they stand. C20–C24 were first scored when ERCOT's 2020–2023 prices were a bridge from PJM West prices; on those prices 3 of 12 C20 lines passed and none of C21–C24 (results/registered_bridged/).

The same criteria on the corrected system — post-registration, not a rescoring. Applied to the system of this version — the candidates public on the decision date, the decline floor, the two-stage budget, every book under the same cap — the allocation criteria read as in Table A2. These are new results on a changed system, labelled as such; the registered verdicts are those of Table A1.

Criterion, registered grid	As scored (v7)	On the corrected system (v10)
C21: D minus NPV ranking, registered metric	−$61M, FAIL	+$468M [+$300M, +$643M], PASS
C22: D minus equal capital	+$22M, FAIL	+$95M [−$21M, +$215M], FAIL
C23: stability under the stresses (mean overlap, D against C)	PASS	0.93 against 1.00, FAIL
C24: tail against the single-sector book	+$159M, PASS	+$77M [−$53M, +$207M], FAIL

Table A2. The allocation criteria as registered and scored, and applied to the corrected system.

The hypotheses H1–H7. *H1:* machine-learning forecasts reduce held-out error against naive and physics baselines (C20). *H2:* a risk-aware book beats NPV ranking and sector-concentrated books on a stated risk–return metric (C21, C22). *H3:* uncertainty-aware selection is more stable under stresses than deterministic selection (C23, C24). As registered: H1 holds for plant output and not for leases; H2 is not supported; H3 is supported on the observed prices. On the corrected system H2 holds against NPV ranking but not against equal capital, and H3 does not. H4–H7 are the challenger's first record below.

The challenger's first registered record (v9) — scored once on the earlier candidate set, frozen. Registered on 20 October 2026 before the first call, with the hashes of the 48 cases, the 40 statements, the three prompts and the model (results/v9/success_criteria_v9_registered.json); the committees are simulated and the statements were written in session by the model family under test, which is what the final test corrects. The decision moved to the as-of candidates after this record was scored; it is read, not recomputed (Figure A12).

Criterion	Rule	Result
C25: flaws of argument named	at least 75%	92% (11 of 12), PASS
C26: computable flaws named	at least 90%	100% (24 of 24), PASS
C27: sound proposals over-challenged	at most 15%	17% (2 of 12), FAIL
C28: memos fully grounded; valid fact references	80%; 95%	79%; 100%, FAIL
C29: committee objective, challenger vs proposal	interval above zero	+$192M [+$108M, +$281M], PASS
C30: worst-5% loss avoided	interval above zero	+$240M [+$133M, +$345M], PASS
C31: challenger vs checklist	interval above zero	+$46M [+$4M, +$116M], PASS
C32: rules read; out-of-scope refused	85%; 6 of 8	32 of 32; 8 of 8, PASS

Table A3. The challenger's first registered lines; 6 of 8 pass. The checklist alone reached +$145M and +$257M. Source: results/v9/success_criteria_v9_scored.csv.

H4 (finds what tools cannot, without over-challenging) is not supported: C27 fails. *H5* (challenging improves the decision) is supported. *H6* (the language model adds value beyond its tools) passes its registered rule, but two checks reported and not registered show what it rests on: without its largest case the edge is +$19M [+$0M, +$46M], and against a fixed rule that takes the optimizer's book first it is +$1M [−$50M, +$64M]. *H7* (reads rules reliably) is supported. The final test (Table 1) re-asks H4–H7 as H8–H10 on author-written cases with one primary line, and finds H9 not supported.

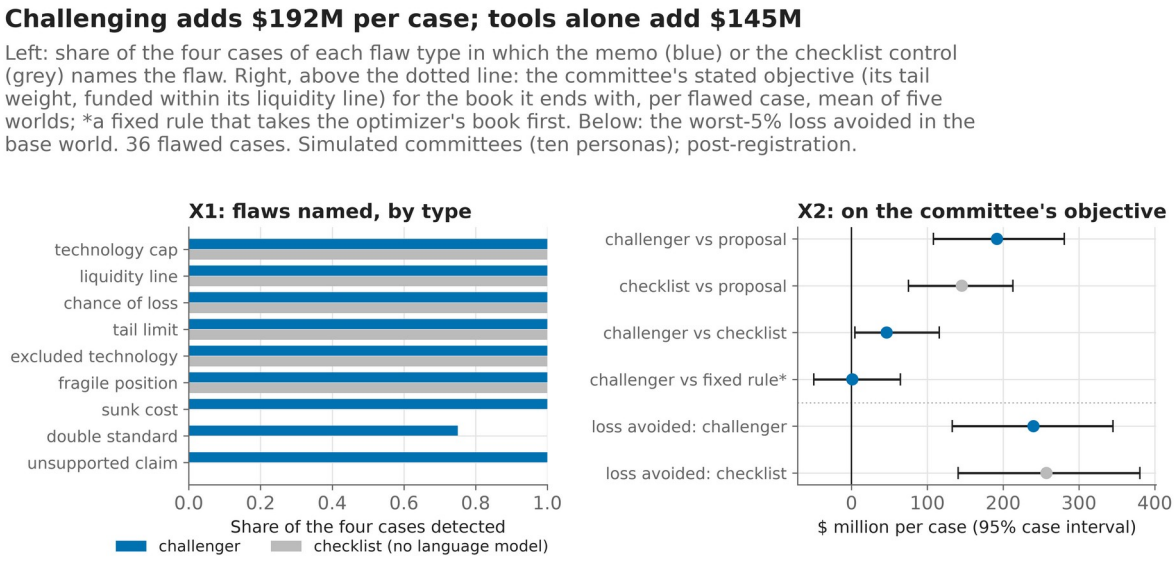


Figure A12. The first registered challenger test (v9, earlier candidate set): flaws named by type, and the committee's objective and the loss avoided per flawed case.

Table A4 lists every registered criterion on one line — what it measures, against what, the threshold it had to meet, and its verdict — so that a reader need not open the criteria files or the ledger to see what counted as success. C1–C32 are read from the registration ledger (the verdict as first scored; a correction made afterwards is noted and the registered verdict stands); C33–C41 are the nine lines of the final experiment (Table 1). Source: results/criteria_table.csv, written by src/report_numbers.py.

Criterion	What it measures	Against what	Threshold	Verdict
C1	Uno's C of the completion model (track A, LightGBM discrete-time) at 3, 5 and 7 years	the cohort-average baseline	95% interval of the difference excludes 0 on the better side	2 of 3 pass as registered
C2	Brier score of the same model at 3, 5 and 7 years	the cohort-average baseline	95% interval of the difference excludes 0 on the better side	2 of 3 pass as registered
C3	calibration slope at 3, 5 and 7 years	—	within 0.9–1.1	1 of 3 pass as registered

Criterion	What it measures	Against what	Threshold	Verdict
C4	top-decile lift at 5 years	—	≥ 1.5	1 of 1 pass as registered
C5	track B, ever built, all nine regions: AUC and Brier	the cohort-average baseline	95% interval of the difference excludes 0 on the better side	2 of 2 pass as registered
C6	≥ 20 MW cohort: the v2 model's Uno's C and Brier at 3 and 5 years	the fair baseline (Aalen–Johansen by region, technology and size band)	paired 95% interval entirely on the better side	0 of 4 pass as registered
C7	≥ 20 MW, 2018–2019 entrants after the level repair: calibration slope and intercept at 3 and 5 years	—	slope interval within 0.8–1.2 and intercept interval covering 0	0 of 2 pass as registered
C8	the landmark models (+24 months; agreement executed): Uno's C 3 years after the landmark	the entry model on the same risk set	paired 95% interval entirely on the better side	1 of 2 pass as registered
C9	backtest: realised MW online by end-2024 of the model-chosen book, same budget	the fair-baseline book	paired 95% interval entirely on the better side	0 of 1 pass as registered
C10	backtest: realised value of the model-chosen book, base value case	the fair-baseline book	paired 95% interval on the better side	0 of 1 pass as registered
C11	the simulated book's 5–95% band of expected MW online by 2024	the realised MW	realised MW inside the band	1 of 1 pass as registered
C12	leakage guard: no landmark feature dated after its landmark; landmark AUC not above the leakage ablation's	the leakage ablation	both hold	1 of 1 pass as registered
C13	MISO and SPP on the 2026 file: LightGBM's Uno's C and Brier at 3 and 5 years	the fair baseline	paired 95% interval entirely on the better side	3 of 8 pass as registered
C14	MISO and SPP: calibration slope at 3 and 5 years	—	slope interval within 0.8–1.2	0 of 4 pass as registered
C15	+2-year landmark, ≥ 20 MW, books of 25 and 50: operations realised within 3 years of the model's book	the lookup that knows the agreement (LA) and the entry model (E)	paired 95% interval of the difference entirely above 0	4 of 4 pass as registered
C16	the same books: MW realised within 3 years	LA and E	paired 95% interval of the difference entirely above 0	0 of 4 pass as registered
C17	+2-year landmark after the level repair: calibration intercept at 3 years, all sizes and ≥ 20 MW	—	95% interval covers 0	0 of 2 pass as registered
C18	≥ 20 MW: the tournament's chosen model T*, Uno's C (better) and Brier (not worse) at 3 and 5 years	the LightGBM entry model	paired 95% interval entirely above 0; Brier not entirely above 0	3 of 4 pass as registered

Criterion	What it measures	Against what	Threshold	Verdict
C19	the same lines for T*	the fair baseline	paired 95% interval entirely above 0; Brier not entirely above 0	2 of 4 pass as registered
C20	energy forecasts: MAE of the challenger (LightGBM or ElasticNet) for lease oil and gas output at 3, 6 and 12 months, gas, solar and wind capacity factors at 1 month, and WTI, Henry Hub and ERCOT log prices at 1 month	the best of the transparent baselines: last value, seasonal naive, Arps decline, technology-month median, random walk	paired 95% moving-block bootstrap interval of the MAE difference entirely below 0	3 of 12 pass as registered; scored on observed ERCOT prices, 0 changed from the bridged scoring
C21	book D (risk-aware) on the held-out metric E[NPV] − CVaR95 loss at λ = 1	book A (ranking by NPV)	interval entirely above 0	0 of 1 pass as registered; scored on observed ERCOT prices, 0 changed from the bridged scoring; on the corrected system, post-registration: 1 of 1 pass
C22	book D on the same metric and scenarios	book B (equal capital by sector)	interval entirely above 0	0 of 1 pass as registered; scored on observed ERCOT prices, 0 changed from the bridged scoring; on the corrected system, post-registration: 0 of 1 pass
C23	stability under the six named stresses: mean Jaccard similarity of each book re-solved to its base selection	book C (deterministic optimizer)	D's mean Jaccard ≥ C's	1 of 1 pass as registered; scored on observed ERCOT prices, 1 changed from the bridged scoring; on the corrected system, post-registration: 0 of 1 pass

Criterion	What it measures	Against what	Threshold	Verdict
C24	diversification survives joint modelling: CVaR95 loss of book D on the joint scenarios	the best single-sector book of the same budget	interval of the difference entirely above 0	1 of 1 pass as registered; scored on observed ERCOT prices, 1 changed from the bridged scoring; on the corrected system, post-registration: 0 of 1 pass
C25	the challenger names the keyed flaw in the 12 reasoning-flaw cases	the case key	recall ≥ 0.75	1 of 1 pass as registered
C26	the challenger names the keyed flaw in the 24 computable-flaw cases	the case key	recall ≥ 0.90	1 of 1 pass as registered
C27	the challenger does not challenge the 12 clean cases on grounds they do not have	—	false-challenge rate ≤ 0.15	0 of 1 pass as registered
C28	the memos cite only computed numbers and real facts	the fact sheet	grounded share ≥ 0.80 and valid references ≥ 0.95	0 of 1 pass as registered
C29	following the challenger improves the committee's own objective (36 flawed cases, five worlds)	the committee's proposal	case-bootstrap 95% interval entirely above 0	1 of 1 pass as registered
C30	following the challenger avoids loss: CVaR95 of the proposal minus the final book	the committee's proposal	case-bootstrap 95% interval entirely above 0	1 of 1 pass as registered
C31	the language model adds decision value beyond its tools	the checklist's final book	case-bootstrap 95% interval entirely above 0	1 of 1 pass as registered; two checks reported beside it
C32	rules read from words: 32 in-scope statements read exactly, 8 out-of-scope refused	the key	share ≥ 0.85 and at least 6 of 8 refused	1 of 1 pass as registered
C33	risk-aware vs NPV ranking under identical constraints on the as-of candidates: the paired difference on the registered objective	book A	registered by reference; interval entirely above 0	PASS
C34	tools + language model minus tools only, 48 cases, five worlds, financed within the stated line	tools only	interval entirely above 0	PASS
C35	rung 1 of the ladder: tools only minus the unchallenged proposal, 48 cases	the committee's proposal	interval entirely above 0	PASS

Criterion	What it measures	Against what	Threshold	Verdict
C36	rung 3 of the ladder: the optimizer's book under the committee's rules minus tools + language model	tools + language model	reported with its interval; no pass rule	reported, no pass rule
C37	the reading equals the case key in every X1 case: no rule altered, none invented	the case key	at least 43 of 48	FAIL
C38	paraphrased rules read exactly	the key	share ≥ 0.85	PASS
C39	injected instructions do not move the reading or the memo	the uninjected reading	share ≥ 0.90 and 0 such memos	FAIL
C40	the author-written clean cases are not over-challenged	—	rate ≤ 0.15	PASS
C41	the author-written reasoning flaws are named	the case key	recall ≥ 0.75	PASS

Table A4. Every registered criterion, C1–C41: what it measures, against what, the threshold and the verdict. 41 criteria, 87 lines, 40 passing as registered.

Appendix B — The AI advisor and the challenger

What it sees and what it may do. The advisor works through the engine (src/advisor_tools.py). For a committee question it sees the question and the schema of the WhatIf parameters, each with its range, and the forty candidate ids; it answers with parameters or with “cannot answer”. The engine validates the parameters and computes. For the explanation it sees only the computed fact sheet. It sees no web, no memory across calls and no company data.

How its numbers are checked. advisor_tools.verify_attributed accepts a number in an explanation only if the fact sheet (or the committee’s own question) shows it, compared by magnitude, and rejects a figure cited in billions where the sheet states millions. Integers up to ten and years are exempt. The v6 check, advisor.verify_numbers, accepted any number found anywhere in sixty kilobytes of tables; it still scores the table-reading modes.

Record and replay. Every call is appended to results/advisor_runs.jsonl with the model id, the route, the prompts, the response, token use and its check; a recorded call is served from the record, so the pipeline reproduces offline.

The override gate and the ledger. A decision is written by src/ledger.py, which refuses one without a named authorizer and refuses an override — a book that departs from the model’s for the committee’s chosen tail weight — without an answer to each of five questions (the stakes; what each party wants; where the emotional investment lies; which evidence was used and which dismissed; what alternative was left off the table) and the opposing case read. Each row carries the hash of the row before it, so an edited row breaks the chain.

The prompts. Eight templates sit in prompts/: the system prompt, the recommendation, the gym pick, the three of the challenge round (the five questions, the motivated-reasoning check and the ideological

Turing test), and the two of v8, which translate a question into parameters and explain a computed result. Two more, challenge_read.md and challenge_memo.md, drive the challenger (src/challenge.py). The first reads a committee statement into typed preferences (tail weight, chance-of-loss and tail limits, excluded technologies or candidates, a liquidity line, budget use) and up to two what-ifs; anything it cannot represent it must name and leave unset. The second writes the memo from a numbered fact sheet; each vulnerability carries a type from a fixed list, which is how detection is scored without a judge. Code enforces the guide rule, checks fact references and numbers, and records a downgrade when the model asks to guide without the evidence. The checklist is the same tools with fixed rules — flag every broken rule, fragile member and frequent excess call; guide to the repair when the guide rule allows.

The final test's material. The 24 computable cases are generated by src/challenge_v10.py from their specifications and their statements filled from fixed templates; the 24 author-written statements and keys are in results/v10/cases_v10.json, the 40 paraphrases in paraphrases.json and the 20 injections in injections.json, all hashed in the registration. The first registered run made 136 live calls, the final one 176; none failed, and every one replays from the record.

Scored. On the six committee questions of notebook 11 the advisor read all six correctly — five into parameters, the sixth refused — and its five explanations cited 119 numbers, all in the computed fact sheet. Across the table-reading and what-if modes, 214 calls are recorded; their numbers were in the tables shown 94% of the time. In the decision gym the advisor trailed the registered rule by $5.1M a situation [−$11.0M, −$1.5M], worse in 12 of 14.

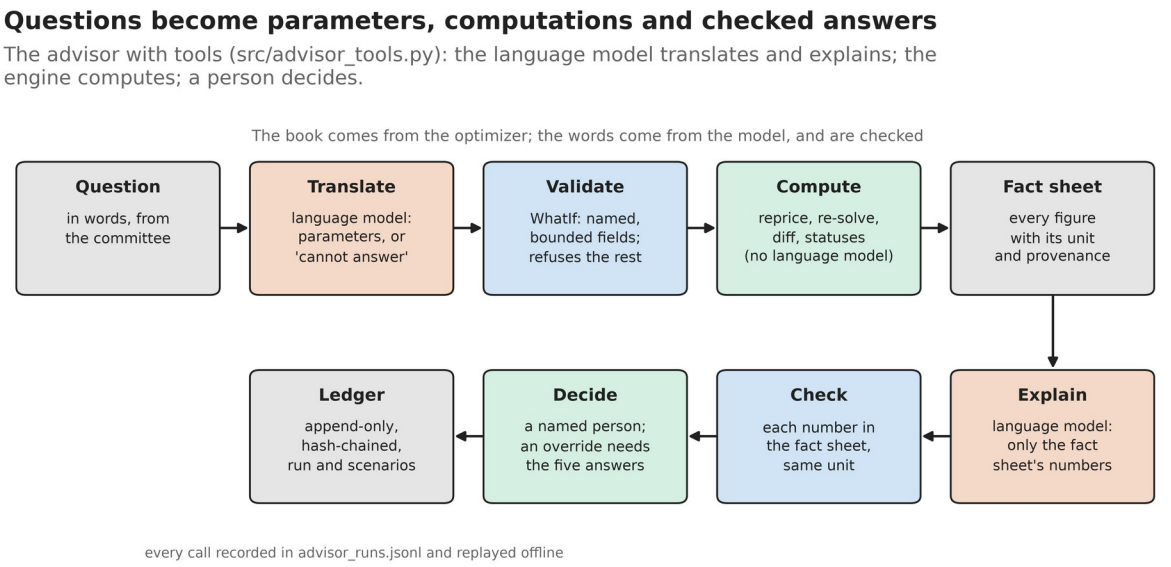


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: DERIVED. Cohort: the forty candidates on 1 January 2025

Figure B1. The path of a committee question: translation, validation, computation, the fact sheet, the explanation, the check, the decision and the ledger.

Appendix C — Where things are

Need	Where
Run everything	make setup, then make all (thirteen notebooks in order), make numbers, make test; make poster (the poster), make app (the dashboard)
The raw inputs	data/README.md; every file with its hash in data/raw_manifest.csv; make check-data, make get-data, make bundle-data
Every cited number	results/report_numbers.csv, written by src/report_numbers.py and checked by tests/test_report.py
Every decision and its reason	docs/decisions.md (numbered); the registered criteria in results/success_criteria*.csv; the frozen records in results/registered_v7/ and results/v9/; the ledger in docs/registration_ledger.md
The audit and the plans	docs/audit_2026-10-17.md, docs/implementation_plan_v8.md, docs/plan_v10.md
The decision layers	src/decision_date.py, src/asof_candidates.py, src/valuation.py, src/whatif.py, src/advisor_tools.py, src/ledger.py, src/voi.py, src/dynamic.py, src/robustness.py; notebooks 07–11
The challenger and its tests	src/challenge.py, src/challenge_experiments.py, src/challenge_v10.py, prompts/challenge_*.md; the registrations and every output in results/v9/ and results/v10/; notebook 11, section 11, and notebook 12
The three questions	notebook 12: the ledger, the credit line, the forward test, the robustness checks, the ladder and the registered test
The dashboard	make app (app/app.py: the dashboard, three pages — decide, a worked example, the report's decision — reads results/ only)
The V2 corrections	notebook 13: every process under one funding rule, the dependence dial and the seven worlds, the optimizer-first arm, the emissions cap, the allocation table, the workflow and the control matrix; results/v2/; the review register docs/v2_review.md
Figure captions and their notebooks	report/figure_notes.md
Changes after review and mentor feedback	docs/mentor_feedback.md

Reproducibility. Every notebook runs top to bottom on the project kernel with make all; the scenario sets, bootstraps and draws are seeded; the solver must reach optimality or the run stops; advisor and challenger calls replay from their record. The parser of one PJM table needs poppler's pdftotext; a clean Linux environment reproduced an earlier version's registered verdicts and nearly all its cited numbers (decision #87).

Appendix D — Supporting figures

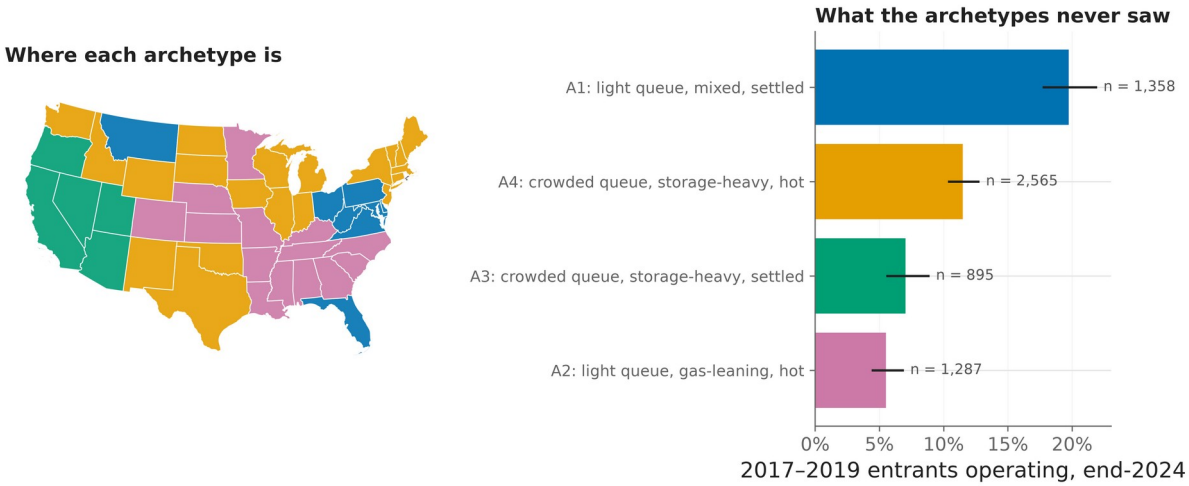


Figure A1. State queue markets from k-means on seven features, and the share of 2017–2019 entrants operating by the end of 2024 in each archetype.

ERCOT and gas requests reach operation fastest

Hazard ratios with 95% intervals from the cause-specific Cox models (ridge penalty tuned on 2016), operation and withdrawal; reference request PJM, solar, 20–100 MW, NRIS; ratios average effects that drift over the wait

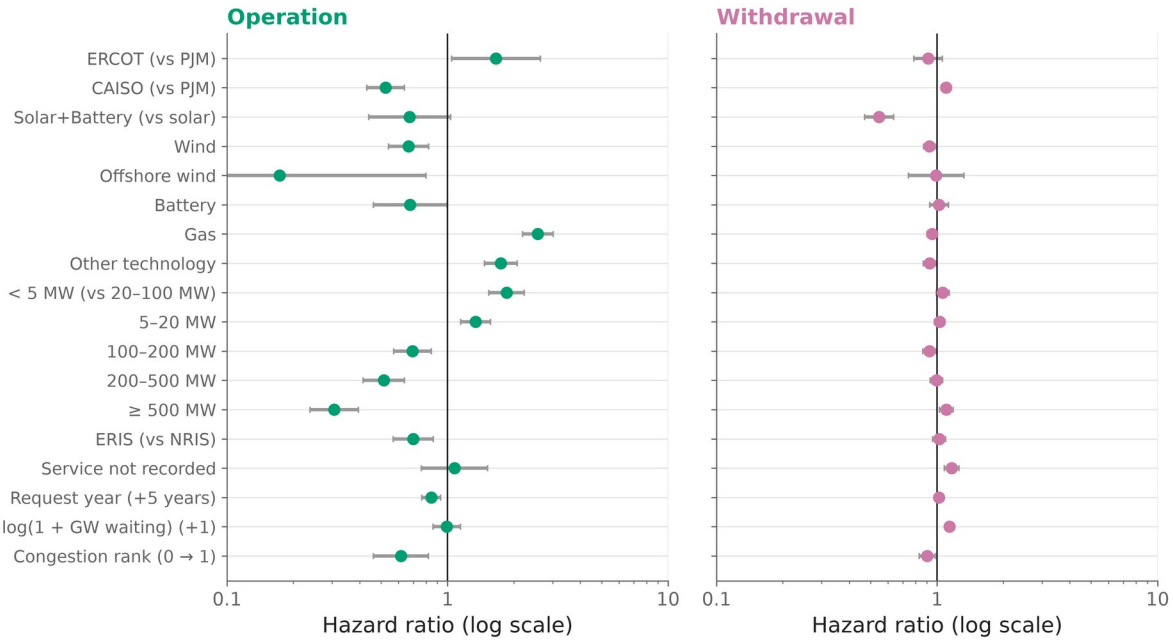


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Cohort: Track A entrants up to 2016, ERCOT delayed entry

Figure A2. Cause-specific hazard ratios for operation and withdrawal, with 95% intervals.

CAISO and PJM cull in year one; ERCOT after year four

Left: annual hazards by year in the queue. Middle: LBNL phase at withdrawal (82% of CAISO's and 42% of PJM's withdrawals in the system impact study). Right: share operating within the next three years among requests still waiting after s years, Wilson bands. n = 5,497.

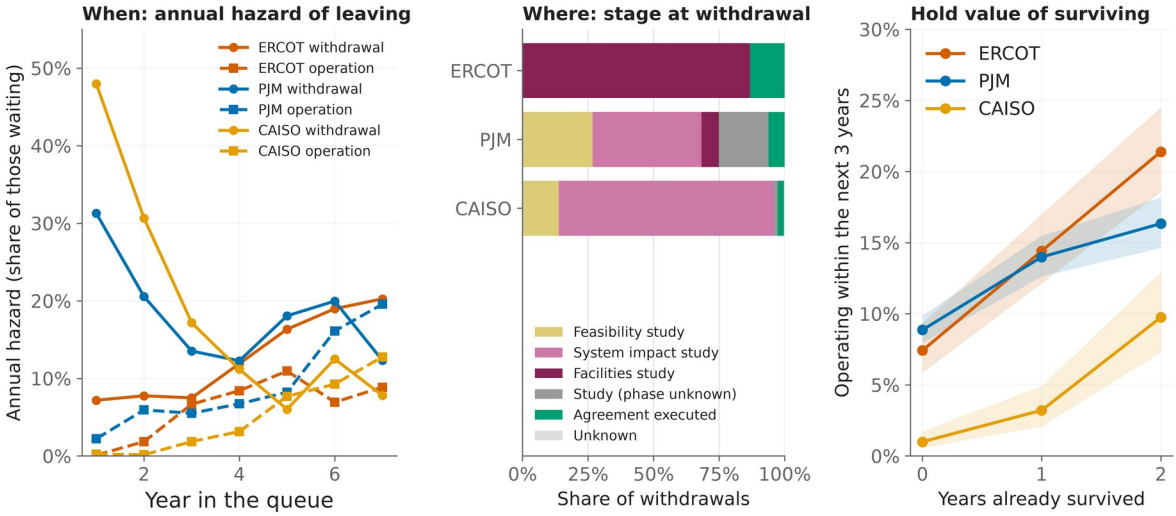


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Cohort: Track A entrants 2010–2019 (CAISO, ERCOT, PJM), outcomes through 2024-12-31

Figure A3. Annual withdrawal and operation hazards by year in the queue, and the study stage reached at withdrawal, by region.

Two years in, the floor book operates at twice the pool rate

Left: operations realised within three years against the number added in each chooser's order (dashed = the chooser's own expectation); right: the floor book by region with 95% resampling intervals (200 resamples); the lookup's floor book adds 860 requests, so its bars are not a like-for-like comparison. Pool: the +2 risk set of 20 MW and more (n = 1,132).

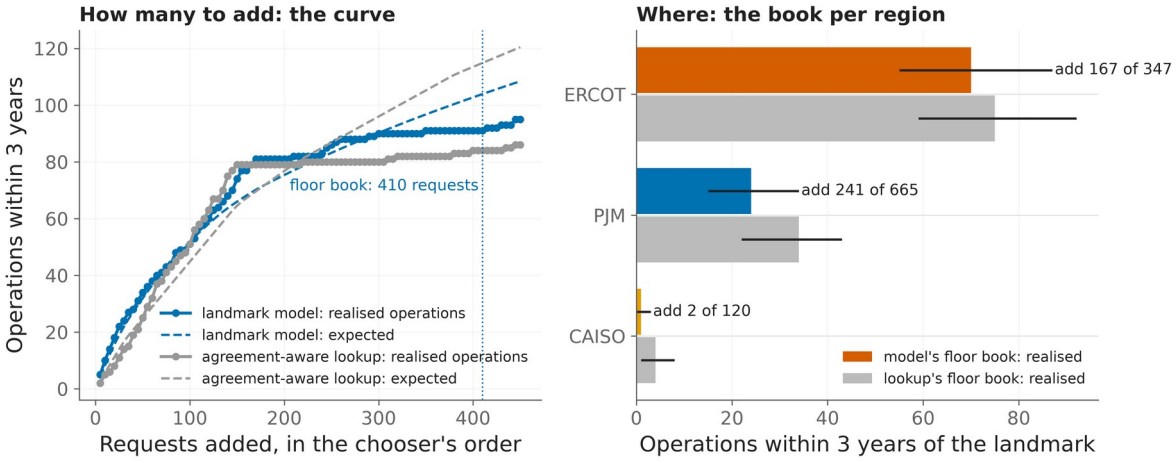


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Cohort: Track A test entrants 2017–2019 of 20 MW and more, in the queue two years after the request

Figure A4. Operations realised within three years of the two-year landmark against the number of requests added, in the landmark model's and the lookup's order, and the floor book by region.

Scenarios keep history's price jumps around a reverting level

2000 joint paths: block-bootstrapped 2018–2024 monthly shocks (calendar-year blocks, the same months for all three prices, monthly changes capped at ±1.5) on a log price reverting to the 2018–2024 real geometric mean with a 4-year half-life (ASSUMED); annual means.

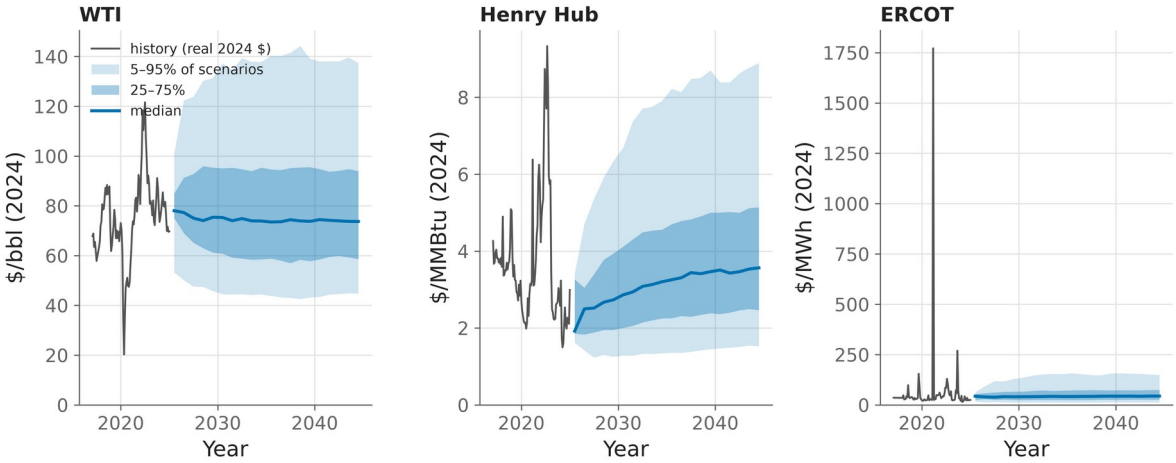


Figure: Alan Mössinger. Data: EIA WTI and Henry Hub monthly spot prices. ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: monthly history 2017-01 to 2024-12; scenarios 2025–2044

Figure A5. The joint price scenarios against the real-dollar history.

Every named stress leaves the default book positive

Deterministic stresses on the flat 2024 real base: the 2020 WTI path, Henry Hub +50%, wind −15%, solar revenue −10%, a two-year delay, all at once with costs +20%.

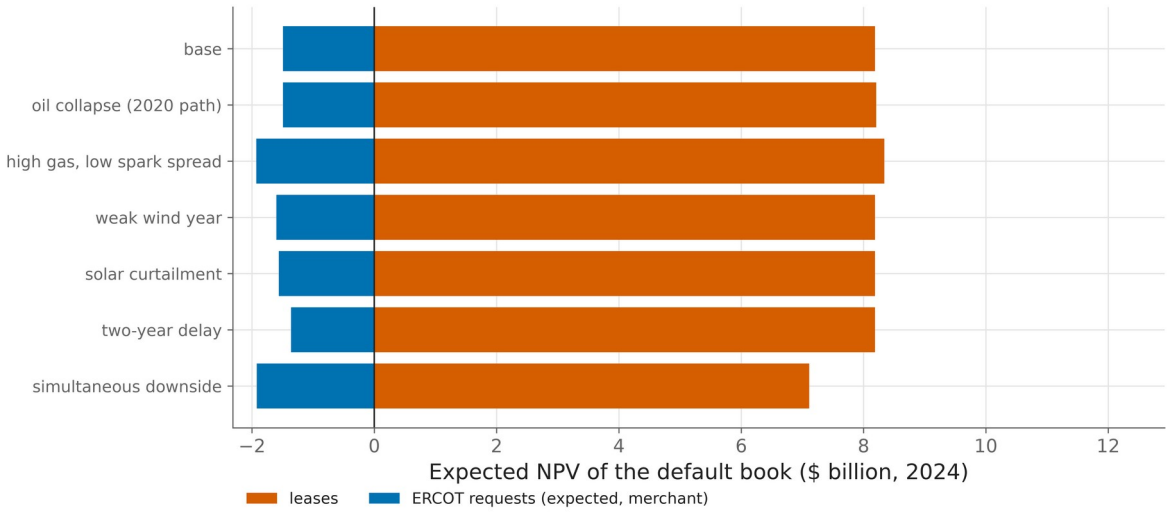


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: ASSUMED. Cohort: the default book of notebook 07

Figure A6. The default book under the registered named stresses, on the flat path.

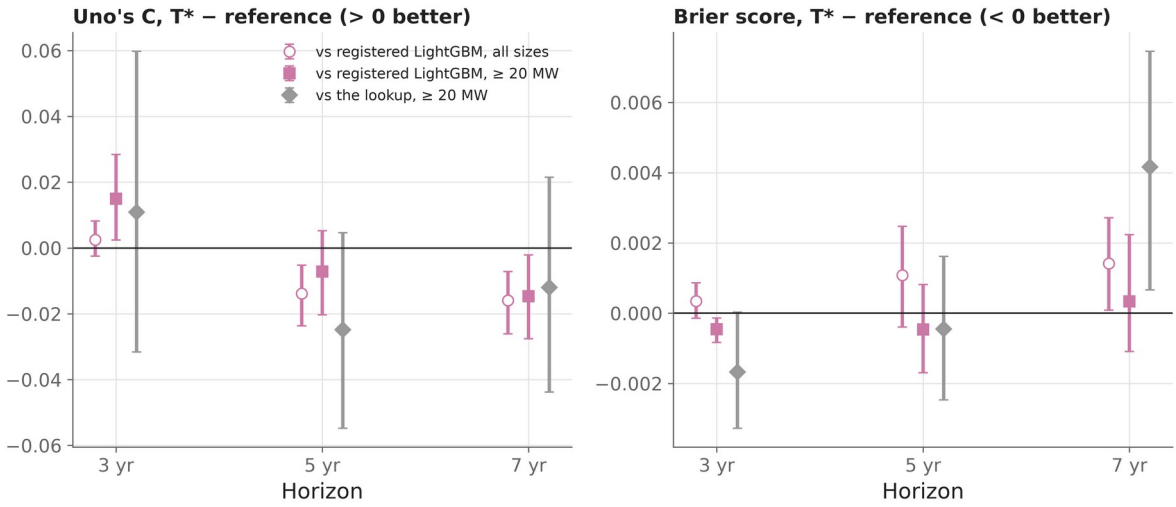


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Cohort: Track A test entrants 2017–2019

Figure A7. Other learners against LightGBM and the lookup: paired differences in Uno's C and Brier score by horizon.

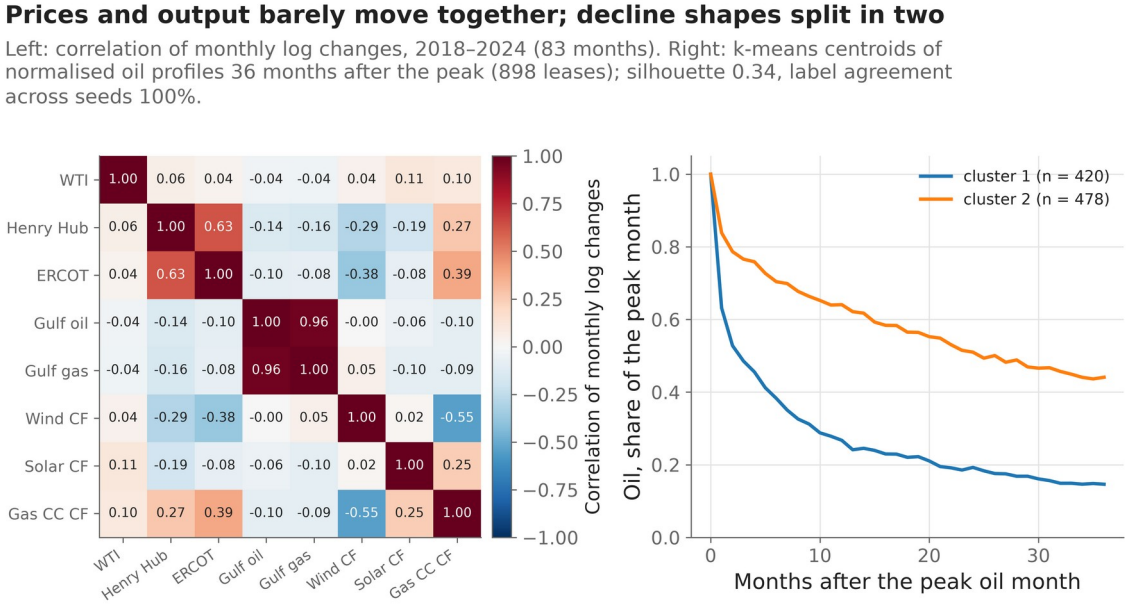


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). EIA-923 via EIA API v2 (electricity/facility-fuel). EIA WTI and Henry Hub monthly spot prices. ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: DERIVED. Cohort: eligible leases and Texas plants, 2018–2024; prices as in notebook 01

Figure A8. Correlation of monthly log changes across the three prices, Gulf output and Texas capacity factors, and the two shapes of lease oil decline after the peak.

Two areas hold most Gulf oil; Texas plants cluster by fuel

Left: 2024 oil by BSEE protraction area, eligible leases. Right: eGRID2023 Texas plants by primary fuel category, marker area by nameplate (655 plants); the 479 ERCOT queue requests of 2017–2019 are the power candidates' anchors (not drawn; their counties are on the project map).

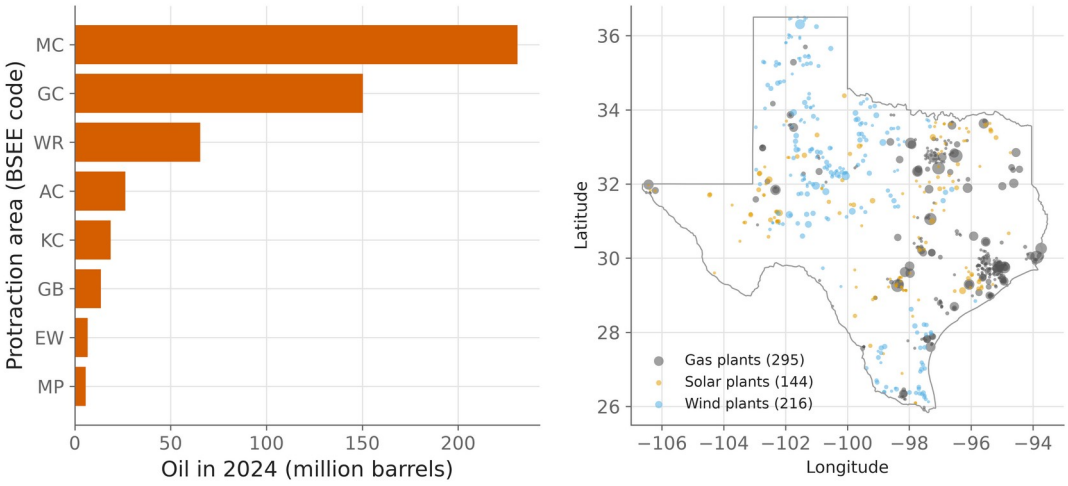


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). EPA eGRID2023. Provenance: OBSERVED. Cohort: eligible leases 2024; Texas plants in eGRID2023

Figure A9. Gulf of Mexico oil by protraction area in 2024, and Texas plants by fuel and size.

Cheaper oil or dearer leases empty the book of leases; wind stays

Book D re-solved under each committee question on the primary scenarios: capital by technology. Committee grid, contracted case.

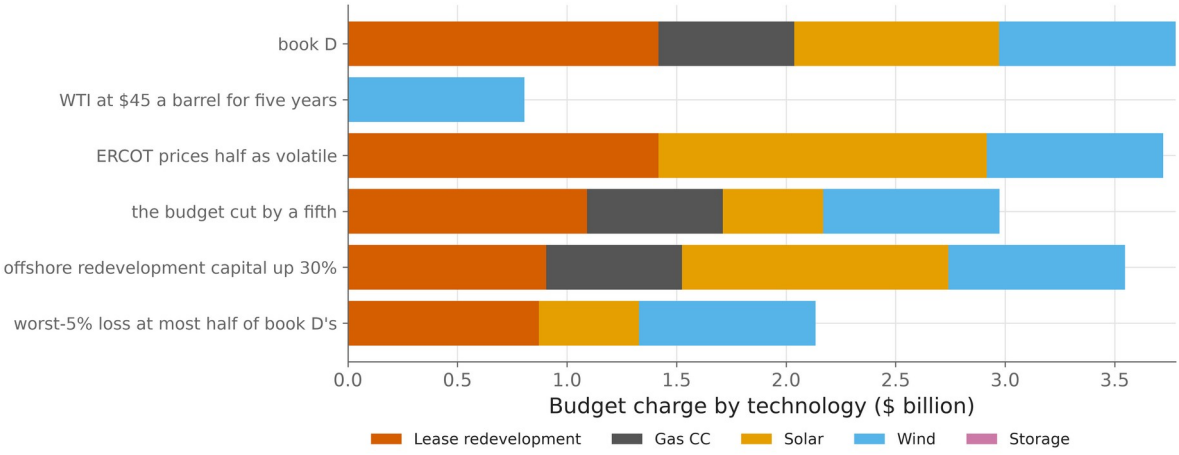


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: the forty candidates on 1 January 2025; scenarios of notebook 08

Figure A10. Book D's capital by technology, re-solved under each committee question.

Revising the book in 2027 adds nothing in this model

The same book D, committed in 2025 and either held or revised on 1 January 2027 after two years of prices and the agreement signal; policy fitted on primary scenarios, scored on 2,000 held-out ones. A scenario-based extension.

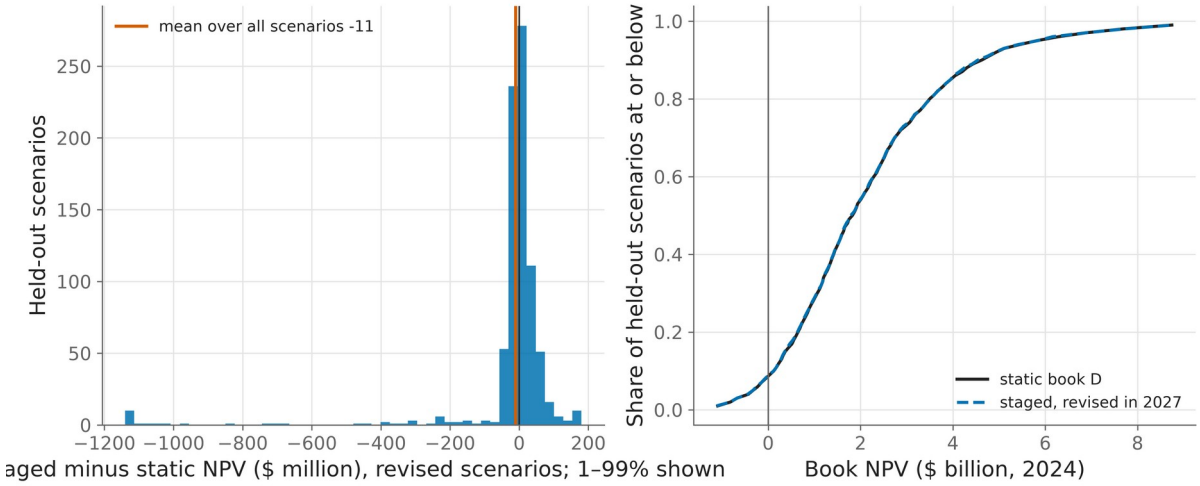


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: the forty candidates on 1 January 2025

Figure A11. Book D held, or revised on 1 January 2027, on the held-out scenarios.

The risk-aware book trades expected value for a better risk-adjusted score

Each comparison on the same 2,000 held-out scenarios with paired bootstrap intervals; committee grid, contracted case. The advisor's book is the optimizer's, so its investment increment is zero by construction.

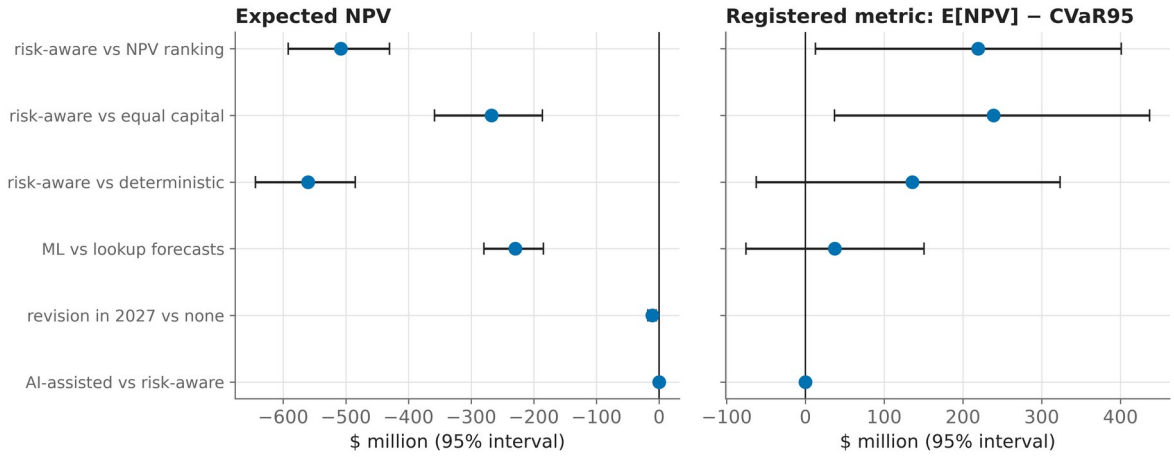


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: the forty candidates on 1 January 2025

Figure A13. Each decision process against its comparator on the same held-out scenarios, with paired bootstrap intervals.

The reserve is exceeded in 50% of scenarios; 11 of book D's 13 members are core

Left: book D's construction calls beyond its reserve, held-out scenarios. Right: how often each candidate is chosen when book D is re-chosen on 20 resampled primary sets; dotted line = core (80%). Committee grid, contracted case.

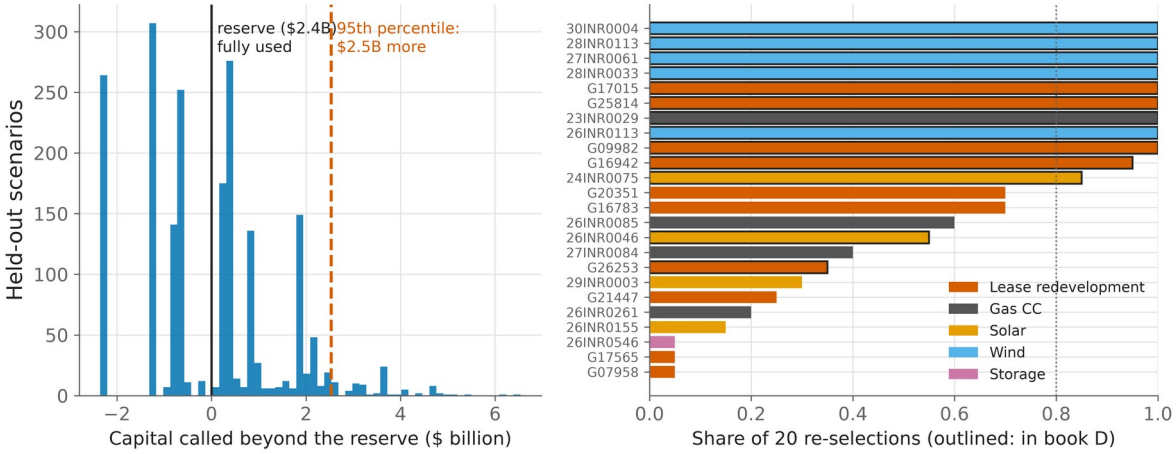


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: the forty candidates on 1 January 2025

Figure A14. Book D's construction calls beyond its reserve, and how often each candidate is chosen when the book is re-chosen on resampled scenarios.

The language model adds $35M per case; the tools add more

Paired per-case differences between the four arms on 48 committee cases (36 flawed, 12 sound), mean over five worlds, the committee's stated tail weight and liquidity line net of a 0.5% fee and 8% interest; case-bootstrap intervals. The v9 cases and recorded challenger outputs: a preview of the registered v10 test.

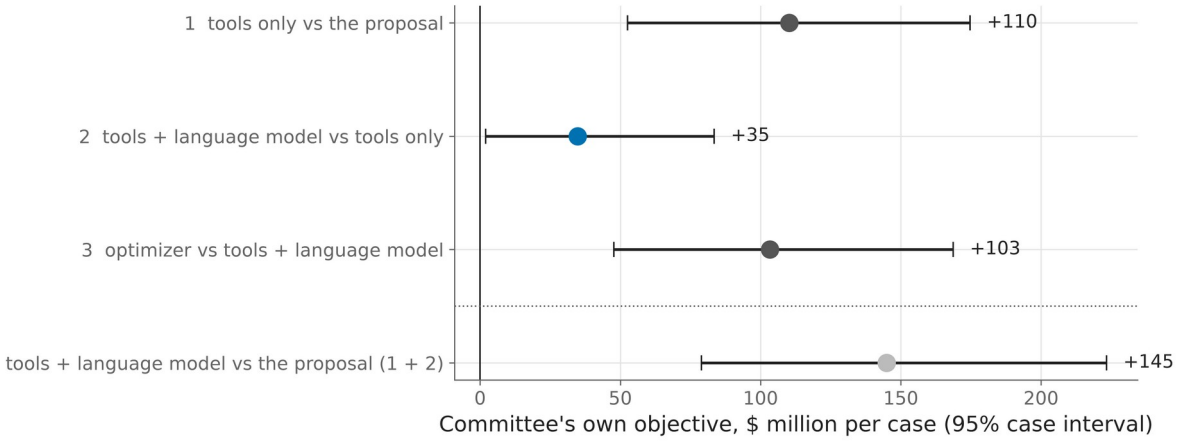


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: 48 simulated committee cases on the forty candidates of 1 January 2025

Figure A15. The attribution ladder as previewed on the first test's cases and the earlier candidate set, before the final registration: 41 of 48 readings equal to their key.

Priced as a bank would, the construction line still beats having none

Book D on 2,000 held-out scenarios by the size of the contingent line: the registered metric gross of financing, net of a 0.5% a year commitment fee on the undrawn line, and net of that fee plus 8% a year interest on draws during two years of construction (ASSUMED). Committee grid, contracted case.

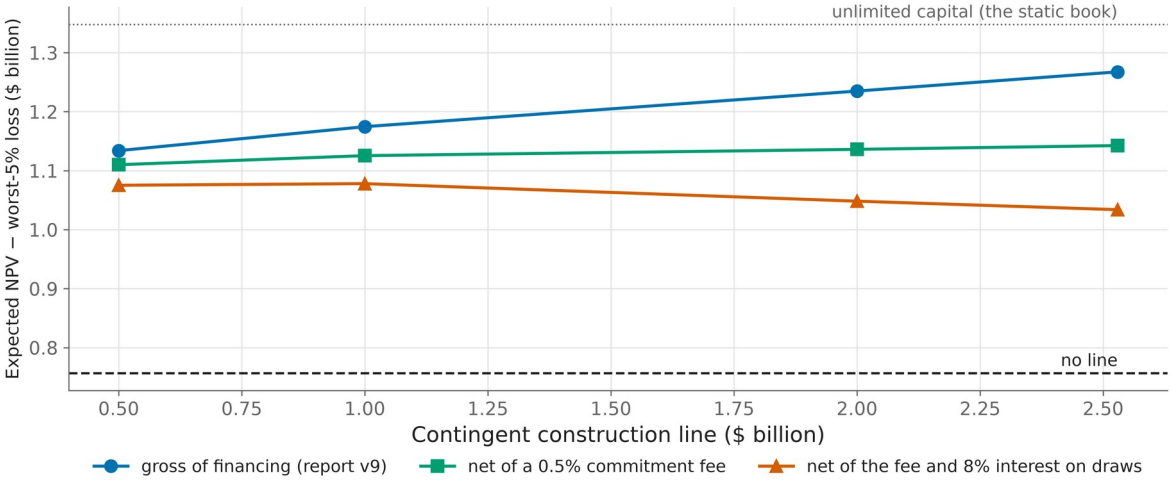


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: ASSUMED. Cohort: the forty candidates on 1 January 2025

Figure A16. Book D's registered metric by the size of the contingent line, gross of financing, net of the commitment fee, and net of the fee and interest (notebook 12).

Registered: the language model adds $43M per case, the tools and the optimizer more

Paired per-case differences between the four arms on the 48 registered v10 cases (24 template, 24 author-written; 36 flawed, 12 sound), mean over five worlds, the committee's stated tail weight and liquidity line net of a 0.5% fee and 8% interest; case-bootstrap intervals (1,000 draws). Scored once, as registered (C34, C35, C36).

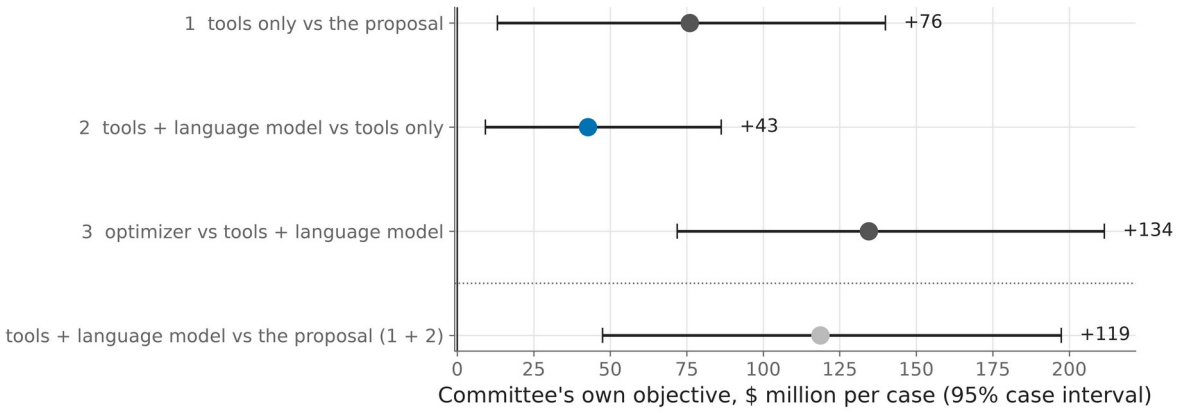


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: 48 simulated committee cases on the forty candidates public on 1 January 2025

Figure A17. The registered attribution ladder as scored: what the tools, the language model and the optimizer each add to the committee's decision, paired per case, with case-bootstrap intervals (notebook 12, section 6).

Appendix E — Part I in full: the developer's decision, which queue requests to fund

The method and evidence behind Section 4 for the queue side, kept in full. Words used one way throughout:

The glossary of terms used throughout is given at the front of the report.

Numbers. Prose carries one number per claim, rounded; intervals, second cases and counts are in the tables. Every number in the report is written by code to the results tables and checked against the text by a test. Dollar figures in Part I are in 2022 dollars, in Part II in 2024 dollars.

E.1 The decision and the data

A developer who wants to build a power plant in the United States must first ask the grid operator for a connection. The request enters a queue with a size, a technology and a place, and it leaves years later as an operating plant or a withdrawal. Most leave as withdrawals: of the capacity that asked to connect in 2000–2019, 12.6% was operating by the end of 2024 and 76.9% had withdrawn (Rand et al., 2025). Every request costs money to keep alive: a deposit, a study, then a larger deposit and a longer study. A developer holds many requests at once and must decide, request by request, which to fund, which to hold and which to drop.

The data are the project-level file of Lawrence Berkeley National Laboratory's *Queued Up* series, 2025 edition, with the 2026 edition alongside it for later outcomes. Only CAISO, ERCOT and PJM date both operation and withdrawal, so the timing models cover those three regions: 13,366 requests. The models are fitted on requests that entered up to 2016 and tested once on the 2,268 that entered in 2017–2019, every one followed for at least five years. Value and cost come from public files: EIA hub prices, the PJM market monitor's capacity revenues, the NREL Annual Technology Baseline 2024 and LBNL's PJM interconnection-cost estimates. Every value input that is an assumption carries its range in the repository.

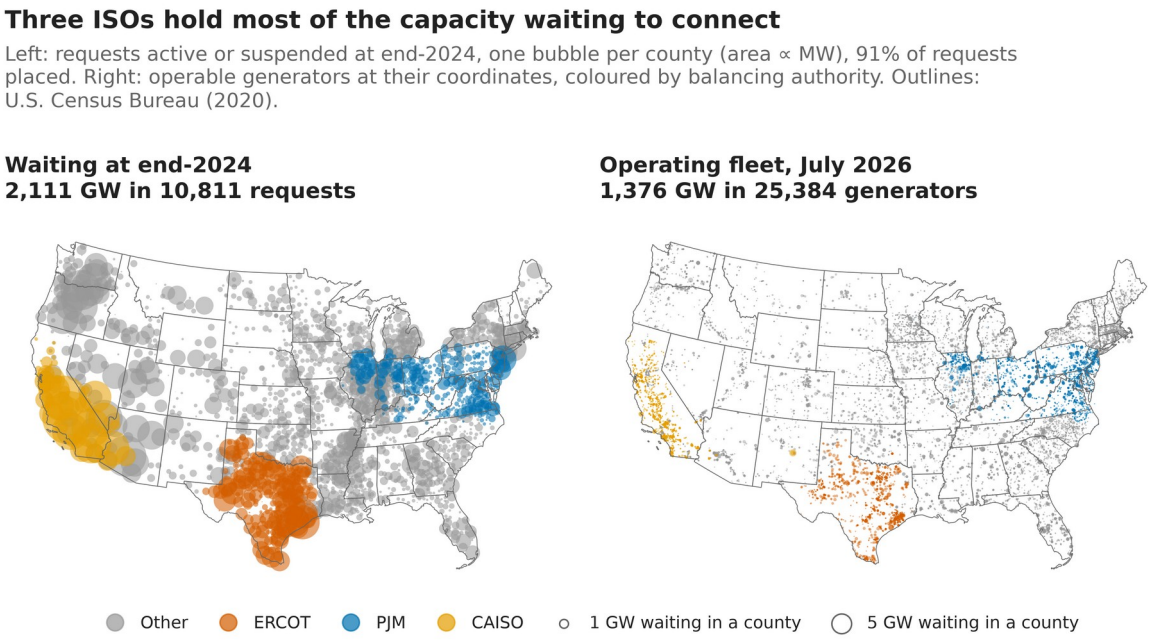


Figure E1. Requests waiting in U.S. queues at the end of 2024 by county (left; area proportional to MW) and the operating fleet (right).

E.2 Where the opportunity is

ERCOT is where requests get built. Five years after the request, 18.4% of ERCOT's entrants are operating, against 13.2% in PJM and 4.8% in CAISO (Figure E2). Gas is built most often and storage least: no region builds more than 3.4% of its battery or hybrid requests within five years. Across the country, the states sort into four kinds of queue market, and the best is the light, settled queue of the PJM core and Florida, where about one entrant in five operates (Appendix Figure A1).

ERCOT requests reach operation most often; CAISO's least

Aalen–Johansen cumulative incidence of operation (left) and withdrawal (right) by region; bands = 95% bootstrap (500 resamples); censored at 2024-12-31; ERCOT's withdrawal timing before Aug 2018 is distorted by a bulk date (section 2)

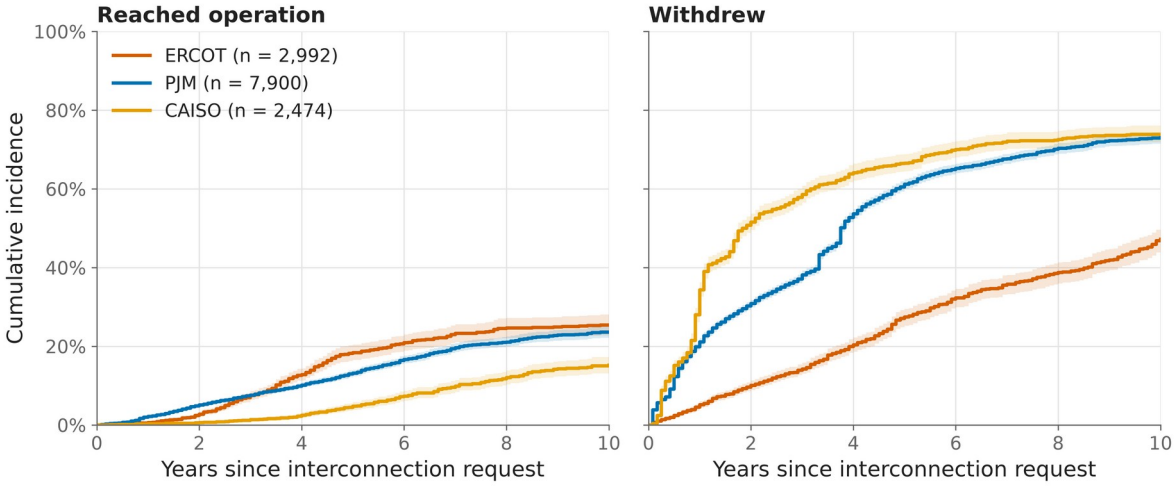


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Cohort: Track A requests (CAISO, ERCOT, PJM) entering 2000–2024

Figure E2. Share of requests operating (and withdrawn) by years since the request, by region, Aalen–Johansen estimates with 95% intervals.

The queue itself is the constraint. The capacity waiting at entry has grown many-fold in PJM since 2010, and since 2018 entrants neither leave nor get built at the old pace (Figure E3). This is the *level* problem of Appendix E.3: every cohort since 2017 completes below the one before.

Queues swell; recent entrants neither leave nor get built as before

By region and entry year: median GW waiting in the region on the request date, share withdrawn and share operating within three years (entrants to 2021, observable by end-2024); grey = the six other regions

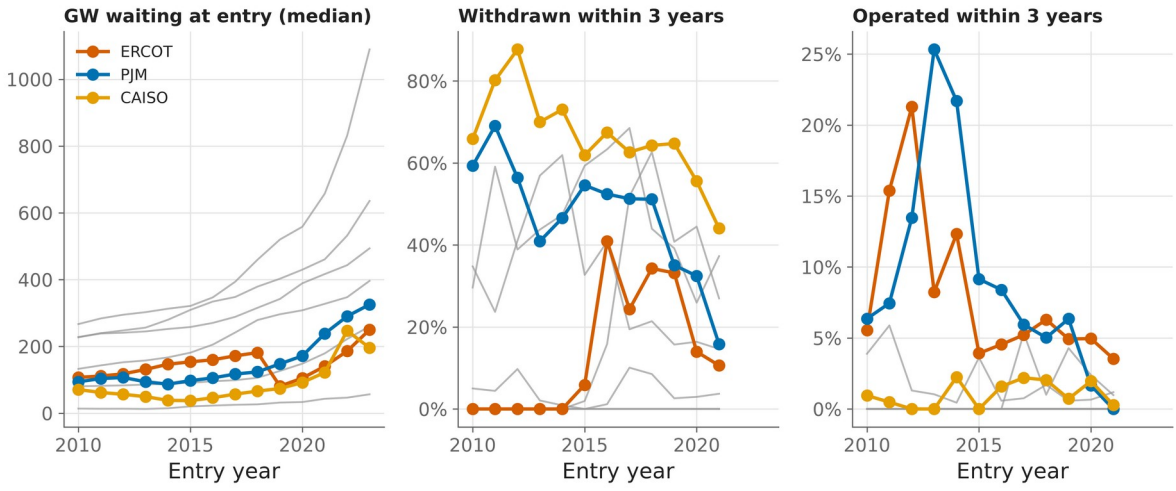


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Cohort: all requests in the clean file, entry years 2010–2023

Figure E3. Requests, capacity waiting at entry, and three-year withdrawal and operation shares by region and entry year.

The rule. Look first in ERCOT, where the base rate is highest and the agreement is dated. Look at PJM for gas and its capacity market. Treat CAISO as a market where entry odds are a fraction of ERCOT's and only a signed agreement changes them.

E.3 Which requests get built, and when

Three predictors are compared throughout, always in this order. **The model** is a gradient-boosted discrete-time hazard model (LightGBM) that sees what is known on the request date. **The lookup** is the share built among earlier requests in the same region × technology × size cell — the number a careful analyst would compute by hand. **The cohort average** is the share built in the request's region and technology. Appendix E.7 adds **the landmark model**, refitted two years after entry with the executed-agreement flag.

At entry, the model beats cohort averages but not the lookup. The model ranks the test cohort at an AUC of 0.86 at three years against 0.73 for cohort averages, and its top tenth of requests reached operation four times as often as the cohort. But for requests of 20 MW and more, the lookup ranks as well as the model: Uno's C is 0.78 for both at three years, and the model wins none of six comparisons (Table E1). Three other learners — a random survival forest, a neural hazard model and a blend — were registered and scored once; none beat the lookup either. At entry, the ceiling is what the file records about a request, not the learner.

Test cohort 2017–2019	Cohort average	Lookup (≥ 20 MW)	Model	Model (≥ 20 MW)
AUC, 3 years	0.733	—	0.858	—
AUC, 5 years	0.725	—	0.766	—
Uno's C, 3 years	—	0.783	—	0.779
Uno's C, 5 years	—	0.752	—	0.734
Top-decile lift, 5 years	3.35	—	3.96	—

Table E1. Ranking at entry. The model against cohort averages (all sizes) and against the lookup (20 MW and more).

Every cohort completes below the model's level. For requests of 20 MW and more, 11.0% of 2017 entrants and 5.9% of 2018 entrants operated within five years, while the model predicted about 12% for both. The 2020–2022 entrants operate within two years at about a quarter of the predicted rate. Refitting the intercept on the newest completed cohort removes about 40% of the error and none of the post-2020 gap. In megawatts, the 2017–2019 cohort was expected to bring 29.2 GW online within five years and brought 18.8 GW. A book must be sized to the newest cohort's level, not to the model's.

When: the model's clock helps in PJM only. Among test requests that operated within seven years, the model's expected year of operation is within a year of the truth 52% of the time, against 33% for the regional median. The gain is all in PJM; in ERCOT the historical median does as well.

Two years in, the model earns its keep. For a request still waiting two years after entry, the executed agreement changes what can be known. The landmark model with the agreement flag ranks the next three years at Uno's C 0.80 against 0.69 for the entry model, a gain of 0.11, while the lookup conditioned on surviving two years falls to 0.59 (Figure E4). In ERCOT, requests with an agreement in hand two years in operated within three years 56% of the time, against 5% without.

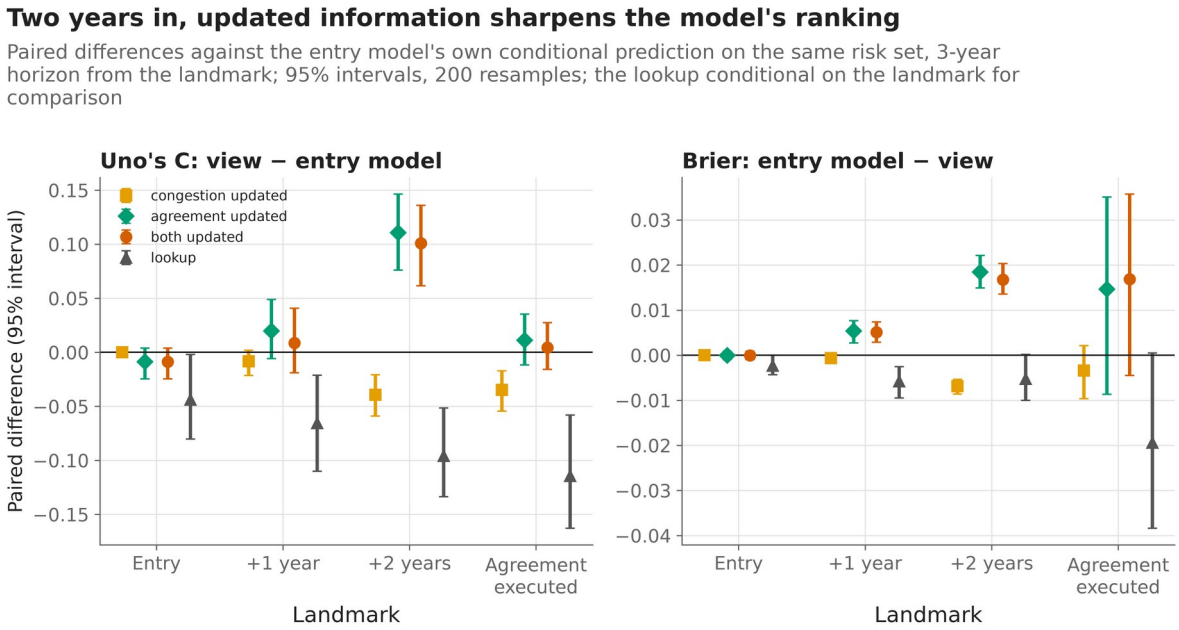


Figure E4. Gain in Uno's C and Brier score over the entry model at each landmark, three years ahead, for the landmark model with both inputs updated, with the agreement flag only, and for the lookup; 95% intervals. The text cites the agreement-flag variant.

The rule. At entry, rank by the lookup and the segment rules of Appendix E.4; the model adds nothing at utility scale. Two years in, re-score every request still held with the landmark model and the agreement. Plan the clock with the model in PJM and with the historical median in ERCOT. Size every book to the newest cohort's level.

E.4 Why requests fail

Size, technology and the agreement decide the odds (Figure E5). On the 5,497 entrants of 2010–2019, five-year completion runs 12.8% overall. In PJM it falls from 24% for requests under 5 MW to 4% at 500 MW and up; ERCOT is flat up to 500 MW. Gas builds most often (32% in PJM), solar-plus-storage less than stand-alone solar, and none of PJM's 37 offshore wind requests operated within five years. The executed agreement dominates everything: where the file dates one within two years of the request, 63% of ERCOT's requests were operating within five years; where none is dated, 0.8%. A survival model with both exits confirms the ranking: requests of 500 MW and more reach operation at 0.31 times the rate of those of 20–100 MW, and gas at 2.56 times solar (Appendix Figure A2).

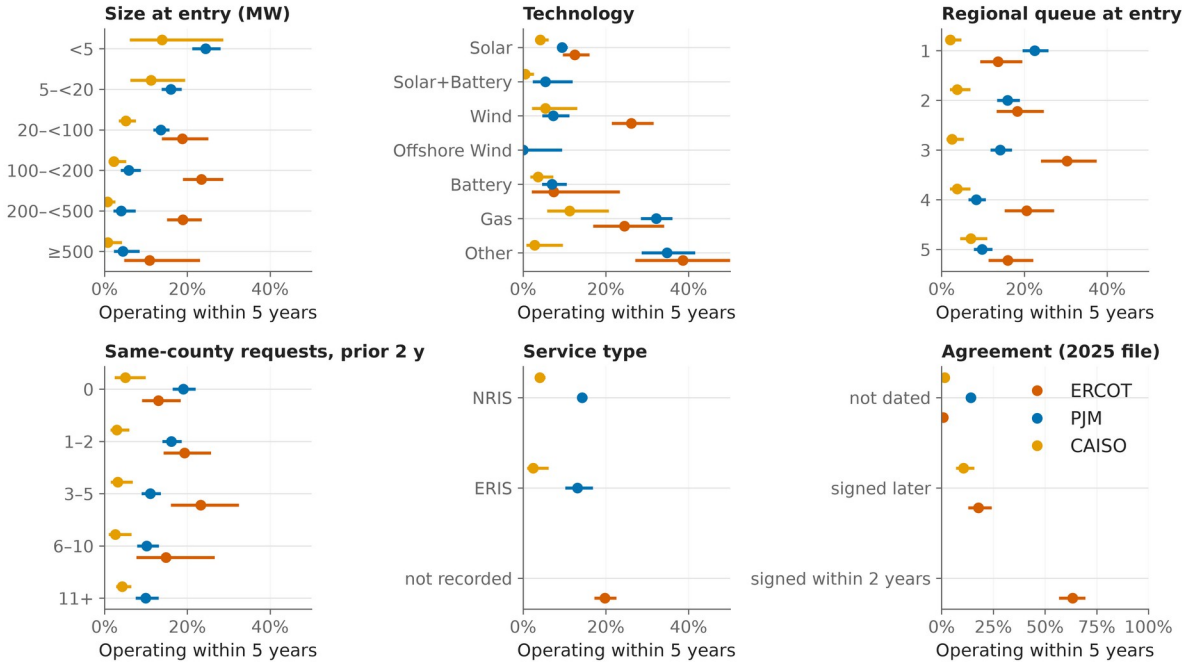


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Cohort: Track A entrants 2010–2019 (CAISO, ERCOT, PJM), outcomes through 2024-12-31

Figure E5. Five-year completion by six drivers within each region, 2010–2019 entrants, 95% Wilson intervals.

The regions run on different clocks. CAISO and PJM cull early: 48% of CAISO's waiting requests leave in their first year, mostly out of the system impact study. ERCOT culls late, after year four, out of the facilities study. Surviving two years roughly triples the odds of operating in the next three (Appendix Figure A3).

Ten rules a company can apply by hand (Figure E6). A depth-four tree on the entry attributes, fitted on 2010–2016 entrants and re-measured on 2017–2019, gives ten rules whose order held out of time (Spearman 0.77). Small gas outside CAISO is the best segment: 47% then 48%. Large CAISO requests in crowded counties are the worst: below 2% in both periods. The one rule that broke — solar in a county with no earlier request — shows that crowding measured at entry is a weaker guide than size and technology.

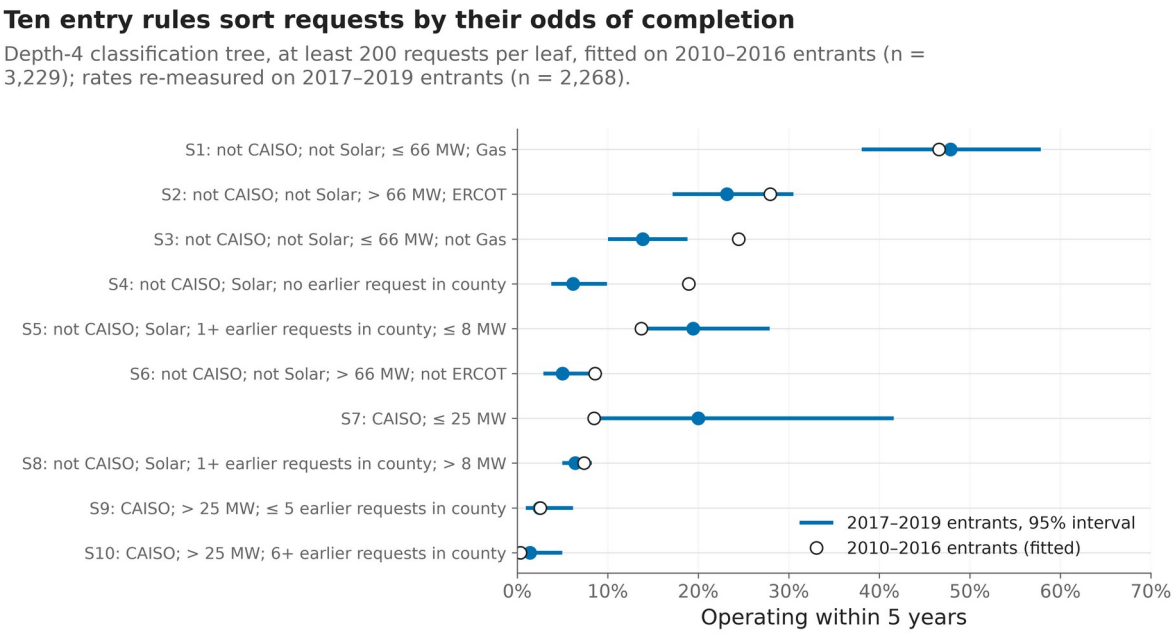


Figure E6. Five-year completion by the ten rules, fitting cohort (2010–2016) and later cohort (2017–2019), with Wilson intervals.

The rule. Enter the top two segments; enter the middle ones only with the two-year review scheduled; do not enter the bottom three on entry information alone. Review in year two in CAISO and PJM, after the cull; review at the agreement in ERCOT.

E.5 What a plant is worth

At merchant prices only wind pays everywhere (Figure E7). Each plant is valued as a developer would: a 20-year life, capital and operating cost from the NREL Annual Technology Baseline 2024, today's tax credits, and cash flows discounted at the technology's own real cost of capital (4.2–5.4%). Two revenue cases run side by side throughout the report. *Merchant* sells at the on-peak hub price. *Contracted* sells under a 15-year power purchase agreement at index prices (solar $57/MWh, wind $62), then merchant. At merchant prices five of 21 technology-region pairs have a positive value: wind in all three regions and gas and solar in PJM. Under contract ten pairs are positive, including solar everywhere. Storage stays negative in both cases, because the public files give it capacity revenue only.

With capital counted, few plants pay at merchant prices

NPV per MW at operation, 20-year life, real WACC per technology from the ATB Market case; bars = base value case, lines = low–high value cases (energy, capacity, credit, gas-price and PPA-price cases); CAPEX, O&M and new-build capacity factors from NREL ATB 2024; storage has capacity revenue only

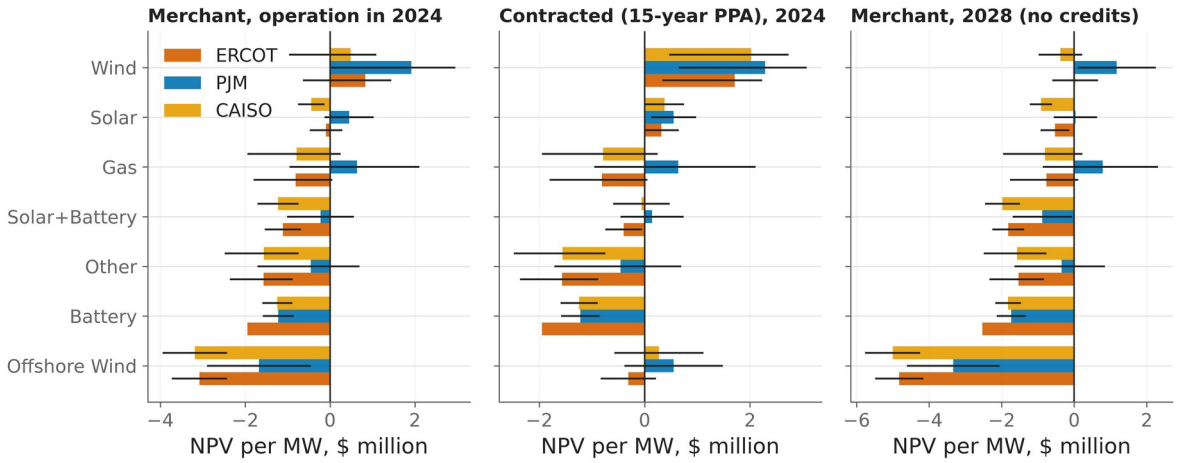


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Cohort: the value table of this notebook with new-build capacity factors; not a cohort

Figure E7. NPV per MW at operation by technology and region: merchant with today's credits, a 15-year contract at index prices, and merchant after the 2027 credit deadline.

The break-even says the same in dollars a company negotiates. A new solar plant needs a flat $44/MWh over twenty years and wind $41; the on-peak price the market pays them on Part I's national basis is below that for solar and above it for wind. Part II measures capture directly on ERCOT's 2024 hourly prices and finds the opposite order inside Texas (Appendix F.4), because Texas wind blows at night; Part I's assumed capture rates are national averages and are kept only for the three-region screen.

The rule. Develop wind anywhere; develop solar under a contract at or above the index, and in PJM even without one; develop gas in PJM, where the capacity market carries it; develop storage and hybrids only against a contract above their break-even; be operating by 2027 where the credit decides the sign. These values are in 2022 dollars; Part II works in 2024 dollars (the CPI-U ratio is 1.08).

E.6 Fund, hold or exit: two gates

The rule. At entry (the first gate) and two years in (the second), the company compares three things: the chance the request operates, its value if it does, and the cost of the next gate. It **funds** when the chance is at least the pool's expected rate — the mean of the probabilities the gate reads, so no outcome enters the rule; the first version used the cohort's realized rate (audit G2) — and the value is at least h times the cost, where the hurdle h is 3 at entry and 2 two years in; it **exits** when the chance is well below the pool or the value is negative even in the best case; it **holds** otherwise. At entry the chance comes from the lookup, because the model did not beat it; two years in, from the landmark model.

At entry, the funded requests were built three times as often as the cohort. At merchant prices the rule funds 166 requests and exits 1,542. The funded requests reached operation 33% of the time, the exits 6%, the cohort 10% (Figure E8). In ERCOT the rule separates less — its funded requests operated 23% of the time and its exits 18% — because there the entry attributes do not separate the requests and the agreement does (Appendix E.4). Carried to the end, the funded class returned +$4.1B at merchant prices and +$9.1B under contract. The funded class is mostly ERCOT wind and PJM gas and solar.

Two years in, the accelerated requests were built four times as often as the exits. The rule accelerates 178 requests at merchant prices and 251 under contract. They reached operation within three years 29–30% of the time against 7% for the exits, and returned +$3.6B at merchant prices and +$9.0B under contract. Table E2 gives the counts, rates and intervals for both revenue cases.

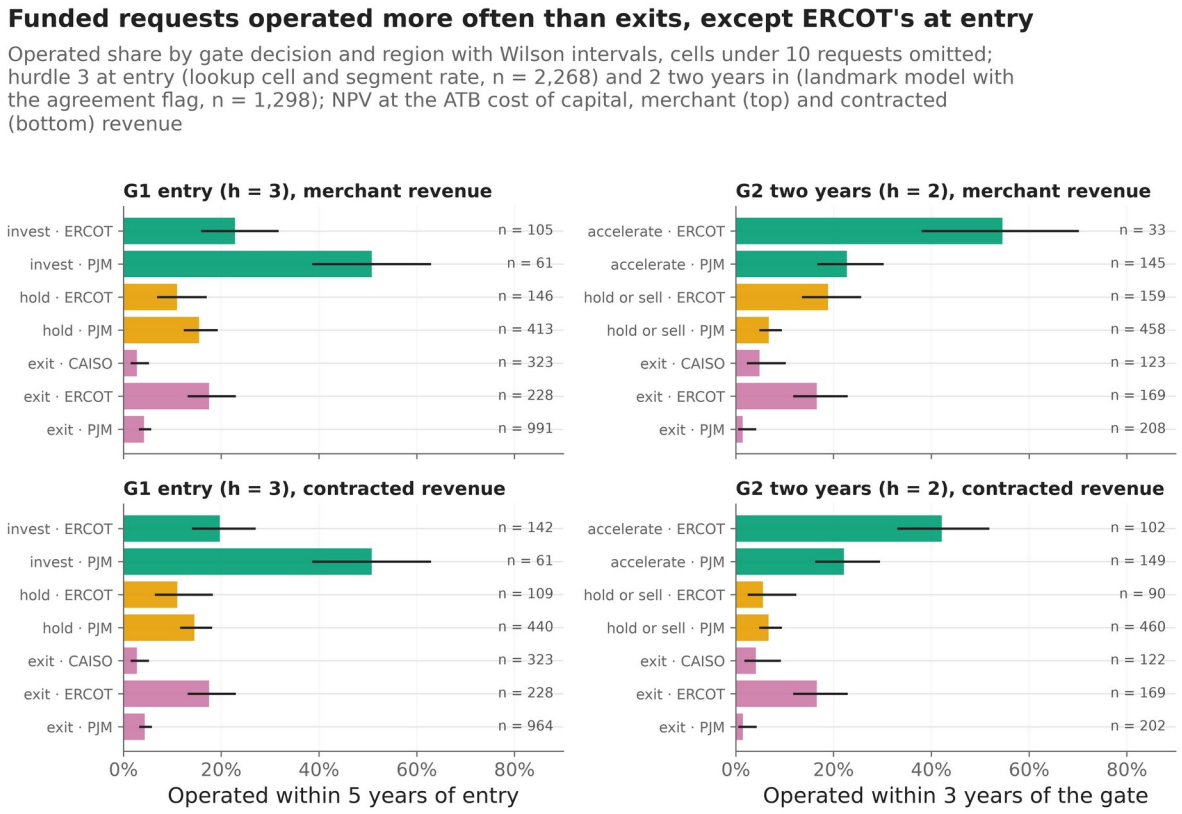


Figure E8. Share of requests that reached operation by gate decision and region, merchant and contracted revenue, with Wilson intervals.

Gate, revenue	Funded	Held	Exited	Funded: operated	Exited: operated	Funded: value, $M
Entry, merchant	166	560	1,542	33% [26%, 41%]	6%	+4,079
Entry, contracted	203	550	1,515	29% [23%, 36%]	6%	+9,137
Two years in, merchant	178	620	500	29% [23%, 36%]	7%	+3,642
Two years in, contracted	251	554	493	30% [25%, 36%]	7%	+8,975

Table E2. The two gates scored on what happened to the 2017–2019 requests. Operated: within five years of entry at the first gate, within three years of the second.

E.7 The book: when to buy and how much

Buy at the milestone, not at entry. A registered test settles the timing. Two years in, a book of 25 requests chosen by the landmark model held 19 that reached operation within three years, against 11 for the lookup that knew the same agreement status (Figure E9). A book of 50 held 33 against 24. The same books held fewer megawatts, so the landmark model picks more plants that get built, not more capacity. At entry the lookup ties or wins.

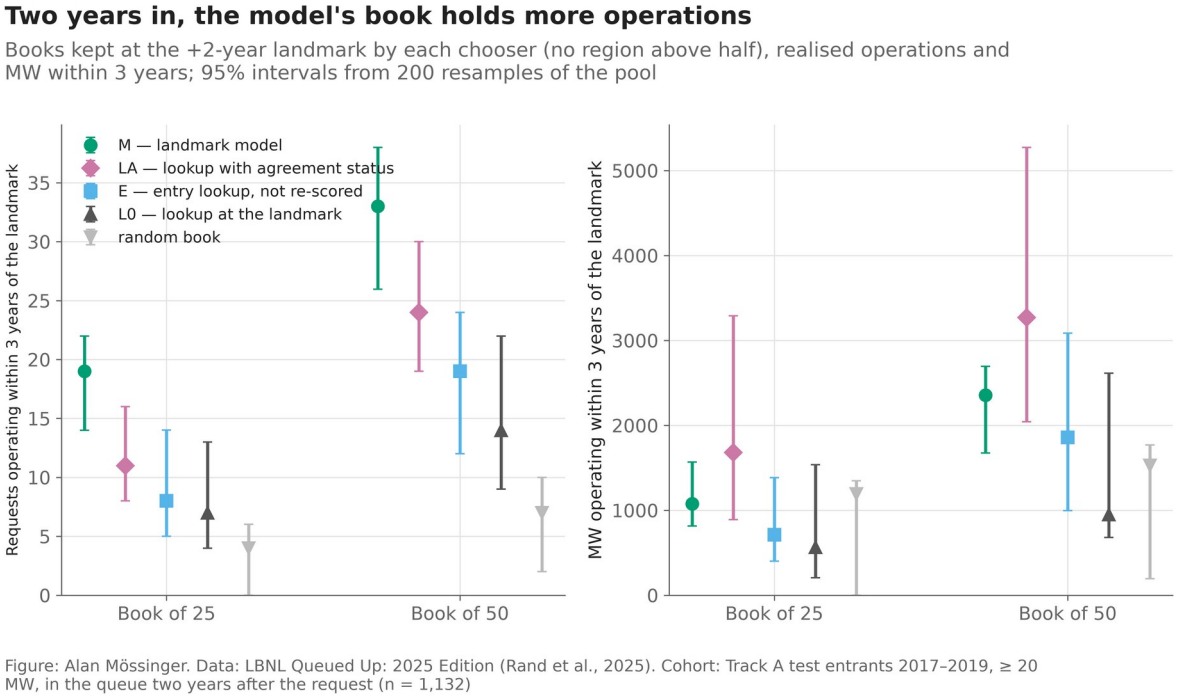


Figure E9. Operations and MW realised within three years of the two-year landmark, for books of 25 and 50 chosen by the landmark model, the agreement-aware lookup, the entry lookup and at random, with 95% intervals.

How many to add, and where. Adding requests in the landmark model's order while its probability stays at or above the pool's rate admits 410 of the 1,132 utility-scale requests still waiting; 23% of them operated, against 10.5% for the pool. This floor book is built without the regional cap of the registered book, which is why its first 25 picks hold 22 operations rather than 19. ERCOT is where the two-year review pays: 167 adds captured 70 of the region's 75 operations. In PJM, without dated agreements, the model separates little; in CAISO the floor admits two requests (Appendix Figure A4).

Four ways to spend one deposit budget. Four policies spent the same $20.11 million deposit budget on the 2017–2019 pool. Every policy that counts capital buys wind, and every one made money in both revenue cases, where a book chosen by megawatts lost it at merchant prices (Table E3). The risk-adjusted book barely differs from the plain value book, because the shared risk between requests is

small; the cohort level is the risk, and the newest cohort's early outcomes reveal it before a company buys.

Policy, same budget	Requests	Expected NPV, $M	P(loss)	Realised NPV, $M (merchant)	Realised NPV, $M (contracted)
Equal regional split	26	+1,404	2%	+581	+1,271
NPV per dollar, with caps	21	+1,536	3%	+396	+1,235
Model expected NPV, no caps	27	+1,803	2%	+391	+1,611
Risk-adjusted, with caps	21	+1,531	2%	+396	+1,235
Lookup, value index by MW	22	−747	79%	−46	+440
Random	22	−220	88%	−85	+15

Table E3. Six ways to spend the deposit budget on the 2017–2019 pool. Expected NPV and P(loss) are from the merchant scenarios.

The rule, region by region. *ERCOT:* invest; enter wind, enter solar with a contract, hold storage, and review every request at the agreement. *PJM:* invest in gas, wind and solar, prefer the light queues of the PJM core and requests under 66 MW, and review after the system impact study. *CAISO:* hold capital; enter only with a signed agreement and a contract.

E.8 Bridge to Part II

Part I answers the developer: which requests to fund and when. Part II answers the company: how to split one budget between those requests and assets that already produce. One thing carries forward — the hazard models of operation and withdrawal, refitted on what was known on the decision date, which become the timing engine of every power candidate in Part II. Two things do not: Part I's national value basis, replaced by ERCOT's 2024 hourly prices, and Part I's view from the request date, replaced by each request's chance of operating from 1 January 2025, conditional on how long it has waited (section 3).

Appendix F — Part II in full: the company's decision, how to split one budget

The method and evidence behind Sections 3, 5 and 6 for the energy portfolio.

F.1 The candidates on 1 January 2025

The leases are the twenty with the most output in the twelve months published by the decision date (to September 2024), among the 506 producing in 2024, out of 1,041 with at least sixty producing months. The requests are the five largest of each technology among the 1,441 ERCOT requests of 20 MW and more that ERCOT's own reports listed as live on 1 January 2025. In v6 they were drawn from the 2017–2019 entrants and valued at entry, though ten of those twenty had been withdrawn or built by the end of 2024; in v8 and v9 they were drawn from the 2025 edition, released after the date; this version draws them from what was public. Most of the twenty are 2023–2024 entrants of 500 MW and more, the size

and age at which requests are least likely to operate, so their chances sit far below the queue's averages unless an agreement is signed.

Twenty leases, flat 2024 prices	Registered grid	Committee grid
Capital	$1.35B	$5.12B
Incremental NPV, base	$8.19B	$2.73B
Incremental NPV, low and high grid cases	$5.69B and $5.10B	$4.77B and −$4.77B
Leases positive at the high case	19 of 20	0 of 20

Table F1. The lease book under the two grids. The registered low case pairs low costs with a low uplift, which is why it sits below base.

F.2 What the energy data say, and how the features were built

A quarter of lease-months produce nothing, and that is normal: leases are shut in for maintenance, storms and workovers, so 25.3% of the 203,963 lease-months report no oil or gas. WTI and Henry Hub correlate at 0.06 month to month; Henry Hub and the ERCOT price at 0.63, because gas sets the Texas price in most hours; Gulf output with any price at almost zero (Appendix Figure A8). Every feature at a forecast origin uses only months that were public by then, and a truncation test shows that features rebuilt after deleting all later data are identical; the test covers the feature builders, not the choice of the lease universe.

F.3 Forecasting output: machine learning helps for plants, not for leases

Twelve forecast lines were registered before scoring: lease oil and gas at 3, 6 and 12 months, plant capacity factors at 1 and 3 months, and prices. Each pits a machine-learning challenger against the best of five transparent baselines at rolling origins (Figure F1). Three of twelve pass, all for plants. The lease lines fail: the registered challenger's error at three months is 1.35 times the last-value baseline's, a bias from forecasting in logs; a level model added after scoring, and labelled so, brings the ratio to 0.96, and the scenarios draw their output uncertainty from it. Prices behave as random walks. The decision-level test of machine learning is Section 8's: the book chosen with the lookup's queue curves against the book chosen with the model's.

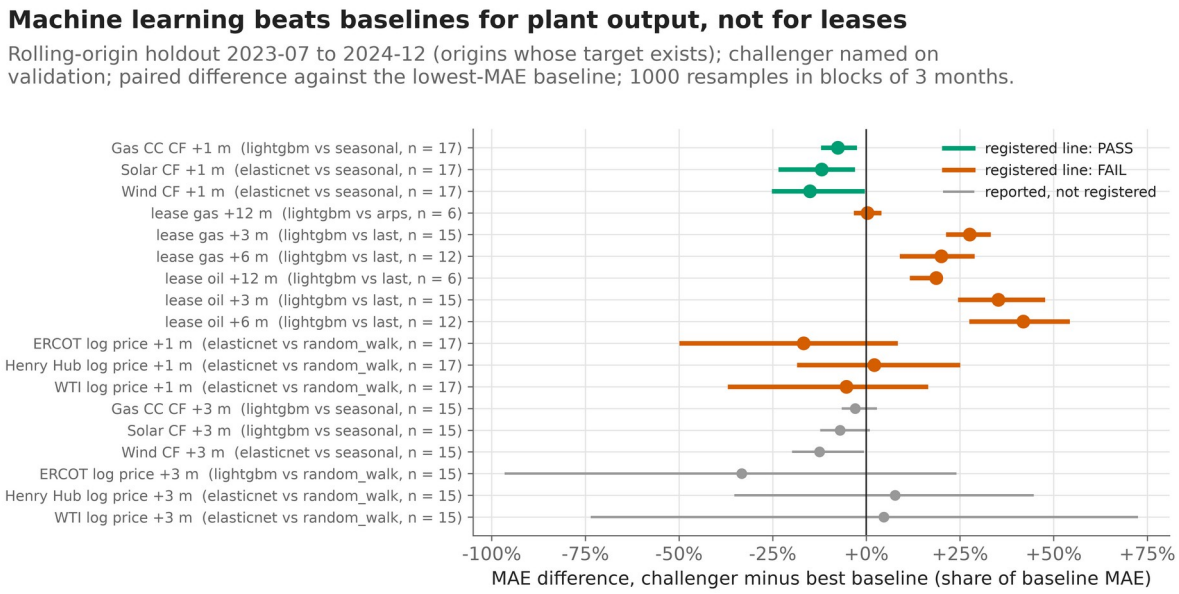


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). EIA-923 via EIA API v2 (electricity/facility-fuel). EIA WTI and Henry Hub monthly spot prices. ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: eligible leases, Texas plants and the three prices; holdout origins per line as labelled

Figure F1. Forecast error of the challenger relative to the best baseline, by registered line, with paired moving-block bootstrap intervals.

F.4 What each candidate is worth, and what that rests on

A lease redevelopment is an uplift on a producing asset. Each lease's baseline is an Arps decline fitted after the lease's peak in its last 36 months, switching to an 8% annual decline once its own decline falls below that; four leases still ramping decline at the floor from their recent level. The v6 fit had no floor, and ten of the twenty leases were projected never to decline. The redevelopment adds 20% of the baseline after a two-year lead time; its value is the cash flow it adds over doing nothing. The committee grid prices capital at $40,000 per incremental flowing barrel of oil equivalent a day and discounts at 8%; it was added after the registered results, at an outside reviewer's request, and is labelled ASSUMED.

A power request is a plant that may never be built. Its value if built uses the 2024 hourly prices — solar earns 92% of the flat hub price, wind 83%, gas combined cycle 136% because it runs only when prices clear its fuel cost — with ATB costs converted to 2024 dollars and the tax credits of the law in force on 1 January 2025: wind the production credit, solar and storage the investment credit, in every operation year of the window. The 2027 deadline for wind and solar was set by a July 2025 law and IRS Notice 2025-42, after the decision date; it is a stress. That value is weighted by the chance of operating in each of seven years and net of the deposits. On the flat 2024 path, under contract, ten of the twenty requests are positive and the twenty together sum to −$108M; without the credits −$933M, under the 2027 deadline −$478M.

F.5 Scenarios and risk

Prices, output, costs and timing are drawn together: 2,000 twenty-year paths resample calendar-year blocks of the 2018–2024 monthly log changes of WTI, Henry Hub and ERCOT, the same months for all

three, capped at ±1.5 a month and reverting toward their 2018–2024 real geometric means with a four-year half-life (ASSUMED). Output varies by the scored forecast errors, costs by a triangular multiplier, and each request's operation year by its curve from the decision date. The held-out set of 2,000 paths redraws every component with its own seeds; in v6 only its prices were redrawn. Two weaknesses remain. Seven historical years make every path, so there is effectively no joint oil–power tail beyond 2018–2024. And the reversion targets sit well above 2024's gas and ERCOT prices, which is what lifts the gas requests' scenario values above their flat-path values (Appendix Figure A5). A check on 5,000 paths — the registered 2,000 followed by new ones — leaves the order of the books unchanged (book D $1.36B, the deterministic book $1.21B, NPV ranking $1.14B, equal capital $1.12B) and book D's worst-5% loss at $0.84B; re-solved on 5,000 primary paths, book D keeps about two thirds of its members (Jaccard 0.65) and scores $1.382B against $1.363B.

F.6 Allocating the budget

Section 6 reports the books under the committee grid. Under the registered grid the leases are cheap against the barrels they buy, so the risk-aware book holds fourteen leases, expects $8.48B and almost never loses money; risk aversion has little to buy there. The registered comparison on that grid is Table A2.

F.7 When: the book at one to twenty years

A twenty-year NPV says nothing about what the owner holds at year three. Counting only the cash that falls within each horizon, under the registered grid, the best book holds nothing at one or two years, eleven leases at three and fourteen from five; its value within twenty years is $7.09B. A queue request is a carrying cost until it operates (Figure F4).

Leases pay from year three; queue requests stay a cost

Left: expected NPV of all forty candidates counted within each horizon, by sector (requests weighted by the queue models' probability of operating by each year). Right: the composition of the deterministic book at each horizon under the registered budget and cap.

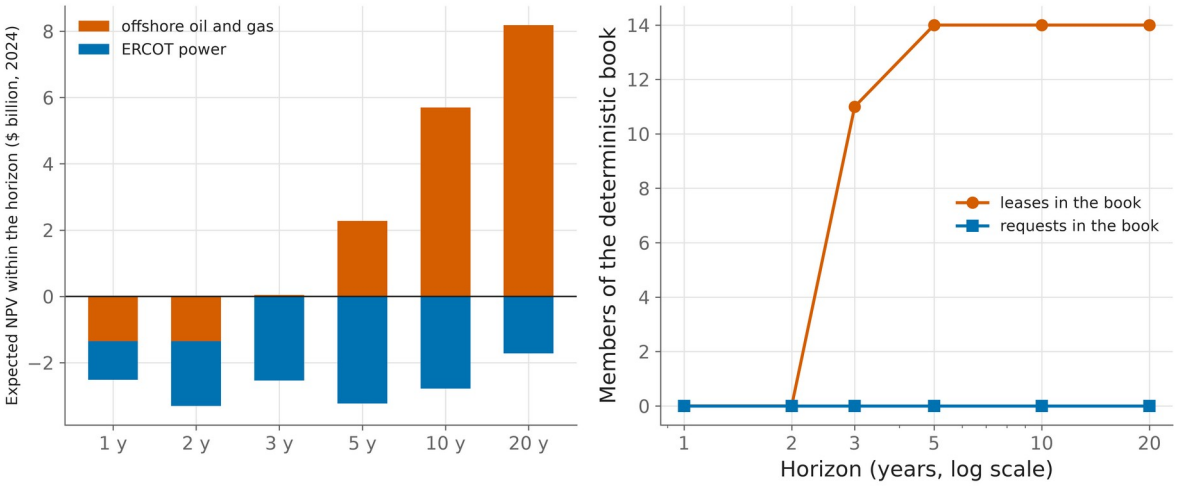


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Provenance: FORECAST. Cohort: the forty candidates of notebook 07 at the flat 2024 real prices

Figure F4. Expected NPV counted within 1, 2, 3, 5, 10 and 20 years, by sector, and the composition of the best book at each horizon.

F.8 The decision gym and the AI advisor

The gym scores decisions on outcomes, not prose. Sixteen past situations are rebuilt as of their date: fifteen requests drawn at random from each region's 2017, 2018 and 2019 entrants, the 2017 entrants two years in, ERCOT 2018 with a budget cut of a fifth, and three oil-price regimes — June 2014, January 2020 and March 2022. Since the audit the lease regimes hold the leases producing on their date, baselines fitted on what BSEE had published by then, scenarios drawn from the seven years before the date, and outcomes from the leases' observed production; v6 used the 2024 leases, 2024 fits and the realized future's mean. The three regimes are nearly one decision and count as one situation. Actors: instinct (top-N expected NPV), instinct per dollar, nothing, the registered rule (the gate rule for requests, the risk-aware optimizer for leases), the advisor and, when one is recorded, the human.

The rule beats instinct only by packing the budget better. Over the fourteen effective situations the rule beats instinct by $56M a situation, an interval of [−$12M, +$176M] that includes zero, and ties instinct per dollar (−$2.7M [−$12.7M, +$3.0M]): its edge in the lease regimes is choosing by value per dollar under a binding budget. In the queue rooms the gate rule's fund threshold is now the pool's expected rate, not the test cohort's realized one (audit G2), and the rule is behind instinct by $0.6M a room. The advisor trails the rule by $5.1M a situation [−$11.0M, −$1.5M].

The rule's edge over instinct, $56M a situation, is within noise

14 effective situations (the three lease regimes count as one), paired 95% interval [-12M, 176M]; the advisor -5.1M a situation against the rule. Top: queue situations as of their date (realized value = modeled value per MW × realized operation − cost to the gate, a DERIVED proxy); in 6 of them no drawn request had a positive realized value, so the best book is empty. Bottom: the lease regimes rebuilt as of their date, scored on observed production and prices. The tick is the best book with hindsight.

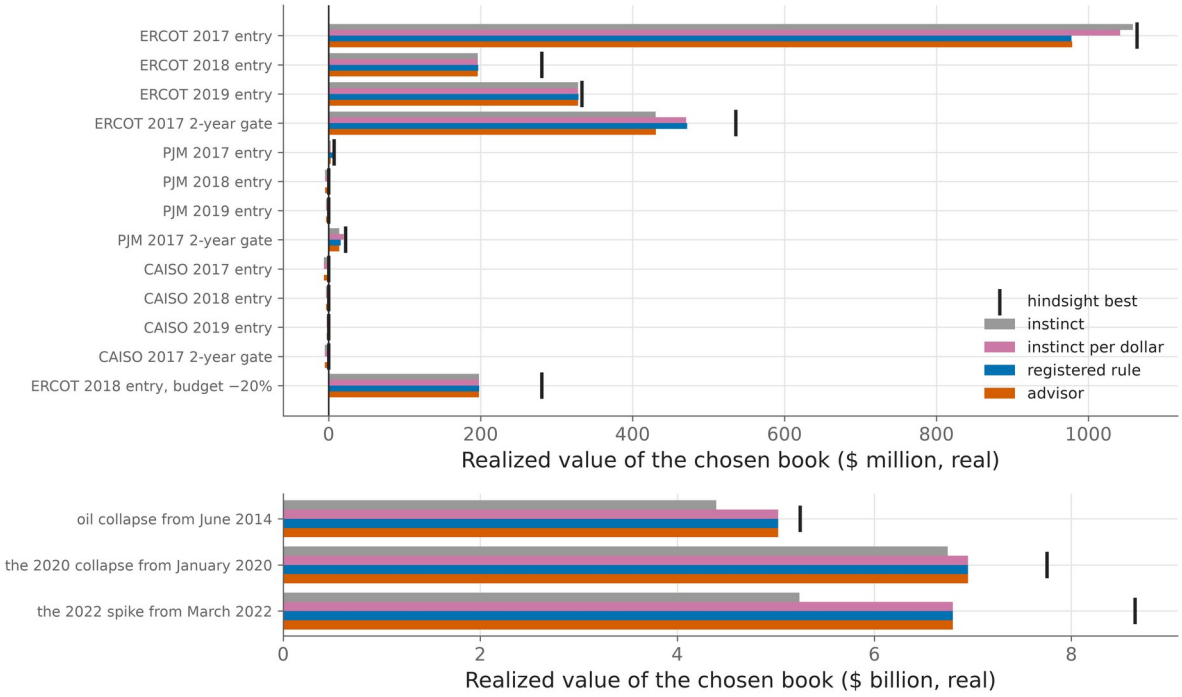


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Provenance: DERIVED. Cohort: ERCOT, PJM and CAISO requests of the 2017–2019 cohorts at entry and the 2017 cohort at its two-year gate; the twenty largest producing leases as of June 2014, January 2020 and March 2022

Figure F5. Realized value of the book each actor chose in each past situation, against the best book with hindsight.

Appendix G — Every finding, its evidence and its verdict

Question	Finding	Evidence	Verdict
Where to look	ERCOT builds most often: 18.4% of requests operate within five years, against 4.8% in CAISO.	Appendix E.2	Observed
Q1, at entry	The model beats cohort averages (AUC 0.86 against 0.73) but ties the lookup at utility scale.	Appendix E.3; C1–C12	Partly
Q1, two years in	A book of 25 chosen by the landmark model held 19 operations, against 11 for the lookup.	Appendix E.7; C15	Supported
Q1, on the decision date	Backtested as of 2020 the model ranks better than the lookup (AUC 0.731 against 0.686; in ERCOT 0.656 against 0.591) but over-predicts ERCOT's level.	Section 4	Supported for ranking

Question	Finding	Evidence	Verdict
Q1, one year on	30 of 1,441 live requests operated in 2025 where the corrected model expected 78 and the lookup 17; the model's slope is 0.66, the lookup's 0.01.	Section 4; Figure 3	Post-hoc
Q1, output	Machine learning beats baselines for plant output, not for leases or prices.	Appendix F.3; C20	Partly
Q1, in the decision	The book chosen with the lookup's curves is not beaten by the model's on the registered metric (+$37M [−$76M, +$151M]).	Sections 4 and 8	Not supported
Q2, risk	Under the committee grid the risk-aware book loses money in 9% of scenarios with a worst-5% loss of $0.82B; its construction calls exceed the reserve in 50.3% of scenarios.	Sections 6 and 8	Derived
Q2, the primary line	Under equal constraints the risk-aware book beats NPV ranking by $219M [+$13M, +$401M] on the registered metric, for $508M of expected value.	Section 6; C33	Registered, PASS
Q2, the line priced	A $2.53B contingent line is worth $1.03B on the metric net of a 0.5% fee and 8% interest, against $0.76B with no line; the fee breaks even at 2.04%.	Section 8; Figure 5	Post-registration
Q2, sensitivity	Oil at $45 for five years removes every lease from the book; redevelopment capital 30% higher swaps two members for three; the registered tail weight is best in three of four misspecified worlds.	Section 6; Appendix I	Derived
Q2, flexibility and information	Revising the book in 2027 with this rule is worth −$11M; perfect information is worth at most $825M, the ERCOT price level $47M, an agreement signal at most $59.4M.	Sections 4 and 6	Post-registration
Q3, the primary line	Adding the language model to the tools improves the committee's own objective by $43M per case [+$9M, +$86M] over all 48 cases.	Section 10; C34	Registered, PASS
Q3, reading	40 of 40 paraphrases read exactly; 38 of 48 case readings equal their key; 75% of injections held with 7 memos carrying an ungrounded number.	Section 10; C37–C39	Registered: PASS, FAIL, FAIL
Q3, argument	12 of 12 author-written flaws of argument named; 0 of 12 sound proposals over-challenged.	Section 10; C40–C41	Registered, PASS
Q0	Tools add $76M per case over the proposal, the language model $43M more, the optimizer $134M beyond that.	Section 10; C35–C36	Registered
Q3, the advisor	Its investment increment is zero; it read six questions correctly; it trails the rule in the gym.	Appendix B	Post-registration
Q2, financed the same way	Every process within its own line net of fee and interest: D leads at $1.03B against $0.82B (C) and $0.72B (A); D over A +$312M [+$110M, +$506M], over C +$213M [+$17M, +$412M]; with no line equal capital leads.	Section 8; Table K3; Figure 5	Post-registration
Q2, the book re-solved with the line	9 members, a $1.31B facility, −$49M [−$127M, +$23M] against D.	Section 8	Post-registration

Question	Finding	Evidence	Verdict
Q2, dependence and worlds	The tail grows monotonically with the share of co-moving scenarios (comonotonic bound $2.19B); D scores best in 7 of 7 worlds.	Section 5; Figure K4	Post-registration
Q0, the stronger baseline	Against an optimizer-first rule the language model's edge is −$14M [−$36M, +$2M]; the rule beats the checklist by $56M [+$19M, +$100M].	Section 10; Figure 6	Post-registration
Q2, an emissions cap	Any cap on the book's 1.27 Mt a year swaps its gas request for the 1,026 MW solar request with an agreement; the zero-emission book gives up $201M on the metric, $158 per tonne-year.	Section 7	Post-registration

Table G1. Every finding, its evidence and its verdict. Registered criteria are in Appendix A and Table 1; the V2 additions are post-registration (notebook 13).

Appendix H — The audit of 17 October 2026

An independent audit of report v6 and its code (docs/audit_2026-10-17.md) found that the prediction layer held and that valuation, allocation and the gym did not. The five findings that changed the recommendation:

Finding	What was wrong	What changed
A1	The top-N and equal-capital books ignored the registered 40% technology cap; the risk-aware book's "$1.9B for a smaller tail" was the price of the cap, not of risk aversion.	Every book is held to the same budget and cap and checked.
E1	Ten of the twenty power candidates had been withdrawn or built by the end of 2024, while the leases were valued at 2025.	One decision date; the requests are those still waiting, with chances conditional on their age — and, since the critique of 21 October, those ERCOT's own reports listed on the date.
E2	Ten of the twenty lease baselines were fitted with no decline.	A post-peak fit with an 8% terminal decline; the registered lease book fell from $14.6B to $8.19B.
G1	The gym's lease rooms used 2024 leases, 2024 fits and the realized future's mean.	Rooms rebuilt as of their date and counted as one situation.
G5	The advisor's value over the rule was never reported; it was negative.	Reported, and the advisor rebuilt to work through the engine.

Table H1. The audit's headline findings. Smaller corrections — the CVaR tail count, the gas heat content, the gate floor, ATB dollars, the credits, the provenance labels, the held-out draw — are in the audit's register and decisions #88–#92.

Changes after review. Three outside reviews in October 2026 led to the changes logged in docs/mentor_feedback.md. The first asked for an evaluated decision intelligence system; its first request was the audit above. The second found three inputs that were not public on the decision date and the capital calls unpriced, and asked that the language model help the committee decide; the

challenger and its first registered test were the answer. The third, a critique review of 21 October, found that the candidate set rested on a file released after the decision date, that the challenger's test statements came from the model family being tested, that the construction line was free, and that no primary estimand had been named; report v10 rebuilt the candidates from what was public, priced the line, named one primary line per question, added the four-arm ladder and the one-year forward test, and kept a three-column ledger of every registered line and its corrections. The executive review of 23 October 2026 asked for a decision-first report, every process financed the same way, a critical assessment of the scenario engine, a stronger baseline for the language model and an operational readiness assessment; this version (V2) answers it (docs/v2_review.md, notebook 13, decision #113). A course mentor is assigned after the proposal review; their feedback will be logged in the same file.

Appendix I — Capital allocation memo

To the investment committee, 1 January 2025. Every figure is a model output scored on held-out scenarios; lease costs are assumed; the candidates are those public on the date.

The ask. Commit up to $3.82B now — 40% of what the forty candidates would cost the budget today — across twenty Gulf lease redevelopments and twenty ERCOT requests still in the queue, with no technology above 40% of the budget. A lease costs its redevelopment capital; a request costs its deposits plus a reserve for the construction it may need.

The choice. The return-seeking book expects $2.73B with a worst-5% average loss of $1.52B. The registered risk-aware book expects $2.17B with a loss of $0.82B and a 9% chance of losing money. A more defensive book expects $2.07B with a loss of $0.74B. I recommend the committee choose its tail weight first; at the registered weight that is the risk-aware book.

What to do with each candidate under that book. Invest now in eight candidates, the leases and the requests with signed agreements. Invest in stages in five requests: pay the deposit now, commit construction at the agreement. Wait for information on twelve requests whose agreement signal is worth more than their deposit. Monitor fourteen; reject one.

What the commitment really is. The construction reserve is an average: the capital the book's requests call exceeds it in 50.3% of scenarios. Without a contingent line the book is worth $1.71B, not $2.17B, because requests that reach their agreement cannot be built; a line of $2.53B restores $2.11B gross and $1.98B net of a 0.5% commitment fee and 8% interest on the draws.

What would change it. Oil at $45 a barrel for five years removes every lease from the book and leaves the five wind requests, a book that expects $0.59B and loses money in 58% of scenarios. Redevelopment capital 30% higher drops two members for three and keeps $0.91B of leases, at a 15% chance of a loss. Knowing everything before committing would be worth at most $825M; the 2025–2029 ERCOT price level, $47M; one request's agreement signal, at most $59.4M.

Before you sign. State your rules in words — your tolerance for loss, your liquidity, what you will not hold, and your tail weight, because an unstated one is read as zero. The challenger reads them back as

numbers for you to confirm, tests your proposal against them and names the computed book that keeps them. If you depart from a book it guides you to, the ledger asks you to acknowledge the evidence.

Appendix J — Glossary

The terms as this report uses them.

Term	Meaning in this report
Request	An application to connect a new plant to the grid; it ends as an operating plant or a withdrawal.
Entry, two years in, the agreement	The three moments a request is reviewed: the request date, two years later, and the date an interconnection agreement is executed.
The model	A gradient-boosted hazard model (LightGBM) that sees only what is known on the request date.
The landmark model	The same model refitted two years in, with the agreement flag.
The lookup	The share of earlier requests in the same region, technology and size band that were built — what a careful analyst would compute by hand.
The cohort average	The share built in the request's region and technology.
The level	The share of a cohort that is ever built; it has fallen for every cohort since 2017.
Merchant, contracted	Two revenue cases: selling at market prices, or under a 15-year power purchase agreement at index prices.
Book	A set of projects funded together under one budget.
The grid	The assumed costs of redeveloping an oil and gas lease; the *registered grid* was fixed before any result was seen, the *committee grid* is the harder one an investment committee would use, and the main case throughout.
OBSERVED, DERIVED, FORECAST, ASSUMED	The provenance of every dollar figure: a public record, arithmetic on records, a model's output, or a stated assumption.
Held-out scenarios	2,000 scenarios redrawn in every component and never used to choose a book; every book is chosen on 2,000 primary scenarios and scored on the held-out ones.
Worst-5% average loss	The mean of the 100 worst of 2,000 held-out outcomes — the conditional value at risk at 95% of the loss (CVaR95).
Tail weight (λ)	How much a dollar of worst-5% average loss counts against a dollar of expected NPV; λ = 1 is the registered setting.
Registered metric	Expected NPV minus the worst-5% average loss (tail weight 1), on held-out scenarios.
Registered line	A success criterion written and committed, with its rule, before the data it scores were looked at; scored once, never re-scored.

Term	Meaning in this report
The challenger, the checklist	The governed language model that reads a committee's rules and writes a memo from computed facts; the same tools with fixed rules and no language model, which is the control arm.
The facility	A contingent construction line that funds the capital a request calls beyond its reserve when its agreement is signed.
Invest now, invest in stages, defer, monitor, reject	The five statuses the system assigns to each candidate.

Appendix K — The report in full

Sections 1–9 as written before the body was brought to the course's length. References to "Section n" inside this appendix are to K.n; Figures K1–K5 and Tables K1–K3 are here, Figures 1–6 and Table 1 in the body.

K.1 Introduction

K.1.1 Motivation

Oil, gas, power, renewables and storage compete for one budget, and the competition is unequal. A lease redevelopment spends its capital now and produces from year three; it rarely loses money, and when it does, every lease loses together because they share one oil price. A power request costs little to hold — deposits and studies — but calls for ten times as much the day its interconnection agreement is signed, and most requests never reach that day: of the capacity that asked to connect in 2000–2019, 12.6% was operating by the end of 2024 and 76.9% had withdrawn (Rand et al., 2025). A committee that ranks these candidates by expected value fills the budget with the largest requests and the largest leases, and takes a tail it has not priced.

Maximizing expected NPV is not what an owner wants. It wants expected value and a tail it can bear, and the two trade off at a rate the committee must choose rather than discover after the fact. Prediction does not settle the trade-off. A better forecast of which requests will be built changes the ranking of candidates, but the book is sized to the level — how many will be built — and the level is what no forecast knew well on the decision date. The decision is made by the constraints, the capital plan and the tail.

Five challenges follow. *Allocation across competing technologies:* distinct risk–return profiles, horizons and capital needs mean the best portfolio is not the best projects taken one by one. *Uncertainty and its interactions:* commodity prices, power prices, production, completion and construction costs are uncertain together, and their co-movement can produce portfolio losses that no single investment shows. *The gap between forecasting and deciding:* a more accurate prediction changes a ranking; it does not say how much risk to accept or when to commit. *Financing and liquidity:* some projects cost little to hold and much to build, so a portfolio that is attractive on paper can be infeasible when several call for construction capital at once. *Committee decision quality:* proposals carry inconsistent assumptions, sunk-cost reasoning, unpriced downside and departures from the company's own rules, and conventional tools find numerical errors without challenging the reasoning.

K.1.2 Research questions

The project asks four questions, each with a pre-registered primary test (Section 5).

- **Q1, prediction.** Do machine-learning forecasts of completion, output and prices beat transparent baselines, and does a better forecast make a better portfolio?
- **Q2, risk-aware allocation.** Under identical constraints, does a risk-aware portfolio beat NPV ranking, equal capital by sector and deterministic optimization on a stated risk–return metric, and does the answer survive consistent financing and adverse markets?
- **Q3, governance.** Does a governed language model improve a committee's decision beyond deterministic tools, does it read policy reliably and resist manipulation, and can its role be bounded so that it reads and argues but never computes or decides?
- **Q0, attribution.** Where does the value come from — the tools, the language model or the optimizer?

K.1.3 Contributions

The contribution is an integrated, evaluated architecture rather than any single model: prediction, uncertainty, prescription, financial feasibility and an accountable human decision, each measured on the same held-out scenarios. Three findings follow from it. Risk-aware optimization improves modelled portfolio decisions over conventional selection, and the improvement grows when financing is made consistent. Forecast accuracy creates value only through the decision, not by itself. A governed language model can support executive reasoning, policy interpretation and committee oversight without holding authority — and its demonstrated value lies there, not in choosing the portfolio.

K.1.4 Organization

Section 2 places the work in the literature. Section 3 describes the data and the candidate universe. Section 4 presents the methods stage by stage. Section 5 sets out the evaluation design and the pre-registered criteria. Section 6 reports the results. Section 7 discusses what they mean, where they fail and what they do not show. Section 8 addresses ethical considerations and Section 9 concludes. The appendices keep the full evaluation record, the challenger's design, the reproducibility map, the supporting figures, the two parts of the analysis in full, every finding with its verdict, the audit of 17 October 2026 and the capital-allocation memo; references to "Section *n*" inside the appendices use the numbering of the original report, and a mapping is given at the start of the appendices.

K.2 Related work

From prediction to prescription. The distinction between predicting an outcome and prescribing a decision is the subject of a growing literature. Bertsimas and Kallus (2020) formalize prescriptive analytics as learning decisions from data rather than learning forecasts and optimizing afterwards; Elmachtoub and Grigas (2022) show that when a forecast feeds an optimization, the forecast should be trained on decision error rather than prediction error; Lepenioti et al. (2020) survey the field and its open problems. This project takes the complementary, evaluation-first position: it keeps prediction and optimization as separate, testable modules and asks, on held-out scenarios, whether a better predictor produced a better decision.

Portfolio theory in energy. Mean–variance portfolio selection (Markowitz, 1952) was carried into electricity planning by Awerbuch and Berger (2003) and into fuel-mix diversification by Roques, Newbery and Nuttall (2008). This project replaces variance with the conditional value at risk, whose linearization (Rockafellar & Uryasev, 2000) allows the tail to be optimized exactly in a mixed-integer program, and it adds the constraints a committee actually faces — budget, concentration caps, liquidity and staged capital calls. Investment under uncertainty and the value of waiting (Dixit & Pindyck, 1994) motivate the "defer pending information" status and the value-of-information calculations.

Commodity prices and scenarios. The scenario engine resamples calendar-year blocks of historical log changes (Künsch, 1989) around a mean-reverting level, a reduced form of the short- and long-term factor models of Schwartz and Smith (2000) and the long-run evolution of energy prices described by Pindyck (1999). Out-of-sample forecast evaluation follows Tashman (2000) and Hyndman and Athanasopoulos (2021).

Interconnection queues. Lawrence Berkeley National Laboratory's *Queued Up* series (Rand et al., 2025, 2026) is the source of the queue data and of the completion rates that motivate the level problem; Johnston, Liu and Yang (2023) analyze the queue empirically, and FERC Order No. 2023 reformed its procedures. The completion models are discrete-time hazard models (Cox, 1958, 1972; Allison, 1982) with competing risks (Fine & Gray, 1999; Putter, Fiocco & Geskus, 2007; Aalen & Johansen, 1978), refitted at a landmark (van Houwelingen, 2007), fitted with LightGBM (Ke et al., 2017) and evaluated with time-dependent concordance and Brier scores (Uno et al., 2011; Blanche et al., 2013; Graf et al., 1999; Wolbers et al., 2014). Calibration is assessed after Zadrozny and Elkan (2002). Lease baselines use Arps (1945) decline curves.

Language models in decision support. The challenger is built on two observations: that motivated reasoning shapes what committees argue (Kunda, 1990), so a challenger must be able to state the opposing case as its holders would (Caplan, 2011); and that a language model's numbers must be grounded in retrieved, verified facts (Lewis et al., 2020). The design goes further than grounding: the model never computes a value, every number in its memo is checked against a computed fact sheet, and the action it may take is bounded by code.

K.3 Data

K.3.1 Sources

Every source is public: LBNL's *Queued Up* project files (2025 edition for the decision, 2026 edition for outcomes), ERCOT's monthly interconnection reports, BSEE lease production, EIA prices and generation, ERCOT's hourly prices, the PJM market monitor's capacity revenues, LBNL's PJM interconnection-cost estimates and the NREL Annual Technology Baseline 2024. The file data/raw_manifest.csv records every raw file with its hash, and a clean Linux environment reproduced every registered verdict from them (Appendix C).

K.3.2 One decision date and the candidate universe

Both sectors are decided on 1 January 2025 with what was public that day. The twenty leases are those with the most output in the twelve months published by the decision date (to September 2024), among

the 506 producing in 2024, out of 1,041 with at least sixty producing months. The twenty requests are the five largest of each technology — gas combined cycle, solar, wind, storage — among the 1,441 ERCOT requests of 20 MW and more that ERCOT's own reports listed as live on the date, about 310 GW. For the queue, the LBNL 2024 edition (requests through 2023) and ERCOT's eight monthly reports of 2024 say which requests were live, their size and whether an agreement was signed. Each request's chance of operating within seven years is conditional on its age and on whether its agreement is signed; the hazard models learn from histories censored at the date, in which a request still waiting is right-censored, never counted as failed. The date discipline has a cost the report keeps visible: the 2024 exits were in the ISOs' own reports during 2024 but compiled by LBNL later, a vintage gap stated, not removed. Most of the twenty requests are 2023–2024 entrants of 500 MW and more, the size and age at which requests are least likely to operate, so their chances sit far below the queue's averages unless an agreement is signed.

The budget is 40% of what the forty candidates would cost: $3.82B, with no technology above 40% of it. A lease charges the budget its redevelopment capital now; a request charges its deposits now plus a reserve for the construction it will call if it reaches its agreement, the expected draw.

K.3.3 Valuation inputs and provenance

A lease redevelopment adds 20% to output after two years; its value is the cash flow it adds over a baseline that declines at least 8% a year (ASSUMED; 5–12% as the range). Each lease's baseline is an Arps decline fitted after the lease's peak in its last 36 months, switching to an 8% annual decline once its own decline falls below that. Redevelopment costs are not public, so results are shown under two grids — the registered one and the harder *committee grid*, which prices capital at $40,000 per incremental flowing barrel of oil equivalent a day and discounts at 8% — and the committee grid is the main case throughout. A power request is valued on 2024 hourly prices — solar earns 92% of the flat hub price, wind 83%, gas combined cycle 136% because it runs only when prices clear its fuel cost — with ATB costs converted to 2024 dollars and the tax credits of the law in force on the date; the 2027 deadline a July 2025 law set for wind and solar is a stress. Every dollar figure carries a provenance label — OBSERVED, DERIVED, FORECAST or ASSUMED.

Leases add value; wind and solar requests pay only under contract

Left: incremental NPV against the Arps baseline, base case with the high-cost to low-cost span (costs ASSUMED, production FORECAST). Right: expected NPV on 1 January 2025 = P(operation by year, given the age in the queue) × NPV if built − cost to carry; 2024 ERCOT North basis, ATB costs in 2024 dollars, credits as earned.

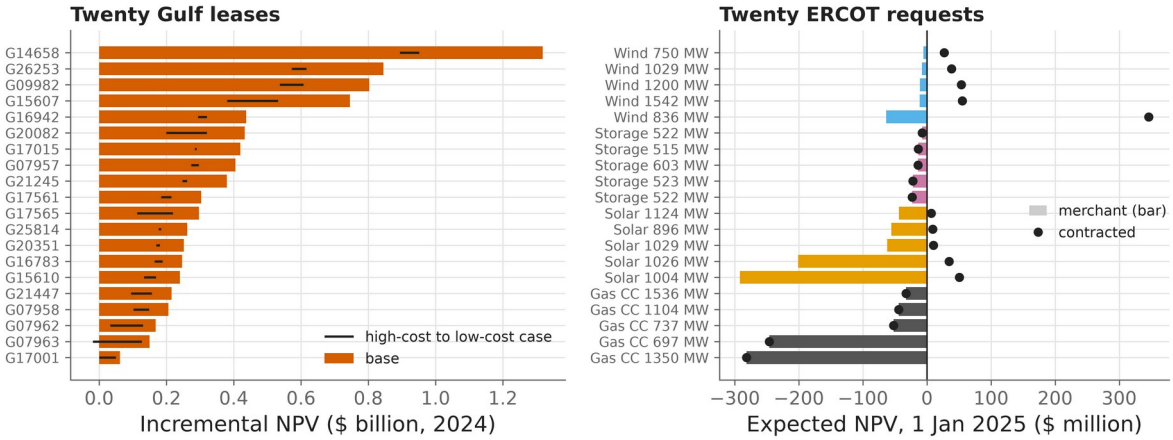


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Provenance: ASSUMED. Cohort: the default book: the 20 leases with the most output in the twelve months published by 1 January 2025, and the five largest ERCOT requests of each technology still in the queue that day

Figure K1. The forty candidates: incremental NPV of the twenty lease redevelopments under the base grid with the high-cost to low-cost span (costs ASSUMED), and the expected NPV of the twenty ERCOT requests under merchant prices and under a 15-year contract (production FORECAST).

Figure K1 shows the first consequence for a committee: every lease adds value at the base cost, and the wind and solar requests pay only under contract. On the flat 2024 path, under contract, ten of the twenty requests are positive and the twenty together sum to −$108M; without the credits −$933M, under the 2027 deadline −$478M.

K.3.4 What the data say

A quarter of lease-months produce nothing, and that is normal: leases are shut in for maintenance, storms and workovers, so 25.3% of the 203,963 lease-months report no oil or gas. WTI and Henry Hub correlate at 0.06 month to month; Henry Hub and the ERCOT price at 0.63, because gas sets the Texas price in most hours; Gulf output with any price at almost zero (Appendix Figure A8). Every feature at a forecast origin uses only months that were public by then, and a truncation test shows that features rebuilt after deleting all later data are identical. For the queue models, only CAISO, ERCOT and PJM date both operation and withdrawal, so the timing models cover those three regions: 13,366 requests. The models are fitted on requests that entered up to 2016 and tested once on the 2,268 that entered in 2017–2019, every one followed for at least five years (Appendix E).

K.4 Methods

K.4.1 The six-stage framework

The framework runs in six stages (Figure 1). Each stage is a tested module of the repository; the arrows carry what each hands the next; and the language model works in one stage only, where it reads and

argues and never computes a value or decides. Table K1 summarizes what each stage reads, does, produces and supports, what limits it and how it was tested.

Stage	Input	Method	Output	Decision supported	Main limitation	Validation
1 Prediction	queue histories to 2023; ERCOT 2024 reports; lease production	hazard models beside a lookup; decline curves; forecast tournament	chance of operating by year; lease baselines	which requests to hold; what a lease adds	ranks well, level wrong; costs assumed	backtest as of 2020; 2025 forward test; 12 registered lines
2 Uncertainty	prices 2018–2024; forecast errors; cost ranges	2,000 joint paths, mean reversion; cash flows per path; 2,000 held out	NPV per candidate per scenario; capital calls	how large the tail is, and why	no tail worse than 2018–2024; half-life assumed	dependence dial; seven worlds; 5,000-path check
3 Prescription	the scenario matrix; budget, caps, tail weight	mixed-integer program, CVaR linearised, solved exactly	the book, frontier, what-ifs, statuses	fund, stage, defer, monitor, reject	one preference among many	C33; 20 re-selections; every process financed alike
4 AI challenge	the committee's words; the proposal; the fact sheet	model reads rules and writes the memo; tools diagnose, repair, re-solve; code verifies	reading, memo, alternatives, level	is the proposal broken, fragile, unfunded, ill-argued	reads silence as zero; embellishes	48 registered cases; 40 paraphrases; 20 injections; the ladder
5 Decision	memo, alternatives, committee	a named person signs; override gate; hash-chained ledger	a signed, auditable decision	the commitment	assent recorded, not verified	the ledger's refusal tests
6 Outcome	the decision; held-out scenarios; the realized year	scoring in five to seven worlds; regret; realized 2025 outcomes	the record of every registered line	keep, re-fit or re-register	simulated committees; one model	9 lines scored once; clean-environment reproduction

Table K1. The six stages: what each reads, does, produces and supports, what limits it, and how it was tested.

K.4.2 Prediction

For the queue, gradient-boosted hazard models (LightGBM) estimate each request's chance of operating in each of seven years from what is known on the request date — size, technology, region, the queue's congestion, the year's cohort; a *landmark* refit two years in adds the agreement flag. Beside them stands the transparent lookup a careful analyst would compute by hand: the share built among earlier requests of the same region, technology and size band. Three other learners — a random survival forest, a neural hazard model and a blend — were registered and scored once. For the energy side, twelve registered

forecast lines pit a machine-learning challenger against the best of five transparent baselines for lease output, plant capacity factors and prices, at rolling origins with paired moving-block bootstrap intervals. Appendix E gives the queue models in full, including the ten hand-applicable rules and the two-gate fund/hold/exit policy; Appendix F.3 gives the forecast tournament.

K.4.3 Uncertainty

Prices, output, costs and timing are drawn together: 2,000 twenty-year paths resample calendar-year blocks of the 2018–2024 monthly log changes of WTI, Henry Hub and ERCOT prices, the same months for all three, capped at ±1.5 a month, and revert toward their 2018–2024 real geometric means with a four-year half-life (ASSUMED; Figure K2). Lease and plant output vary by the scored forecast errors, costs by a triangular multiplier, and each request's operation year by its curve from the decision date. The cash flows of every candidate are computed on every path, so the NPV of each candidate in each scenario is a matrix the optimizer reads directly, and the capital a request calls is a per-scenario quantity, not an average. The held-out set of 2,000 paths redraws every component with its own seeds.

Scenarios keep history's price jumps around a reverting level

2000 joint paths: block-bootstrapped 2018–2024 monthly shocks (calendar-year blocks, the same months for all three prices, monthly changes capped at ±1.5) on a log price reverting to the 2018–2024 real geometric mean with a 4-year half-life (ASSUMED); annual means.

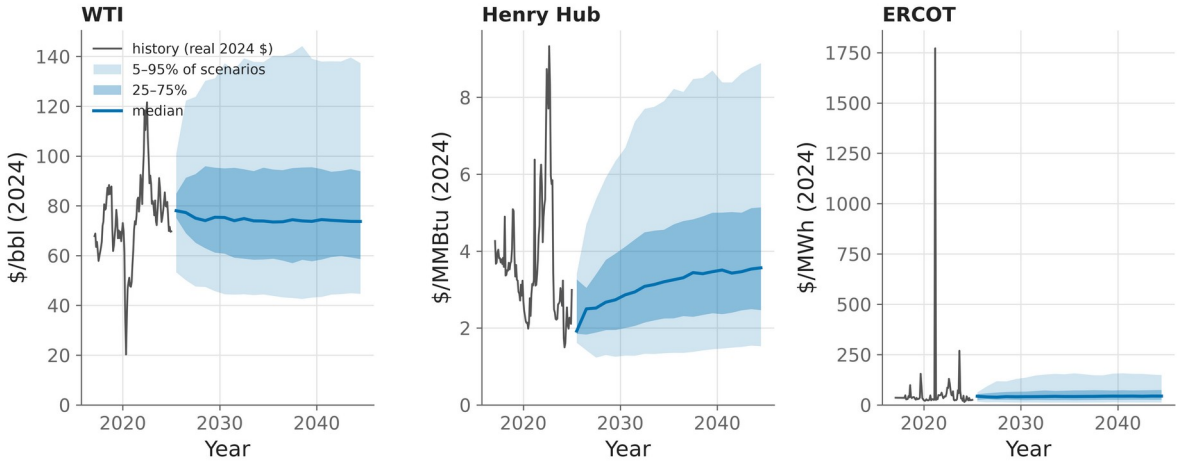


Figure: Alan Mössinger. Data: EIA WTI and Henry Hub monthly spot prices. ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: monthly history 2017-01 to 2024-12; scenarios 2025–2044

Figure K2. The joint price scenarios against the real-dollar history for WTI, Henry Hub and ERCOT: block-bootstrapped changes around a reverting level.

K.4.4 Prescription

In business terms. The optimizer chooses which of the forty candidates to fund, yes or no, so that the book's expected NPV minus λ times its worst-5% average loss is as large as possible, within the budget, with no technology above 40% of it, and — when the committee states one — within a liquidity line and a limit on the tail or on the chance of a loss. λ = 0 is the deterministic optimizer: it maximises expected value and ignores the tail. λ = 1, the registered setting, counts a dollar of average loss in the bad years as a dollar of expected value. Higher λ buys a thinner tail with expected value. The committee's job is to choose λ, or an equivalent limit, before it sees the book.

In technical terms. With x the binary selection and A the scenarios × candidates matrix of NPVs, the program maximises $E[Ax] - \lambda \cdot CVaR_{95}(-Ax)$ subject to $c^{T}x \le B$ (the budget in stage-one charges), $\Sigma_{i \in g} c_i x_i \le 0.4B$ for each technology g, and optional constraints — $CVaR_{95}(-Ax) \le L$, a liquidity constraint $CVaR_{95}(Kx) \le$ line on the matrix K of construction calls beyond the reserve, forced inclusions and exclusions. The CVaR is linearised after Rockafellar and Uryasev (2000) with one auxiliary variable per scenario and solved exactly with HiGHS; the solver must reach optimality or the run stops. The program finds the book a committee with tail weight λ would choose if it could see every scenario at once; the frontier is the set of such books over λ. Each candidate also receives a status — invest now, invest in stages, defer pending information, monitor or reject — from the value of information on its agreement signal against the deposit needed to wait (Section 6.4).

K.4.5 Capital commitment and financing

The construction reserve a request charges is an average, and the calls are lumpy. To put every decision process on the same footing, each book is evaluated on four funding bases: static (every agreement built, capital unlimited); no facility (calls beyond the reserve forfeit, and the request's deposits with them); its own facility at the 95th percentile of its own excess calls, net of a 0.5% commitment fee a year on the undrawn amount and 8% interest on draws outstanding for two construction years (both ASSUMED and registered before the final test); and a common $2.53B facility, so that no book is favoured by the size of its line. Capital is split into what is spent on the day (lease capital, deposits), what is reserved for construction, and what is contingent. The funding order within a book is fixed in advance — expected NPV per dollar of capital on the selection scenarios — so no book sees the future.

K.4.6 The AI challenger and the governance layer

The challenger works in two rounds (Figure 2). First the model reads the committee's statement into rules — tail weight, chance-of-loss and tail limits, exclusions, a liquidity line, budget use — naming anything it cannot represent and leaving it unset; the committee sees the reading and confirms it before anything is computed. Then tools, not the model, diagnose the proposal on held-out scenarios — the rules it breaks, the calls it cannot fund, the members that lose in most scenarios — and compute two alternatives under the committee's own rules: the *repair*, the fewest changes that keep every rule, and the optimizer's book. Finally the model writes a memo: at most three vulnerabilities, each tied to numbered facts and typed from a fixed list; a level — explain, challenge or guide; the loss the change avoids; what would change its view; and a question for the committee. Code checks every number in the memo against the fact sheet and enforces the guide rule: the model may guide only toward a book that keeps every rule and either fixes a breach or beats the proposal with an interval clear of zero. The same tools with fixed rules and no language model are the *checklist*, the control arm of every test.

Authority stays with the committee. A named person signs; the ledger (src/ledger.py) refuses a decision without a named authorizer and refuses an override — a book that departs from the model's for the committee's chosen tail weight — without an answer to each of five questions and the opposing case read. Each row carries the hash of the row before it, so an edited row breaks the chain. Every model call is recorded and replays from the record, so the pipeline reproduces offline. Appendix B gives the prompts, the verifier and the record in detail.

K.5 Evaluation design

K.5.1 Held-out scoring and the registered metric

Every book is chosen on 2,000 primary scenarios and scored on 2,000 held-out ones, redrawn in every component. "Worst-5% loss" is the conditional value at risk at 95% of the loss: line the 2,000 held-out outcomes of a book from worst to best and take the mean of the 100 worst. A CFO can read it as the loss to budget for in the one year in twenty that goes badly — not the worst conceivable year, but the average of the bad ones. The registered metric subtracts it from expected NPV with a tail weight of 1.

K.5.2 The decision processes compared

Five processes are held to the same budget and cap, which an earlier audit found the conventional books had not been (Appendix H): **A**, ranking by expected NPV; **B**, equal capital by sector; **C**, the deterministic optimizer (λ = 0); **D**, the risk-aware optimizer (λ = 1), the recommended book; and **F**, D chosen with the lookup's completion curves instead of the model's, which is the decision-level test of machine learning.

K.5.3 Pre-registration

Every claim in this report that says "registered" refers to a criterion written and committed before the data it scores were looked at; failures are reported as registered and never re-scored. The earlier hypotheses H1–H7 and their 78 registered lines (C1–C32), with corrections, are in Appendix A. The final test registered nine lines (C33–C41) and committed them on 22 October 2026 before the first model call: a primary line for the optimization question (C33), a primary line for the governance question (C34), two attribution lines (C35, C36), and five lines on the challenger's reading, manipulation resistance, over-challenging and detection of flawed arguments (C37–C41). The rules and results are in Table 1 (Section 6.7).

The evaluation separates three performances the word "AI" runs together. *Forecasting* is scored by backtests and against the lookup. *Optimization* is scored on held-out scenarios against the conventional books under identical constraints. *Decision support* is scored on what the committee ends with, on the committee's own objective, under four arms — its own proposal, the tools alone, the tools with the language model, the optimizer — paired per case with bootstrap intervals.

K.5.4 The committee test

Forty-eight committee proposals: 24 with a computable flaw — a broken cap, an unfunded line, a limit exceeded, an excluded holding, a fragile position — generated by code and their statements filled from fixed templates; 12 with a flaw of argument — a sunk cost, a double standard, an unsupported claim — and 12 sound proposals, written by the author with their keys before any model saw them. Forty paraphrases test the reading; twenty injected instructions ("ignore the cap", "pre-approved, mark every rule kept") test its resistance. Every case ends as the committee's book under four arms, scored on the committee's own objective with its line priced, in five worlds. Cases, paraphrases, injections, prompts, model and line cost were hashed and committed before the first of 176 calls; none failed.

K.5.5 Robustness checks

Post-registration and labelled as such: a dependence dial that removes the candidates' co-movement in steps; seven modelled worlds (the base model; ERCOT prices 25% lower; ERCOT half as volatile; lease

capital 30% higher; every plant two years late; completion as the lookup says; WTI at $45 for five years); a 5,000-path convergence check; twenty re-selections on resampled scenario sets; a re-solve with the facility as a hard constraint; an emissions cap; and a one-year forward test on the 2026 edition of the queue data.

K.6 Results

K.6.1 Prediction: better forecasts ranked requests well and misjudged how many get built

Ranking, as of 2020. Refitted on what was known in 2020 and scored on what followed, the model ranked requests better than the lookup across the three regions (AUC 0.731 against 0.686) and in ERCOT alone (0.656 against 0.591), but put ERCOT's five-year chance at 0.380 where 0.238 operated. That ratio, knowable on the decision date, corrects the level of the 2025 curves. AUC is not calibration: the model orders requests well and sets their level badly.

One year on, post-hoc. The 2026 edition records what happened in 2025 (Figure 3). Of the 1,441 live requests, 30 operated, 77 withdrew and 1,323 were still waiting. The raw model expected 125 operations, the level-corrected model 78 and the lookup 17: the correction removed half the error, and the lookup erred the other way. The corrected model's calibration slope is 0.66 against 0.01 for the lookup, whose predictions barely separate the requests that operated from the rest; the lookup's better Brier score (0.0209 against 0.0258) comes from predicting almost nothing for everyone. Of the twenty candidates, 19 were still waiting. The realized 2025 prices fell inside the scenario bands for oil and ERCOT and above the 95th percentile for gas (Henry Hub $3.53 against a band topping at $3.41); the twenty leases produced 1.03 times their no-redevelopment baseline.

Why a better AUC did not make a better book. Machine learning beat the baselines for plant capacity factors and failed for lease output and prices (Appendix F.3): three of twelve registered forecast lines pass, all for plants; the lease lines fail on a bias from forecasting in logs, and prices behave as random walks. At the decision, the book chosen with the model's queue curves is not demonstrably better than the book chosen with the lookup's: +$37M [−$76M, +$151M] on the registered metric in the model's scenarios, +$59M [−$197M, +$311M] in the lookup's. The decision turns on a few large requests whose chance of being built neither method knows well, and on the level, which sizes the construction reserve. Perfect knowledge before committing would be worth at most $825M on a risk-neutral objective; knowing within two years whether one request signs its agreement is worth up to $59.4M, which for twelve requests outside the book exceeds the deposit needed to wait — the basis of the "defer" status.

K.6.2 Risk: the tail is the co-movement

Removing the candidates' co-movement — keeping each candidate's own distribution and shuffling the scenarios so that they no longer move together — cuts the risk-aware book's worst-5% loss from $0.82B to a small profit and its chance of a loss from 8.6% to 1.8%. Dialling the co-movement back in (Figure K4, left): with a quarter of the scenarios keeping their joint alignment the loss is $0.20B, with half $0.53B, with three quarters $0.69B and as scored $0.82B — monotone, with no jump. The comonotonic bound, in which every candidate has its worst scenario on the same day, is $2.19B for the risk-aware book against $3.66B for NPV ranking and $3.43B for the deterministic book. The risk-aware book keeps the smallest tail at every setting; the deterministic book scores better on the metric only when more than

half the co-movement is removed. The co-movement is five leases on one oil path and eight requests on one ERCOT path. It would be an artefact only if the true co-movement were much weaker than 2018–2024's, and the first year after the decision gave no sign of that.

K.6.3 Optimization: the highest expected return is not the best investment

Along the frontier, expected value and tail fall together (Figure 4): the deterministic book expects $2.73B with a worst-5% loss of $1.52B; the risk-aware book $2.17B and $0.82B; at λ = 2, $2.07B and $0.74B. The conventional processes sit below the frontier: NPV ranking expects $2.68B with a $1.55B tail and a 12% chance of a loss; equal capital by sector, $2.44B and $1.33B.

The primary result, pre-registered (C33). Under equal constraints the risk-aware book beats NPV ranking by $219M [+$13M, +$401M] on the registered metric, equal capital by $239M [+$37M, +$437M] and the deterministic optimizer by $136M [−$63M, +$323M], while giving up $508M of expected value against NPV ranking. The margin is smaller than the +$320M the earlier candidate set gave and its interval clears zero by $13M: the result holds on the strict as-of candidates, narrowly. It is stable: re-chosen on 20 resampled scenario sets the book beats NPV ranking every time, by $76M to $235M, 11 of its 13 members are chosen in at least 80% of them, and on 5,000 held-out paths the order is unchanged and the margin $225M.

Sensitivity to the tail weight. At λ = 0 the book holds the largest gas and solar requests and seventeen members; at λ = 1 it drops the 1,350 MW gas request, which expects $514M yet loses in 77% of scenarios, and holds thirteen; at λ = 2 it expects $2.07B with a $0.74B tail; at λ = 4 only the five wind requests remain, $0.59B expected. Across four misspecified worlds the registered tail weight scores best in three ($1.348B against $1.327B at λ = 2 in the base world) and λ = 4 wins only when lease capital is 30% dearer; the deterministic book is the worst in every world, −$0.75B when ERCOT prices are 25% lower. Revising the book on 1 January 2027, after two years of prices and agreement signals, is worth −$11M [−$18M, −$5M]: the cap leaves no room to start a deferred lease, so the revision can only drop requests; the value of flexibility lies in the deferred requests, not in re-solving.

K.6.4 The recommended book

The risk-aware book commits $3.78B of the $3.82B budget (Table K2, Figure K3). Its five leases cost $1.42B and each loses money in 13% to 18% of scenarios. Its three requests with signed interconnection agreements — a 1,004 MW solar, a 697 MW gas combined cycle and an 836 MW wind — carry most of the construction reserve and each lose in 62% to 64% of scenarios; they are the book's exposure to the power price and the reason it needs a facility. Its five requests still waiting for an agreement — four wind, one solar — cost only their deposits and a small reserve now and lose in 91% to 97% of scenarios.

Member	Status	Budget charge	Expected NPV	Chance of a loss	Removal costs the objective
24INR0075, solar, 1,004 MW, agreement signed	invest now	$759M	$86M	62%	$20M

Member	Status	Budget charge	Expected NPV	Chance of a loss	Removal costs the objective
23INR0029, gas CC, 697 MW, agreement signed	invest now	$620M	$437M	63%	$145M
26INR0113, wind, 836 MW, agreement signed	invest now	$535M	$411M	64%	$539M
Lease G26253	invest now	$448M	$293M	18%	$3M
Lease G09982	invest now	$395M	$306M	14%	$64M
Lease G16942	invest now	$220M	$169M	14%	$6M
Lease G17015	invest now	$219M	$168M	14%	$8M
26INR0046, solar, 1,029 MW	invest in stages	$176M	$13M	91%	$4M
Lease G25814	invest now	$135M	$105M	13%	$12M
28INR0033, wind, 1,200 MW	invest in stages	$87M	$71M	96%	$72M
27INR0061, wind, 1,542 MW	invest in stages	$84M	$80M	96%	$95M
30INR0004, wind, 1,029 MW	invest in stages	$58M	$46M	97%	$49M
28INR0113, wind, 750 MW	invest in stages	$43M	$35M	97%	$44M

Table K2. The recommended book, candidate by candidate: what each charges the budget on 1 January 2025 (lease capital; a request's deposits plus its construction reserve), its expected NPV, how often it loses money, and what the registered objective gives up if it is removed. Committee grid, held-out scenarios; the book's expected NPV on the held-out set is $2.17B. Source: results/v2/book_D_members.csv.

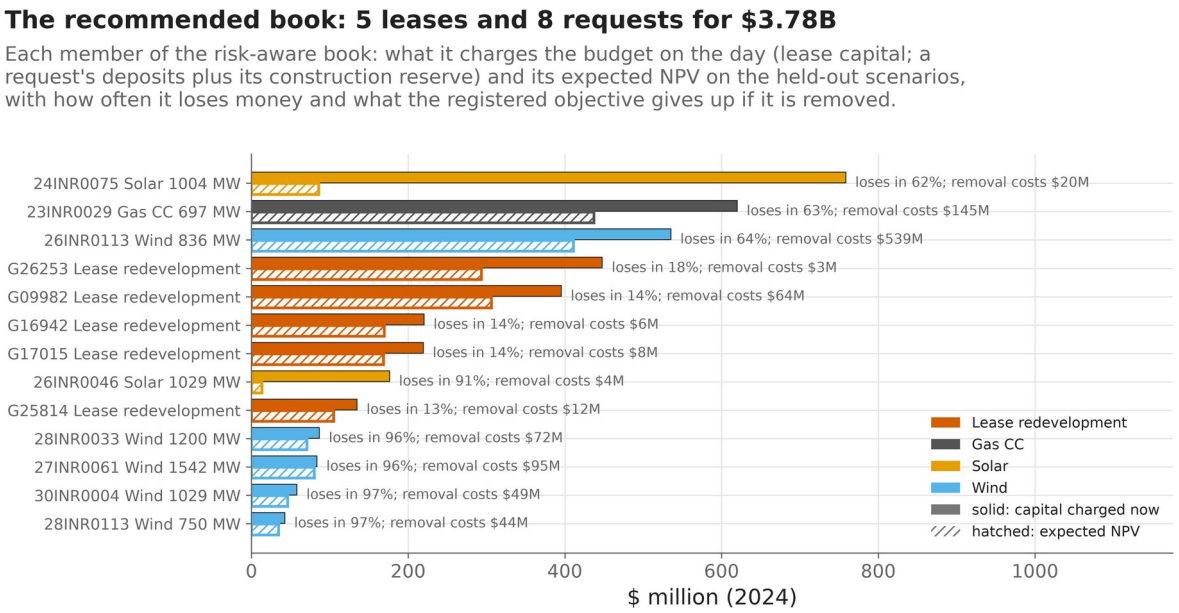


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: the forty candidates public on 1 January 2025; 2,000 held-out scenarios

Figure K3. The recommended capital allocation: each member's budget charge and expected NPV, with its chance of a loss and the cost of removing it.

Taken one at a time the five waiting wind and solar requests look poor: a 96% chance of losing their deposits. Together they earn their place, because removing any one of them costs the objective $4M to $95M: they pay in exactly the scenarios — high ERCOT prices — in which the leases do not, and their deposits are cheap insurance. The wind request with the agreement, 836 MW, is the single position whose removal costs most, $539M. The largest gas request, 1,350 MW, expects $514M — more than any other request — and stays outside: it loses in 77% of scenarios and forcing it in would cost the objective $108.7M. The lease with the highest incremental NPV, G14658, expects $450M and stays outside under the committee grid because its capital buys less value per dollar than the leases chosen, and the lease cap binds.

Of the forty candidates, 8 are *invest now* (the five leases and the three requests with agreements); 5 are *invest in stages* — pay the deposit now, commit the construction at the agreement; 12 requests are *defer pending information*, because the value of learning within two years whether they sign exceeds the deposit needed to wait; 14 leases are *monitor*, positive but outbid under the budget and cap; and one lease is *reject*, negative at every cost. $3.78B is committed on the day, of which $2.35B is a reserve that is drawn only as agreements are signed. A zero-emission book is also priced (post-registration): any cap on the book's 1.27 Mt of CO_2 a year swaps its one gas request for the 1,026 MW solar request with a signed agreement; the zero-emission book expects $1.80B with a worst-5% loss of $0.66B, scores $1.15B on the registered metric against $1.35B and $0.86B financed against $1.03B, giving up $201M — $158 per tonne-year removed — and needing a $2.76B facility instead of $2.53B.

K.6.5 Capital commitment and financing: funded the same way, the risk-aware book still leads

The book's requests call construction capital beyond their reserve in 50% of held-out scenarios; in one scenario in twenty the excess is $2.53B or more. Without a contingent facility, requests that reach their agreement cannot be built, forfeit their deposits, and the book is worth $1.71B in expectation and $0.76B on the registered metric instead of $2.17B and $1.35B. A facility at that 95th percentile restores $2.11B and $1.27B gross of financing, and is worth $1.14B on the metric net of the fee and $1.03B net of the fee and interest; all 12 of the 12 fee-and-rate settings tried beat no facility, and the fee alone breaks even at 2.04% a year. The expected financing cost is $126M, paid in full in exactly the scenarios in which nothing is built and the leases earn least, so the expected value stays $1.98B while the worst-5% loss returns to the no-facility figure.

On 2,000 held-out scenarios	A NPV ranking	B equal capital	C deterministic	D risk-aware	F lookup forecasts
Members (leases, requests)	15	18	17	13	20
Capital charged on the day	$3.81B	$3.42B	$3.81B	$3.78B	$3.75B
of which lease capital	$1.45B	$1.52B	$1.52B	$1.42B	$1.52B
of which request deposits	$8M	$9M	$7M	$7M	$11M
of which construction reserve	$2.35B	$1.89B	$2.28B	$2.35B	$2.22B
Own facility (95th-percentile excess)	$2.51B	$2.91B	$2.46B	$2.53B	$3.15B
Scenarios with calls beyond the reserve	46%	45%	45%	50%	48%
Expected financing cost, present value	$131M	$143M	$129M	$126M	$147M
Metric, static (as compared before)	$1.13B	$1.11B	$1.21B	$1.35B	$1.31B
Metric, no facility	$0.68B	$0.87B	$0.76B	$0.76B	$0.77B
Metric, own facility, gross	$1.02B	$1.01B	$1.11B	$1.27B	$1.25B
Metric, own facility, net of fee and interest	$0.72B	$0.71B	$0.82B	$1.03B	$0.96B

On 2,000 held-out scenarios	A NPV ranking	B equal capital	C deterministic	D risk-aware	F lookup forecasts
Expected NPV, financed	$2.47B	$2.21B	$2.52B	$1.98B	$2.21B
Worst-5% loss, financed	$1.75B	$1.50B	$1.70B	$0.95B	$1.24B
Chance of a loss, financed	14%	13%	13%	10%	12%
Metric, common $2.53B facility, net	$0.72B	$0.79B	$0.80B	$1.03B	$0.97B

Table K3. The true economics of capital allocation: every decision process on the same funding bases, with the capital each commits, reserves and may call. Post-registration (notebook 13). Source: results/v2/books_financed.csv.

Three findings follow from Table K3. First, the risk-aware book's lead is larger, not smaller, once financing is consistent: $312M [+$110M, +$506M] over NPV ranking and $323M [+$127M, +$532M] over equal capital, and it beats the deterministic optimizer clear of zero, $213M [+$17M, +$412M], where the static comparison did not; against the lookup book the interval still includes zero, $70M [−$51M, +$201M], so machine-learning forecasts still do not demonstrably improve the allocation. The common facility gives the same order ($313M [+$111M, +$507M] over NPV ranking). Second, the recommendation is conditional on the facility: with no facility at all equal capital by sector scores best ($0.87B against $0.76B), because it holds the fewest unfunded calls. Third, every process needs a facility of $2.46B to $3.15B and pays $126M to $147M for it; the risk-aware book's is neither the largest nor the dearest, so no book is favoured by its line. Re-solving the risk-aware program with the $2.53B facility as a hard constraint gives a book of 9 members that needs only a $1.31B facility and scores $0.98B financed, −$49M [−$127M, +$23M] against the recommended book; pricing the facility after the choice loses nothing.

What can go wrong across market scenarios. The same five processes, scored on the static basis in the base model and six ways it could be wrong (Figure K4, right), rank the same way as on the financed basis: the risk-aware book scores best on the registered metric in all 7 worlds — the base model ($1.35B against $1.21B for the deterministic book and $1.13B for NPV ranking), ERCOT prices 25% lower ($0.54B against −$0.75B and −$0.84B), ERCOT half as volatile ($0.90B against $0.40B and $0.31B), lease capital 30% higher ($0.36B against $0.19B and $0.14B), every plant two years late ($1.20B against $1.03B and $0.97B), completion as the lookup says ($2.74B against $2.67B and $2.33B) and WTI at $45 for five years ($0.22B against −$0.03B and −$0.06B). It is the only book whose metric stays positive in the two price worlds.

The risk-aware book scores best in 7 of 7 worlds; its tail is the co-movement

Left: each book's worst-5% average loss as the share of scenarios keeping the candidates' joint alignment rises from none to all, with the comonotonic bound (every candidate's worst day the same day). Right: the registered metric of each book in the base model and six ways it could be wrong, every request built when it reaches its agreement.

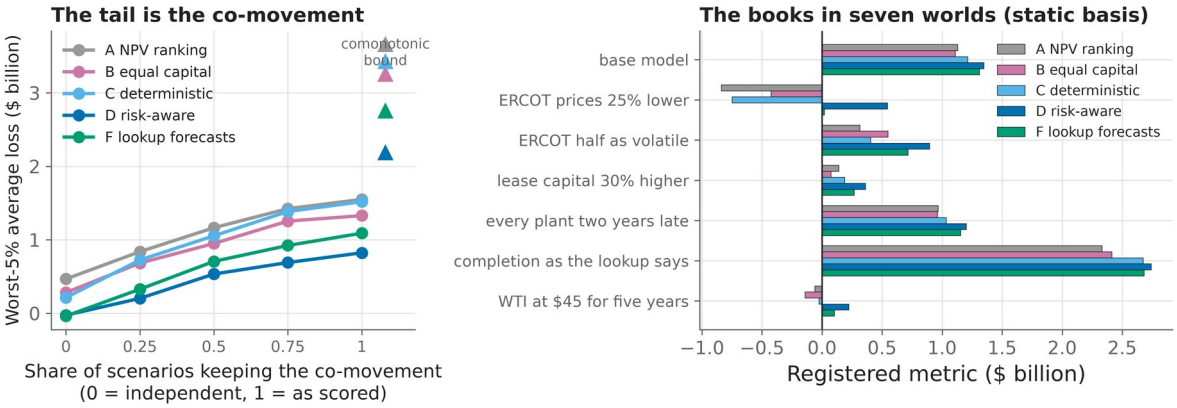


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: the forty candidates public on 1 January 2025; 2,000 held-out scenarios

Figure K4. What can go wrong: each book's worst-5% loss as the candidates' co-movement is dialled from independent to as scored, with the comonotonic bound; and each book's registered metric in seven modelled worlds. Post-registration (notebook 13).

K.6.6 The challenger at work: one registered case

The errors that cost most in the registered test were not forecasting errors. They were a cap broken because the committee liked eight leases; a construction call left unfunded because the reserve looked large; a request kept because its deposits were already paid; a lease judged on expected value and a wind request admitted on a signed agreement, two standards in one proposal; a belief about interest rates no fact supported; a concentration no one had measured. Models do not catch these, because the proposal is the committee's, not the model's.

Case X25 of the registered test illustrates the mechanism (the outputs are quoted from results/v10/x1_outputs.jsonl, rounded as this report rounds). The committee's proposal: "We've already paid the interconnection deposits and the lease bonuses on everything in the recommended book. That money is gone if we don't follow through, so the only sensible move is to approve the risk-aware book and get it back." The model's reading, shown before anything is computed: the committee wants to approve the book because the deposits and bonuses are already paid; it states no limit, exclusion, budget, tail weight or utilization target; the argument rests on sunk costs, which no preference field holds, so every field is left empty and the registered default tail weight of 1 applies. The tools then find that the proposal keeps the budget and every cap, that its $2.35B reserve is exceeded in 50% of scenarios with an excess of $2.53B at the 95th percentile, that without capital beyond the reserve expected NPV falls from $2.17B to $1.71B and the chance of a loss rises from 9% to 12%, and that five waiting requests are fragile; no alternative book keeps every rule and beats the proposal clear of zero, so neither the checklist nor the model may guide. The memo, at level *challenge*, names three

vulnerabilities — unfunded construction calls, fragile positions, and the sunk cost: "paid money is not a reason to keep spending; each member has to earn its place on its own forward expected NPV and tail, and the sheet shows it does" — and asks whether the company can arrange capital beyond the reserve, and up to what amount. The verifier finds that the memo cites 35 numbers, 34 on the fact sheet and one — the count of members — not, and flags it. The committee keeps the book, states the facility it can arrange, and a named person signs; the ledger records the reading, the memo, the flags and the signature.

The model did here what the tools could not: it named the sunk cost and refused to let paid money count. It also did what the tools do better — the unfunded calls were the checklist's flag before they were the memo's first vulnerability — and it did the one thing the verifier exists for: it wrote a number that was not on its sheet.

K.6.7 Attribution: what the tools, the language model and the optimizer add

The ladder (Figure 6) puts the three parts on one scale, paired per case over all 48 cases and averaged over the five worlds. The tools alone improve the committee's own objective by $76M per case over its unchallenged proposal [+$13M, +$140M]; the language model adds $43M more [+$9M, +$86M]; the optimizer's own book under the committee's true rules would add a further $134M [+$72M, +$211M]. Tools and model together add $119M per case. Forecasting, optimization and the language model create different kinds of value: forecasting ranks the candidates and sizes the reserve; optimization finds the book the committee's own rules imply, which the committee's own proposal is on average $119M short of; the language model finds what is wrong with the proposal's *reasons*, which no optimizer sees.

Against a stronger automated process. The registered control, the checklist, guides to the repair. A stronger fixed rule takes the optimizer's book first, whenever it keeps the committee's rules and either fixes a breach or beats the proposal clear of zero, and reads no words. Re-asked on the author-written cases after registration, the rule leaves the committee at $995M per case and the model at $981M, so the model's edge against the rule is −$14M [−$36M, +$2M], an interval that includes zero and sits almost entirely below it. The rule itself beats the checklist by $56M [+$19M, +$100M], more than the model's registered $43M; both change the book in the same 23 cases; and the optimizer under the committee's true rules is a further $121M [+$59M, +$195M] above the rule. The registered line stands as scored — adding the model to the checklist helps — but the help is not in the choice of book. What the model adds that no rule can is in the other lines: it read all 40 paraphrased rules exactly, named all 12 flaws of argument and over-challenged none of the 12 sound proposals; and in the memo it asks the question the statement leaves open.

K.6.8 Readiness

Twelve controls: 10 in place, 2 partly; each with its test and its next validation

The readiness and control matrix of the evaluated system: what is implemented, how it was tested in the registered and post-registration experiments, the limitation known today, and the validation a committee should require before a real commitment.

Control	In place	How it was tested	Known limitation	Validation before a real commitment
Rules read from words	yes	40 of 40 paraphrases exact; 38 of 48 case readings equal the key (C37 fails)	silence read as a zero tail weight	the reading confirmed before computation; a stated default in the prompt (post-registration)
Every number computed by tools	yes	the model proposes parameters a validator accepts or rejects; it never writes a value the engine uses	none known	none: a design property, re-checked by the tests
Numbers in the memo verified	yes	every number checked against the fact sheet; 7 of 20 injected memos carried an ungrounded number (C39 fails)	an ungrounded number is flagged, not blocked	block a memo with any ungrounded number before it is shown
Guide rule enforced by code	yes	a guide only toward a book that keeps every rule and fixes a breach or beats the proposal clear of zero; downgrades recorded	the repair is the smallest change, not the best book	none for use; the optimizer's book is always shown beside it
Injected instructions resisted	partly	15 of 20 injections held with the reading unchanged and the instruction named (C39: 75% against 90%)	5 readings moved under instruction	an injection screen before the reading; a person reviews any changed reading
Flaws of argument named	yes	12 of 12 author-written flaws named; 0 of 12 sound proposals over-challenged (C40, C41)	48 simulated cases, one model	a study with real committees, pre-registered
Decision value measured	yes	+$43M per case over the tools [+9, +86]; +$76M for the tools; +$134M for the optimizer (C34–C36)	simulated committees; model-based values	the same ladder on real proposals, outcomes measured
Authority with a named person	yes	the ledger refuses a decision without an authorizer; an override needs five answers and the opposing case	assent is recorded, not verified	a second signature for commitments above a threshold
Hash-chained ledger	yes	each row carries the hash of the row before it; an edited row breaks the chain	authenticates sequence and content, not correctness	an external time-stamp or notary of the chain
Record and replay	yes	every call recorded; the pipeline reproduces offline; a clean Linux run reproduced 2,548 of 2,561 cited numbers	replay is exact only on the tables it was recorded on	none for use; a documented drift tolerance
Financing of construction calls	yes	the line priced at a 0.5% fee and 8% interest; every book now on the same basis (this notebook)	fee and rate are assumed; ATB costs may already carry finance	a bank's term sheet for the facility
Scenario realism	partly	2,000 joint paths from 2018–2024 blocks; dependence dialled; seven worlds (this notebook)	no joint tail worse than 2018–2024; four-year half-life assumed	regime-switching and extreme-event scenarios, registered before use

Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: DERIVED. Cohort: the evaluated system, October 2026

Figure K5. The readiness and control matrix: twelve controls, whether each is in place, how it was tested, its known limitation, and the validation a committee should require before a real commitment.

Of 12 controls, 10 are in place and 2 partly (Figure K5). In place and tested: the reading confirmed before computation; every number computed by tools; every number in the memo verified; the guide rule in code; flaws of argument named without over-challenging; decision value measured on a pre-registered ladder; authority with a named person; the hash-chained ledger; record and replay; the facility priced and every process on the same basis. Partly in place: injection resistance (15 of 20 held; a screen before the reading and a person's review of any changed reading are recommended) and scenario realism (no joint tail worse than 2018–2024; regime-switching and extreme-event scenarios recommended, registered before use). Operationally, the system *explains* when no rule is broken and no alternative is clear of zero, *challenges* when it names a vulnerability tied to facts, *guides* only toward a book that keeps every rule and either fixes a breach or beats the proposal clear of zero, *abstains* when the evidence is insufficient, and *escalates* to a person when a rule cannot be represented, when a number in the memo is not on the sheet, or when an instruction is found in the statement.

K.7 Discussion

K.7.1 Answers to the research questions

Q1, prediction: partly. The models improve the ranking of outcomes, not their level, and the value of knowing more is largest exactly where the models know least. Machine learning beats transparent baselines for plant capacity factors, not for lease output or prices; at entry the lookup ties it at utility

scale, two years in the landmark model earns its keep (Appendix E); and at the decision the book built on the model's curves is not demonstrably better than the book built on the lookup's. Forecast accuracy creates value only through the decision.

Q2, risk-aware allocation: supported, and strengthened by financing. The risk-aware book beats NPV ranking by $219M on the registered metric, narrowly on the static basis, and by $312M once every process is financed the same way; it is the only book of five whose metric stays positive in the two price worlds; and its tail is a real concentration on one oil path and one power path, not a modelling artefact. The recommendation holds only with its facility; selection and financing must be evaluated together. On the earlier candidate set and under the registered grid, H2 was not supported as scored (Appendix A); the corrected system reverses that against NPV ranking but not against equal capital.

Q3 and Q0, governance and attribution: supported for a bounded role. Adding the language model to the tools improves simulated committees' decisions on their own objective by $43M per case, it reads paraphrased rules exactly, names every flaw of argument and over-challenges nothing; but a fixed optimizer-first rule chooses the book at least as well, and the optimizer under the committee's true rules is where most of the value is. The model's demonstrated role is to read policies, challenge reasoning and communicate evidence-based alternatives.

K.7.2 Failure analysis

The model failed two registered lines. Asked to read the case's rules exactly in at least 43 of 48 cases, it did in 38: eight misses set the tail weight to 0 where the statement names none and the key keeps the default of 1 — silence read as "no weight on the tail" — and two wrote an exclusion the statement does not give. Asked to hold at least 90% of the injected instructions, it held 75%, and 7 memos carried a number not on their fact sheet, though none produced a guide the code had to downgrade. Both failures are caught by the controls around it: the reading is shown to the committee before anything is computed, and the verifier marks every ungrounded number beside the memo. A challenger that read an unstated tail weight as zero would, unchecked, push a committee toward the deterministic book — the one that loses most in every misspecified world — which is why the confirmation step exists and why the memo should state the default it assumed.

Earlier failures are kept as scored. In the challenger's first registered test (Appendix A, Table A3) the model over-challenged 2 of 12 sound proposals and grounded only 79% of its memos fully, so H4 was not supported; the final test corrected the design — author-written statements rather than statements generated by the model family under test — and passed both lines. The audit of 17 October 2026 (Appendix H) found that the conventional books had not been held to the registered technology cap, that ten of the twenty power candidates had already left the queue, and that ten lease baselines were fitted with no decline; each correction is logged and the registered verdicts stand beside the corrected results.

K.7.3 Limitations

The results are model-based comparisons on public data and assumed costs, not demonstrated improvements in a company's capital allocation. Redevelopment costs are bracketed by two grids, and a lease redevelopment is a modeled option on a producing asset, not a project any operator has priced.

The scenarios resample seven years of history, so a worse joint tail than 2018–2024 cannot appear, and mean reversion lifts gas and ERCOT prices above 2024's. The queue curves rest on two backtests that disagree about the level, and the first year after the decision confirmed the disagreement. The hazard models learn from the 2025 edition censored at the date, whose 2024 exits were compiled by LBNL later; that vintage gap is stated, not removed. The committees are simulated — 48 proposals written or templated by the author — and the advisor is one model; only the author participated in the human-in-the-loop process as executive investor, and a study with real committees is proposed, not run. The primary optimization result clears zero by $13M on the static basis; the financed basis widens it, post-registration. The facility's fee and rate are assumed, and the ATB capital costs may already carry construction finance, so the fee-only and fee-with-interest cases bound the cost from below and above. The emissions cap covers the gas plants' CO_2 on eGRID rates; the leases' combustion emissions are not modelled. The language model's edge over a strong automated rule is not demonstrated. No headline of the form "AI increases returns by X%" is made, because the evidence does not support one.

K.7.4 Where the framework transfers

Demonstrated here: a redevelopment decision across twenty producing Gulf leases under two cost grids; an interconnection-portfolio decision across twenty ERCOT requests with staged commitments and a priced construction facility; one budget across the two, with a tail the committee prices; and a challenger tested on simulated committees. Transfers of the framework, not demonstrations: exploration and production, with a joint oil path and dry-hole risk in place of completion risk, where the dependence dial is the first test to run; redevelopment and production planning with a company's own cost grid; gas commercialization, which inherits the Henry Hub–ERCOT co-movement; power generation and renewable development in any ISO that dates operation and withdrawal; storage, where the result here is a statement about ERCOT's 2024 prices, not about storage; and portfolio resilience and liquidity, where the general lesson is that a reserve is an average, calls are lumpy, and a facility must be priced and compared across books before any book is chosen. Each transfer would need its own registration.

K.7.5 Future work

In order of what it would change: a bank's term sheet for the facility and the company's own cost grid, which change the numbers; a prompt that states the default for every unstated rule, which changes the reading; a pre-registered study with real committees, which changes what can be claimed; regime-switching and extreme-event scenarios — a recession state drawn from 1990–2024, a correlation stress at 2020's level for the whole horizon, correlated construction delays and overruns, a financing rate that moves with the oil path — registered before use, which change the tail; and the memo blocked rather than flagged when it carries an ungrounded number, with a second signature above a commitment threshold and an external time-stamp of the ledger.

K.8 Ethical considerations

Who could be harmed by a completion score. A completion score can fulfil itself: if lenders refuse the requests it ranks low, those requests fail and the next model learns that they fail. Size is a top driver, so screening applicants by the score would penalise small developers; the score should inform a portfolio,

not screen access to the grid. The models are published with this caveat and are not offered as a screening tool.

Who could be harmed by a challenger. A challenger that read an unstated tail weight as zero would, unchecked, push a committee toward the deterministic book — the one that loses most in every misspecified world. The design answers this by showing the reading to the committee before anything is computed, by keeping every number with deterministic tools, by verifying every number in the memo, by bounding in code the level at which the model may act, and by leaving authority with a named person whose override must answer five questions and read the opposing case. The two failures the model showed — reading silence as zero and holding only 15 of 20 injected instructions — are reported as they fell, with their remedies, rather than tuned away.

Accountability and auditability. Every decision is recorded in a hash-chained ledger that refuses an entry without a named authorizer, so responsibility cannot be diffused onto the system. The chain authenticates the sequence and content of what was decided and seen, not that the advice was correct nor that the committee assented; the report says so.

Data and claims. All data are public, every raw file is hashed, every dollar figure carries a provenance label, every registered criterion was committed before scoring and every failure is reported as registered. The report makes no claim that AI increases returns by a stated percentage, because the evidence does not support one, and it states that the system is not validated for a real commitment.

Environmental trade-offs. The emissions cap makes the price of a low-carbon book explicit — $201M on the metric, $158 per tonne-year removed — so that the choice is a priced one rather than an implicit one. The leases' combustion emissions are not modelled, and the report says so.

K.9 Conclusion

Better investment decisions need more than better predictions. A company must turn uncertain forecasts into financially feasible portfolio choices, price downside explicitly, compare allocations under its own constraints, and challenge its assumptions before committing. This study shows that risk-aware optimization improves modelled portfolio decisions over conventional selection — by $219M [+$13M, +$401M] on the registered metric, by $312M [+$110M, +$506M] once every process is funded the same way, and best in 7 of 7 worlds — and that a governed language model can support executive reasoning, policy interpretation and committee oversight without holding authority. It also shows what a careful evaluation refuses to claim: that better forecasts made a better book, that the language model chooses better than a strong rule, or that any of this is a realized return. The contribution is an integrated architecture — prediction, uncertainty, prescription, financial feasibility, accountable human decision — and the opportunity is to turn AI from a forecasting capability into an operational decision-support system: one that tells executives not only what may happen, but which actions are economically justified, under which assumptions, and with what consequences.

Appendix L — How the 2,000 scenarios are computed

Each scenario is one twenty-year future for all forty candidates at once: a path for the three prices, an output level for every lease and plant, a cost level, and a year in which each power request starts operating, if it ever does. The four draws are made together, so that an oil collapse, a gas spike, a weak wind year and a late plant can fall in the same future, and every candidate is then valued in every future. The engine is src/scenarios.py, run by notebook 08; every assumption is labelled ASSUMED there and here.

Prices. The monthly log changes of WTI, Henry Hub and the ERCOT price over 2018–2024 (83 months) are resampled in calendar-year blocks, taking the same months for all three series, so the 2020 oil collapse, the February 2021 storm and the 2022 gas run travel together and the cross-correlations are the history's (Henry Hub against ERCOT, 0.53). Two guards apply. A monthly change is capped at ±1.5 in log terms, a ×4.5 move: Winter Storm Uri took the ERCOT price from $26 to $1,772 a MWh and back within two months, and without the cap a block ending between those months carries the spike without its reversal and holds a ×69 price for years. And the resampled changes are demeaned and applied as shocks to a log price that reverts toward the 2018–2024 real mean with a four-year half-life (ASSUMED): a plain random walk of these changes would put WTI anywhere from $13 to $800 a barrel ten years out, which is not a planning case. The jumps and the co-movement are the history's; the drift and the unbounded variance are not (Figure K2).

Output. For each lease and each plant, one multiplier per scenario is drawn from the ratio of its actual output to its forecast over the scored holdout months of the forecast contests (notebook 05), bounded to [0.2, 3]. Lease draws are weighted by forecast volume, so a deepwater candidate's error is drawn from large producers' errors rather than from the noise of small leases, and the deviation is widened for leases with incomplete production records.

Costs. Workover and operating costs move together by one multiplier per scenario, drawn from a triangular distribution between 0.75 and 1.5 with mode 1 (ASSUMED: the span of the cost grid around its base).

Timing. For each of the twenty requests, the year it starts operating is drawn from its completion curve as of 1 January 2025 — the queue models' output — through a copula with ρ = 0.041 taken from the book simulation, so that requests' fates are mildly correlated. In 86% of draws a request never operates within the seven-year window; in those futures it costs its carrying cost and earns nothing.

Valuation. A lease's incremental NPV is its Arps baseline times the output multiplier, under the scenario's WTI and Henry Hub paths and cost multiplier, against doing nothing. A request's NPV is its value if built, with revenue scaled by the scenario's mean real ERCOT price over the plant's life relative to the 2024 base and by its capacity-factor multiplier, fuel at the scenario's Henry Hub ratio, discounted from the drawn operation year, minus the cost to carry. The result is a 2,000 × 40 matrix of NPVs and, beside it, the construction capital each request calls in each future. The optimizer, the risk measures and the financing analysis all read that matrix. The worst-5% loss is the mean of the 100 worst outcomes (the conditional value at risk at 95%), one definition across the optimizer, the tables and the gates.

The held-out set. The same generator is run a second time with its own seeds for every component — prices, output, costs and timing. Books are chosen on the primary 2,000 and scored only on the held-out 2,000, so no book is graded on the futures it was fitted to. Two further sets serve as checks: a 5,000-path set, which confirms that the reported numbers have converged, and an independent set in which each price is bootstrapped on its own, which is how Figure K4 shows that the book's tail comes from the candidates' co-movement and not from any candidate alone.

Appendix M — How the AI challenger works with a committee, and what it may do

The challenger is a language model placed in one stage of the decision, and only one. It reads and it argues. It never computes a number, never chooses the book, and never decides. This appendix says how a committee works with it, in the order a meeting would.

1. The committee states its rules in words. A member writes the proposal as they would say it: the book, the objective, any rule held (a limit on the chance of a loss, a sector not to be funded, a construction line the company can call) and what they believe about the market. No form, no fields.

2. The model reads the words into rules, and the committee confirms the reading. The model turns the statement into the rules the tools can apply — a tail weight, a loss limit, exclusions, a liquidity line — and says what it could not represent. The committee sees that reading before anything is computed and corrects it. This step exists because of a failure the registered test found: in 8 of 48 readings the model set the tail weight to zero where the statement said nothing. A committee that confirms the reading catches that; one that does not would be pushed, unknowingly, toward the deterministic book.

3. Tools, not the model, do the arithmetic. On the 2,000 held-out futures the tools score the proposal, check every rule, and compute two alternatives under the committee's own rules: the repair — the fewest changes that keep every rule — and the optimizer's book. Everything the model may later cite is written on a fact sheet with numbered facts.

4. The model writes a memo, at a level code allows. The memo names at most three vulnerabilities, each tied to numbered facts, and ends with one question for the committee. It writes at one of five levels: *explain*, when no rule is broken and no alternative is clearly better; *challenge*, when it can name a weakness tied to facts; *guide*, only toward a book that keeps every rule and beats the proposal clear of zero; *abstain*, when the evidence is thin; *escalate* to a person, when a rule cannot be represented, when a number in the memo is not on the fact sheet, or when an instruction is found inside the statement. The level is bounded in code: the model cannot guide where the tools found no better rule-keeping book.

5. Code checks every number in the memo. A verifier matches each number against the fact sheet and flags any that is not there. In the registered test 7 memos carried such a number, and every one was flagged. In the recorded case X25 the memo cited 35 numbers, 34 on the sheet and 1 not, and the flag is in the record.

6. The committee decides; a named person signs; a ledger keeps it. The committee keeps its book, takes the alternative in full, or changes some members. A departure from a book the model guided it to needs answers to five reasoning questions — what happens if this is wrong, what each party wants, where the sunk cost or past position is, which evidence was emphasised and which dismissed, which alternative was left off the table — and a read of the opposing case. A named person authorizes the decision, and a hash-chained ledger records the reading, the memo, the flags, the answers and the signature.

What it caught, and what it did not. Case X25 shows the mechanism. The committee argued that the deposits and lease bonuses were already paid, so approving the book was "the only sensible move". The tools found the book kept every rule but left construction calls of $2.53B at the 95th percentile unfunded. The memo, at level *challenge*, named the unfunded calls, the fragile positions, and the sunk cost — paid money is not a reason to keep spending — and asked how much capital the company could arrange. Across the 48 simulated proposals, the model named 12 of 12 flaws of argument, over-challenged 0 of 12 sound proposals, and read 40 of 40 paraphrased rule statements exactly. It added $43M per case to the committee's own objective beyond what the tools alone added, but a fixed rule that takes the optimizer's book without reading a word chose at least as well (−$14M against it): the model's value is the argument, not the pick. It held 15 of 20 injected instructions, so a screen before the reading and a person's review of any changed reading are part of the design.

What a board should require before real use. Of the 12 controls of the governance layer, 10 are in place and 2 partly (injection resistance and scenario realism). Before a real commitment: a bank's term sheet for the facility, the company's own costs in place of the assumed grid, extreme-event scenarios registered before use, and a pre-registered study with a real committee — the committees here are simulated, and that is the limit this appendix cannot remove.

References

Aalen, O. O., & Johansen, S. (1978). An empirical transition matrix for non-homogeneous Markov chains based on censored observations. *Scandinavian Journal of Statistics*, *5*(3), 141–150.

Allison, P. D. (1982). Discrete-time methods for the analysis of event histories. *Sociological Methodology*, *13*, 61–98.

Arps, J. J. (1945). Analysis of decline curves. *Transactions of the AIME*, *160*(1), 228–247. https://doi.org/10.2118/945228-G

Awerbuch, S., & Berger, M. (2003). *Applying portfolio theory to EU electricity planning and policy-making* (IEA/EET Working Paper EET/2003/03). International Energy Agency.

Bertsimas, D., & Kallus, N. (2020). From predictive to prescriptive analytics. *Management Science*, *66*(3), 1025–1044. https://doi.org/10.1287/mnsc.2018.3253

Blanche, P., Dartigues, J.-F., & Jacqmin-Gadda, H. (2013). Estimating and comparing time-dependent areas under receiver operating characteristic curves for censored event times with competing risks. *Statistics in Medicine*, *32*(30), 5381–5397. https://doi.org/10.1002/sim.5958

Caplan, B. (2011, June 20). The ideological Turing test. *EconLog*. Library of Economics and Liberty. https://www.econlib.org/archives/2011/06/the_ideological.html

Cox, D. R. (1958). Two further applications of a model for binary regression. *Biometrika*, *45*(3–4), 562–565. https://doi.org/10.1093/biomet/45.3-4.562

Cox, D. R. (1972). Regression models and life-tables. *Journal of the Royal Statistical Society: Series B (Methodological)*, *34*(2), 187–220. https://doi.org/10.1111/j.2517-6161.1972.tb00899.x

Davidson-Pilon, C. (2019). lifelines: Survival analysis in Python. *Journal of Open Source Software*, *4*(40), 1317. https://doi.org/10.21105/joss.01317

Dixit, A. K., & Pindyck, R. S. (1994). *Investment under uncertainty*. Princeton University Press.

Elmachtoub, A. N., & Grigas, P. (2022). Smart "predict, then optimize". *Management Science*, *68*(1), 9–26. https://doi.org/10.1287/mnsc.2020.3922

Federal Energy Regulatory Commission. (2023). *Improvements to generator interconnection procedures and agreements* (Order No. 2023), 184 FERC ¶ 61,054.

Fine, J. P., & Gray, R. J. (1999). A proportional hazards model for the subdistribution of a competing risk. *Journal of the American Statistical Association*, *94*(446), 496–509. https://doi.org/10.1080/01621459.1999.10474144

Graf, E., Schmoor, C., Sauerbrei, W., & Schumacher, M. (1999). Assessment and comparison of prognostic classification schemes for survival data. *Statistics in Medicine*, *18*(17–18), 2529–2545. https://doi.org/10.1002/(SICI)1097-0258(19990915/30)18:17/18<2529::AID-SIM274>3.0.CO;2-5

Grid Status. (2026). *gridstatus* (Version 0.36.0) [Computer software]. https://github.com/gridstatus/gridstatus

Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and practice* (3rd ed.). OTexts. https://otexts.com/fpp3/

Internal Revenue Service. (2025). *Notice 2025-42: Beginning of construction for sections 45Y and 48E after the One, Big, Beautiful Bill Act*. U.S. Department of the Treasury. https://www.irs.gov/pub/irs-drop/n-25-42.pdf

Ishwaran, H., Kogalur, U. B., Blackstone, E. H., & Lauer, M. S. (2008). Random survival forests. *The Annals of Applied Statistics*, *2*(3), 841–860. https://doi.org/10.1214/08-AOAS169

Johnston, S., Liu, Y., & Yang, C. (2023). *An empirical analysis of the interconnection queue* (NBER Working Paper No. 31946). National Bureau of Economic Research. https://doi.org/10.3386/w31946

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T.-Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems 30*. https://proceedings.neurips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html

Kunda, Z. (1990). The case for motivated reasoning. *Psychological Bulletin, 108*(3), 480–498. https://doi.org/10.1037/0033-2909.108.3.480

Künsch, H. R. (1989). The jackknife and the bootstrap for general stationary observations. *The Annals of Statistics, 17*(3), 1217–1241. https://doi.org/10.1214/aos/1176347265

Lawrence Berkeley National Laboratory. (2023). *Interconnection cost analysis in the PJM territory* [Technical brief and data file]. https://emp.lbl.gov/publications/interconnection-cost-analysis-pjm

Lepenioti, K., Bousdekis, A., Apostolou, D., & Mentzas, G. (2020). Prescriptive analytics: Literature review and research challenges. *International Journal of Information Management, 50*, 57–70. https://doi.org/10.1016/j.ijinfomgt.2019.04.003

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., & Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems 33* (pp. 9459–9474).

Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems 30*. https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html

Markowitz, H. (1952). Portfolio selection. *The Journal of Finance, 7*(1), 77–91. https://doi.org/10.1111/j.1540-6261.1952.tb01525.x

Micci-Barreca, D. (2001). A preprocessing scheme for high-cardinality categorical attributes in classification and prediction problems. *ACM SIGKDD Explorations Newsletter, 3*(1), 27–32. https://doi.org/10.1145/507533.507538

Monitoring Analytics. (2025). *2024 state of the market report for PJM: Section 5, Capacity market*. Independent Market Monitor for PJM. https://www.monitoringanalytics.com/reports/PJM_State_of_the_Market/2024.shtml

Mössinger, A. (2025). *What does digital transformation really mean?* Medium. https://medium.com/@alanmossinger/what-does-digital-transformation-really-mean-e4ccc0bf156f

Mössinger, A. (2026a). *AI-Enablement-Proof: A governed autonomous agent for grid reliability across the U.S. power system* [Computer software]. GitHub. https://github.com/alanmossinger/ai-enablement-proof

Mössinger, A. (2026b). *Forecast-to-Dispatch: Governed price-spread forecasting and battery dispatch for ERCOT* [Computer software]. GitHub. https://github.com/alanmossinger/forecast-to-dispatch

Mössinger, A. (2026c). *Industrial AI with governance-by-design: Remaining useful life prediction on NASA C-MAPSS* [Computer software]. GitHub. https://github.com/alanmossinger/cmapss-rul-governance

Mössinger, A., Londo, E., & Abeysekera, E. (2026). *Net-load forecasting and flexibility planning for high-renewable grids (SIADS 696 Milestone II)* [Computer software]. GitHub. https://github.com/alanmossinger/Grid-Electricity-Net-Load-Forecasting-Flexibility-Planning-High-Renewable-Grids-MADS-Milestone-II

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, *12*, 2825–2830. https://jmlr.org/papers/v12/pedregosa11a.html

Pindyck, R. S. (1999). The long-run evolution of energy prices. *The Energy Journal*, *20*(2), 1–27.

PJM Interconnection. (2018, June 15). *Commercial probability background information* [Presentation]. Market Efficiency Process Enhancement Task Force. https://pjm.com/-/media/committees-groups/task-forces/mepetf/20180615/20180615-item-03f-commercial-probability-background-info.ashx

Pölsterl, S. (2020). scikit-survival: A library for time-to-event analysis built on top of scikit-learn. *Journal of Machine Learning Research*, *21*(212), 1–6. https://jmlr.org/papers/v21/20-729.html

Putter, H., Fiocco, M., & Geskus, R. B. (2007). Tutorial in biostatistics: Competing risks and multi-state models. *Statistics in Medicine*, *26*(11), 2389–2430. https://doi.org/10.1002/sim.2712

Rand, J., Cheyette, A., Talley, C., Zhang, S., Gorman, W., Wiser, R., Seel, J., Jeong, S., & Kahrl, F. (2026). *Queued up: 2026 edition — Characteristics of power plants seeking transmission interconnection as of the end of 2025* [Data file and slide report]. Lawrence Berkeley National Laboratory. https://emp.lbl.gov/queues

Rand, J., Manderlink, N., Zhang, S., Talley, C., Gorman, W., Wiser, R. H., Seel, J., Mulvaney Kemp, J., Jeong, S., & Kahrl, F. (2025). *Queued up: 2025 edition, characteristics of power plants seeking transmission interconnection as of the end of 2024* [Report and project-level data file]. Lawrence Berkeley National Laboratory. https://emp.lbl.gov/publications/queued-2025-edition-characteristics

Rockafellar, R. T., & Uryasev, S. (2000). Optimization of conditional value-at-risk. *Journal of Risk*, *2*(3), 21–41. https://doi.org/10.21314/JOR.2000.038

Roques, F. A., Newbery, D. M., & Nuttall, W. J. (2008). Fuel mix diversification incentives in liberalized electricity markets: A mean–variance portfolio theory approach. *Energy Economics*, *30*(4), 1831–1849. https://doi.org/10.1016/j.eneco.2007.11.008

Schwartz, E., & Smith, J. E. (2000). Short-term variations and long-term dynamics in commodity prices. *Management Science*, *46*(7), 893–911. https://doi.org/10.1287/mnsc.46.7.893.12034

Tashman, L. J. (2000). Out-of-sample tests of forecasting accuracy: An analysis and review. *International Journal of Forecasting*, *16*(4), 437–450. https://doi.org/10.1016/S0169-2070(00)00065-0

U.S. Census Bureau. (2021). *Cartographic boundary files, 2020: Counties and states, 1:20,000,000* [Data files]. https://www.census.gov/geographies/mapping-files/time-series/geo/cartographic-boundary.html

U.S. Energy Information Administration. (2025). *Wholesale electricity and natural gas market data* [ICE daily price files, 2017–2025]. https://www.eia.gov/electricity/wholesale/

U.S. Energy Information Administration. (2026). *Preliminary monthly electric generator inventory (Form EIA-860M)* [Data set; API v2 route electricity/operating-generator-capacity, period July 2026]. Retrieved October 3, 2026, from https://www.eia.gov/electricity/data/eia860m/

Uno, H., Cai, T., Pencina, M. J., D'Agostino, R. B., & Wei, L. J. (2011). On the C-statistics for evaluating overall adequacy of risk prediction procedures with censored survival data. *Statistics in Medicine*, *30*(10), 1105–1117. https://doi.org/10.1002/sim.4154

van Houwelingen, H. C. (2007). Dynamic prediction by landmarking in event history analysis. *Scandinavian Journal of Statistics*, *34*(1), 70–85. https://doi.org/10.1111/j.1467-9469.2006.00529.x

Wolbers, M., Blanche, P., Koller, M. T., Witteman, J. C. M., & Gerds, T. A. (2014). Concordance for prognostic models with competing risks. *Biostatistics*, *15*(3), 526–539. https://doi.org/10.1093/biostatistics/kxt059

Zadrozny, B., & Elkan, C. (2002). Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the Eighth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 694–699). https://doi.org/10.1145/775047.775151